CHAPTER 9

VECTORS IN THE PLANE AND IN SPACE

Mathematics is the gate and key of the sciences.

Roger Bacon
Opus Majus, Part 4 (1897)

PREVIEW

In this chapter, we focus on various algebraic and geometric aspects of vector representations, and then in the next chapter, we see how vectors can be combined with calculus to study motion in space and other applications.

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A Lamborghini Vector M12

PERSPECTIVE

A *scalar* quantity is one that can be described in terms of magnitude alone, whereas a *vector* quantity requires both magnitude and direction. Force, velocity, and acceleration are common vector quantities, and an important goal of this chapter is to develop methods for dealing with such quantities.

This chapter forms the building blocks for an important part of calculus introduced in the next chapter, *vector calculus*. This chapter and the next will complete what is known as single-variable calculus.

CONTENTS

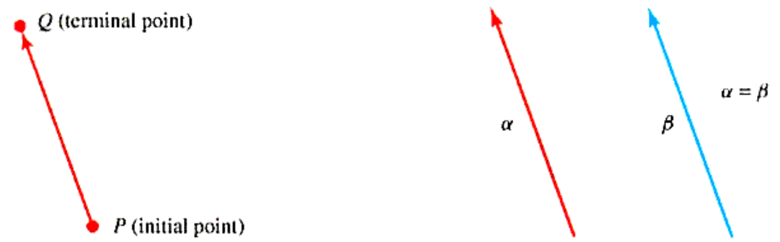
9.1 Vectors in $\mathbb{R}^2$

IN THIS SECTION: *Introduction to vectors, vector operations, standard representation of vectors in the plane*

Many applications of mathematics involve quantities that have both magnitude and direction, such as force, velocity, acceleration, and momentum. Vectors are an important tool in mathematics, and in this section we introduce you to the terminology and notation for vector representation.

Introduction to Vectors

A **vector** is a quantity (such as velocity or force) that has both magnitude and direction. We sometimes represent a vector as a directed line segment, an "arrow" with **initial point** P and **terminal point** Q. The direction of the vector is that of the arrow, and its magnitude is represented by the arrow's length, as shown in Figure 9.1**a**. We shall indicate such a vector by writing **PQ** in boldface print, but in your work you may write an arrow over the designated points: $\overrightarrow{PQ}$. The order of letters you write down is important: **PQ** means that the vector is from P to Q, but **QP** means that the vector is from Q to P. The first letter is always the initial point and the second letter is the terminal point. We shall denote the magnitude (length) of a vector by $\|\mathbf{PQ}\|$. Two vectors are regarded as **equal** (or **equivalent**) if they have the same magnitude and the same direction, even if they do not coincide (see Figure 9.1**b**). Thus, the initial point of a given vector may be moved to any convenient location by translating the vector parallel to its original position.



a. The vector **PQ** has magnitude $\|\mathbf{PQ}\|$

b. Two vectors are equal if they have the same magnitude and direction

Figure 9.1 Vectors in a Plane

A vector with magnitude 0 is called a **zero** (or **null**) **vector** and is denoted by **0**. The **0** vector has no specific direction, and we shall adopt the convention of assigning it any direction that may be convenient in a particular problem.

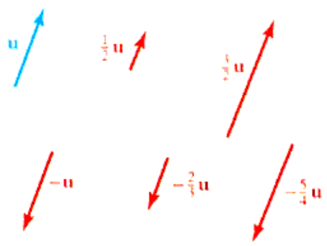
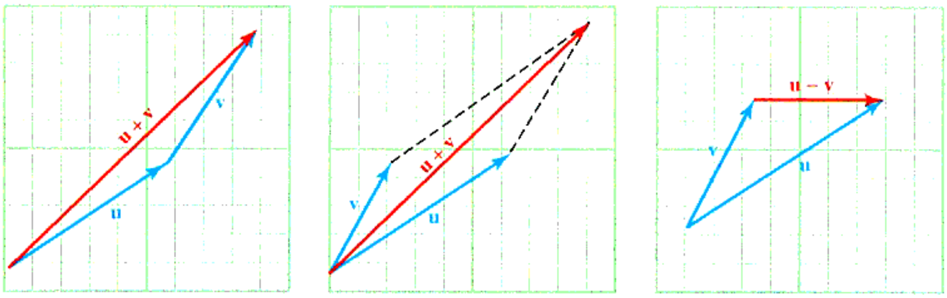


Figure 9.2 Interactive Some multiples of the vector **u**

Vector Operations

Vectors can be multiplied by scalars and added to one another according to the rules developed by physics. If $\mathbf{v}$ is a vector other than $\mathbf{0}$, then any vector $\mathbf{w}$ that is parallel to $\mathbf{v}$ is called a **scalar multiple** of $\mathbf{v}$ and satisfies $\mathbf{w} = s\mathbf{v}$ for some nonzero number s. A **scalar** quantity is one that has only magnitude and, in the context of vectors, is used to describe a real number. The scalar multiple $s\mathbf{v}$ has length $|s|$ times that of $\mathbf{v}$; it points in the same direction as $\mathbf{v}$ if $s > 0$ and the opposite direction if $s < 0$ (see Figure 9.2). Notice that for any distinct points P and Q $\mathbf{PQ} = -\mathbf{QP}$. For the zero vector $\mathbf{0}$, we define $s\mathbf{0} = \mathbf{0}$ for any scalar s.

Physical experiments indicate that force and velocity vectors can be added (or **resolved**) according to a **triangular rule** displayed in Figure 9.3**a**, and we use this rule as our definition of vector addition. In particular, to add the vector $\mathbf{v}$ to the vector $\mathbf{u}$, we place the end (initial point) of $\mathbf{v}$ at the tip (terminal point) of $\mathbf{u}$ and define the sum also called the **resultant** $\mathbf{u}+\mathbf{v}$, to be the vector that extends from the initial point of $\mathbf{u}$ to the terminal point of $\mathbf{v}$.



a. The triangular rule **b.** The parallelogram rule **c.** The difference rule

Figure 9.3 Adding and substracting vectors

Equivalently, the **parallelogram rule** states that $\mathbf{u}+\mathbf{v}$ is the diagonal of the parallelogram formed with sides $\mathbf{u}$ and $\mathbf{v}$, as shown in Figure 9.3**b**. **The difference** $\mathbf{u}-\mathbf{v}$ is just the vector $\mathbf{w}$ that satisfies $\mathbf{v}+\mathbf{w}=\mathbf{u}$, and it may be found by placing the initial points of $\mathbf{u}$ and $\mathbf{v}$ together and extending a vector from the terminal point of $\mathbf{v}$ to the terminal point of $\mathbf{u}$ (see Figure 9.3**c**). Note that $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$. In other words, addition is commutative.

The vector $\mathbf{v}-\mathbf{u}$ may be found similarly and has the same magnitude, but points in the opposite direction to $\mathbf{u}-\mathbf{v}$.

Many issues involving vectors can be simplified by introducing a rectangular coordinate system. Suppose the vector $\mathbf{v}$ is positioned in a rectangular coordinate plane with its initial point at the origin $(0,0)$ and its terminal point at (v_1, v_2), as shown in Figure 9.4. Then v_1 and v_2 are called the **standard components** of $\mathbf{v}$, and we write $\mathbf{v}=\langle v_1, v_2\rangle$. The "angle" brackets "$\langle\ \ \rangle$" are used to distinguish the **component form** $\langle v_1,\ v_2\rangle$ of $\mathbf{v}$ from the point $(v_1,\ v_2)$ in $\mathbb{R}^2$.

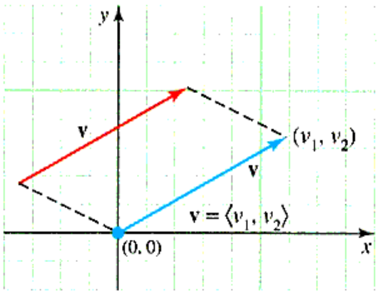


Figure 9.4 The standard component form of a vector v

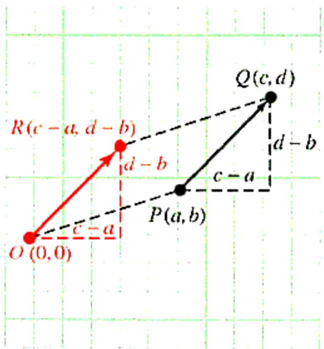


Figure 9.5 Representation of PQ

Using the geometric relationships shown in Figure 9.5, the vector $\mathbf{PQ}$ with initial point $P(a,b)$ and terminal point $Q(c,d)$ equals the vector $\mathbf{OR}$, where $O(0,0)$ is the origin and R is the point with coordinates $(c-a, d-b)$.

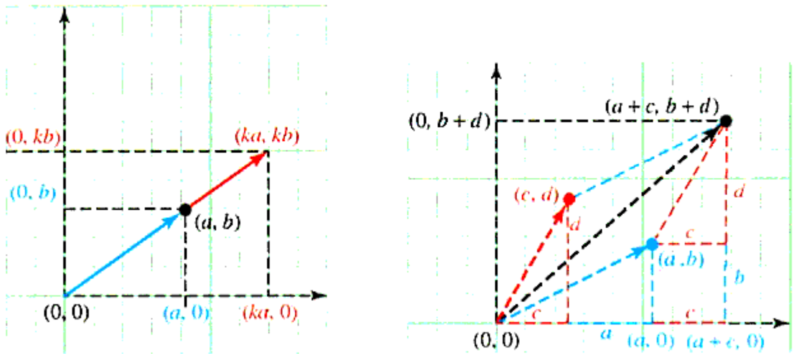
Thus, we can make the following general statement.

STANDARD COMPONENTS OF A VECTOR IN $\mathbb{R}^2$ If $P(a,b)$ and $Q(c,d)$ are points in a coordinate plane, then the vector $\mathbf{PQ}$ has the unique standard component form $\mathbf{PQ}=\langle c-a, d-b\rangle$.

Vector operations are easily represented when vectors are given in component form. In particular, we have

$$\langle a_1, b_1\rangle = \langle a_2, b_2\rangle \qquad \textit{If and only if } a_1 = a_2 \textit{ and } b_1 = b_2$$
$$k\langle a, b\rangle = \langle ka, kb\rangle \qquad \textit{For constant } k$$
$$\langle a, b\rangle + \langle c, d\rangle = \langle a + c, b + d\rangle$$
$$\langle a, b\rangle - \langle c, d\rangle = \langle a - c, b - d\rangle$$

These formulas may be verified by analytic geometry. For instance, the rule for multiplication by a scalar may be obtained by using the relationships in Figure 9.6**a**, and Figure 9.6**b** can be used to obtain the rule for vector addition.



a. Scalar multiplication **b.** Vector addition

Figure 9.6 Vector operations

Example 1 Vector operations

For the vectors $\mathbf{u} = \langle 2, -3\rangle$ and $\mathbf{v} = \langle -1, 7\rangle$, find:

a. $\mathbf{u} + \mathbf{v}$ **b.** $\frac{3}{4}\mathbf{u}$ **c.** $3\mathbf{u} - \frac{1}{2}\mathbf{v}$

Solution

a.
$$\begin{aligned}\mathbf{u} + \mathbf{v} &= \langle 2, -3\rangle + \langle -1, 7\rangle \\ &= \langle 2 + (-1), -3 + 7\rangle \\ &= \langle 1, 4\rangle\end{aligned}$$

b.
$$\begin{aligned}\frac{3}{4}\mathbf{u} &= \frac{3}{4}\langle 2, -3\rangle \\ &= \left\langle \frac{3}{4}(2), \frac{3}{4}(-3)\right\rangle \\ &= \left\langle \frac{3}{2}, \frac{-9}{4}\right\rangle\end{aligned}$$

c.
$$\begin{aligned}3\mathbf{u} - \frac{1}{2}\mathbf{v} &= 3\langle 2, -3\rangle - \frac{1}{2}\langle -1, 7\rangle \\ &= \langle 6, -9\rangle + \left\langle \frac{1}{2}, -\frac{7}{2}\right\rangle && \textit{Scalar multiplication} \\ &= \left\langle 6 + \frac{1}{2}, -9 - \frac{7}{2}\right\rangle && \textit{Add vectors.} \\ &= \left\langle \frac{13}{2}, -\frac{25}{2}\right\rangle && \textit{Simplify.}\end{aligned}$$

In general, an expression of the form $a\mathbf{u} + b\mathbf{v}$ is called a **linear combination** of the vectors $\mathbf{u}$ and $\mathbf{v}$. Note that if $\mathbf{u} = \langle u_1, u_2 \rangle$ and $\mathbf{v} = \langle v_1, v_2 \rangle$, then

$$a\mathbf{u} + b\mathbf{v} = a\langle u_1, u_2 \rangle + b\langle v_1, v_2 \rangle = \langle au_1 + bv_1, au_2 + bv_2 \rangle$$

Vector addition and multiplication of a vector by a scalar behave very much like ordinary addition and multiplication. The following theorem lists several useful properties of these operations.

Theorem 9.1 Properties of vector operations

For any vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in the plane and scalars s and t.

Commutativity of vector addition	$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
Associativity of vector addition	$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
Associativity of scalar multiplication	$(st)\mathbf{u} = s(t\mathbf{u})$
Identity for addition	$\mathbf{u} + \mathbf{0} = \mathbf{u}$
Inverse property for addition	$\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
Vector distributivity	$(s + t)\mathbf{u} = s\mathbf{u} + t\mathbf{u}$
Scalar distributivity	$s(\mathbf{u} + \mathbf{v}) = s\mathbf{u} + s\mathbf{v}$

Proof: Each vector property can be established by using a corresponding property of real numbers. For example, to prove associativity of vector addition, let $\mathbf{u} = \langle u_1, u_2 \rangle$, $\mathbf{v} = \langle v_1, v_2 \rangle$, and $\mathbf{w} = \langle w_1, w_2 \rangle$. Then

$$\begin{aligned}
(\mathbf{u} + \mathbf{v}) + \mathbf{w} &= (\langle u_1, u_2 \rangle + \langle v_1, v_2 \rangle) + \langle w_1, w_2 \rangle \\
&= \langle u_1 + v_1, u_2 + v_2 \rangle + \langle w_1, w_2 \rangle \\
&= \langle (u_1 + v_1) + w_1, (u_2 + v_2) + w_2 \rangle \\
&= \langle u_1 + (v_1 + w_1), u_2 + (v_2 + w_2) \rangle \quad \textit{Associativity of addition for the real numbers} \\
&= \langle u_1, u_2 \rangle + (\langle v_1, v_2 \rangle + \langle w_1, w_2 \rangle) \\
&= \mathbf{u} + (\mathbf{v} + \mathbf{w})
\end{aligned}$$

You are asked to prove the other six properties in the problem set.

Example 2 Vector proof of a geometric property

Show that the line segment joining the midpoints of two sides of a triangle is parallel to the third side and has half its length.

Solution Consider $\triangle ABC$ and let P and Q be the midpoints of sides $\overline{AC}$ and $\overline{BC}$, respectively, as shown in Figure 9.7.

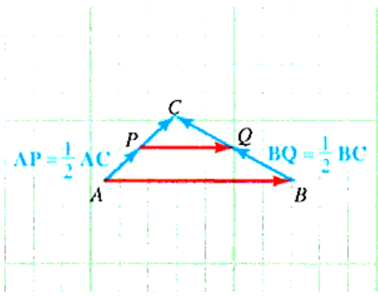


Figure 9.7 Vector proof of a geometric property

Given that $\mathbf{AP} = \frac{1}{2}\mathbf{AC}$ and $\mathbf{BQ} = \frac{1}{2}\mathbf{BC}$ we must prove that $\mathbf{PQ}$ is parallel to $\mathbf{AB}$ and $\|\mathbf{PQ}\| = \frac{1}{2}\|\mathbf{AB}\|$, which means that we must establish the vector equation $\mathbf{PQ} = \frac{1}{2}\mathbf{AB}$. Toward this end, we begin by noting that $\mathbf{AB}$ can be expressed as the following vector sum:

$$\begin{aligned}
\mathbf{AB} &= \mathbf{AP} + \mathbf{PQ} + \mathbf{QB} \\
&= \frac{1}{2}\mathbf{AC} + \mathbf{PQ} - \mathbf{BQ} && AP = \tfrac{1}{2}AC \text{ and } QB = -BQ \\
&= \frac{1}{2}(\mathbf{AB} + \mathbf{BC}) + \mathbf{PQ} - \frac{1}{2}\mathbf{BC} && AC = (AB + BC) \text{ and } BQ = \tfrac{1}{2}BC \\
&= \frac{1}{2}\mathbf{AB} + \frac{1}{2}\mathbf{BC} + \mathbf{PQ} - \frac{1}{2}\mathbf{BC} \\
&= \frac{1}{2}\mathbf{AB} + \mathbf{PQ} \\
\frac{1}{2}\mathbf{AB} &= \mathbf{PQ} && \textit{Subtract } \tfrac{1}{2}AB \textit{ from both sides.}
\end{aligned}$$

Therefore, the theorem is proved.

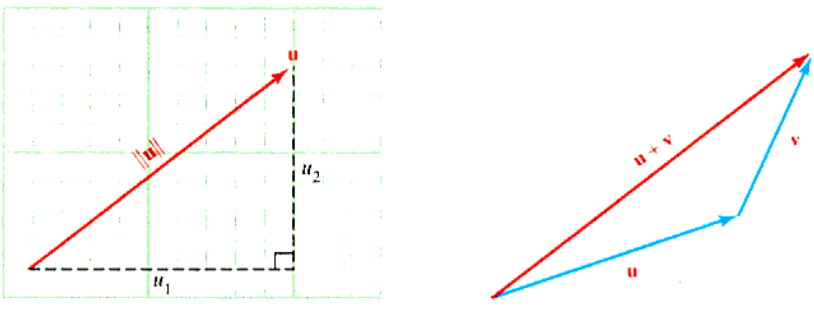
When a vector $\mathbf{u}$ is represented in component form $\mathbf{u} = \langle u_1, u_2 \rangle$, its length is given by the formula

$$\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2}$$

This is a simple application of the Pythagorean theorem as shown in Figure 9.8**a**. Another important relationship involving the length of vectors is the *triangle inequality*

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$$

for any vectors $\mathbf{u}$ and $\mathbf{v}$. Equality will occur precisely when $\mathbf{u}$ and $\mathbf{v}$ are multiples of one another (that is, when $\mathbf{u}$ and $\mathbf{v}$ have the same direction). To establish the inequality, we observe that $\mathbf{u}$ and $\mathbf{v}$ are two sides of a triangle in the plane, the third side has length $\|\mathbf{u} + \mathbf{v}\|$ and is "shorter" than the sum $\|\mathbf{u}\| + \|\mathbf{v}\|$ of the lengths of the other two sides, as shown in Figure 9.8**b**.



a. Length of **u** **b. Interactive** Triangle inequality

Figure 9.8 Geometric representation of two vector properties

In component form, two vectors are equal if their components are equal, as illustrated by the following example.

Example 3 Equal vectors in component form

If $\mathbf{u} = \langle 8, -2\rangle$ and $\mathbf{v} = \langle -3, 5\rangle$, find s and t so that $s\mathbf{u} + t\mathbf{v} = \mathbf{w}$, where $\mathbf{w} = \langle 2, 8\rangle$.

Solution

$$\begin{aligned} s\mathbf{u} + t\mathbf{v} &= s\langle 8, -2\rangle + t\langle -3, 5\rangle \\ &= \langle 8s, -2s\rangle + \langle -3t, 5t\rangle \\ &= \langle 8s - 3t, -2s + 5t\rangle \end{aligned}$$

Thus, if $s\mathbf{u} + t\mathbf{v} = \mathbf{w}$, we see

$$\begin{cases} 8s - 3t = 2 \\ -2s + 5t = 8 \end{cases}$$

Solving this system we find $(s, t) = (1, 2)$.

Example 4 Using vectors in a velocity problem

A special boat, the *Earthrace*, draws attention wherever it goes (see Figure 9.9).

A river 4 mi wide flows south with a current of 5 mi/h, and for an exhibition, the *Earthrace* must travel directly across the river from east to west past a viewing stand in 20 min. What is the desired heading for the boat?

Figure 9.9 The *Earthrace*, a high speed boat

Solution Begin by drawing a diagram, as shown in Figure 9.10.

Let $\mathbf{B}$ be the velocity vector of the boat in the direction of the angle θ. If the river's current has velocity $\mathbf{C}$, the given information tells us that $\|\mathbf{C}\| = 5$ mi/h and that $\mathbf{C}$ points directly south. Moreover, because the boat is to cross the river from east to west in 20 min (that is, $\frac{1}{3}$ h), its *effective velocity* after compensating for the current is a vector $\mathbf{V}$ that points west. We will calculate $\|\mathbf{V}\|$ to find the effective velocity and then use this information to find the magnitude and direction of $\mathbf{B}$.

$$\|\mathbf{V}\| = \frac{\text{WIDTH OF THE RIVER}}{\text{TIME OF CROSSING}} = \frac{4 \text{ mi}}{\frac{1}{3}\text{ hr}} = 12 \text{ mi/h}$$

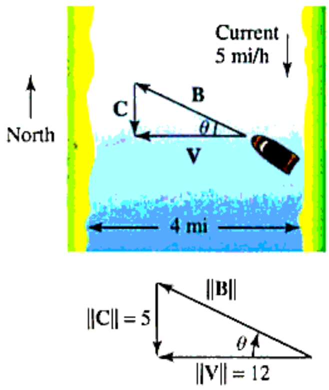


Figure 9.10 A velocity problem

The effective velocity $\mathbf{V}$ is the resultant of $\mathbf{B}$ and $\mathbf{C}$; that is, $\mathbf{V} = \mathbf{B} + \mathbf{C}$. Because $\mathbf{V}$ and $\mathbf{C}$ act in perpendicular directions, we can determine $\mathbf{B}$ by referring to the right triangle with sides $\|\mathbf{V}\| = 12$ and $\|\mathbf{C}\| = 5$ with hypotenuse $\|\mathbf{B}\|$. We find that

$$\|\mathbf{B}\| = \sqrt{\|\mathbf{V}\|^2 + \|\mathbf{C}\|^2} = \sqrt{12^2 + 5^2} = 13$$

The direction of the velocity vector $\mathbf{V}$ is given by the angle θ in Figure 9.10, and we find that

$$\tan\theta = \frac{5}{12} \qquad \text{so that} \qquad \theta = \tan^{-1}\left(\frac{5}{12}\right) \approx 0.3948$$

Thus, the boat should travel at 13 mi/h in a of direction approximately 0.3948. In navigation, it is common to specify direction in degrees rather than radians; this is 22.6° north of west and to measure the direction as an angle from either north or south. In this case it is $90° - 22.6° = 67.4°$, or N67.4°W (that is, 67.4° west of north).

A **unit vector** is a vector with length 1, and a **direction vector** for a given nonzero vector $\mathbf{v}$ is a unit vector $\mathbf{u}$ that points in the same direction as $\mathbf{v}$. Such a vector can be found by simply "dividing" $\mathbf{v}$ by its length $\|\mathbf{v}\|$ $\left(\text{or multiplying by the scalar } \frac{1}{\|\mathbf{v}\|}\right)$; that is, $\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$.

Example 5 Finding a direction vector

Find a direction vector for the vector $\mathbf{v} = \langle 2, -3\rangle$.

Solution The vector $\mathbf{v}$ has length (magnitude) $\|\mathbf{v}\| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$. Thus, the required direction vector is the unit vector

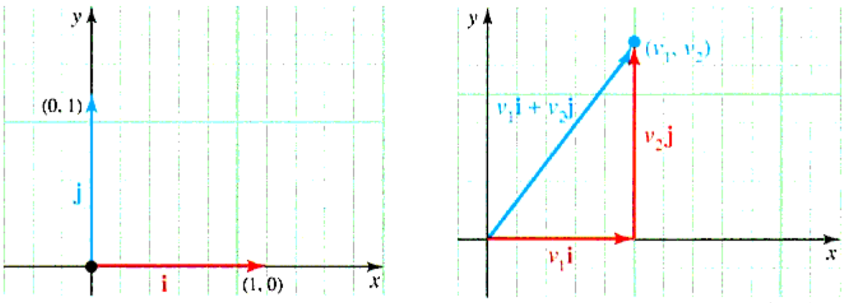
$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\langle 2, -3\rangle}{\sqrt{13}} = \frac{\sqrt{13}}{13}\langle 2, -3\rangle = \left\langle \frac{2\sqrt{13}}{13}, \frac{-3\sqrt{13}}{13} \right\rangle$$

Standard Representation of Vectors in the Plane

The unit vectors $\mathbf{i} = \langle 1, 0\rangle$ and $\mathbf{j} = \langle 0, 1\rangle$ point in the directions of the positive x- and y-axes, respectively, and are called **standard basis vectors**. Any vector $\mathbf{v} = \langle v_1, v_2\rangle$ in the plane can be expressed as a linear combination of the vectors $\mathbf{i}$ and $\mathbf{j}$, because

$$\mathbf{v} = \langle v_1, v_2\rangle = v_1\langle 1, 0\rangle + v_2\langle 0, 1\rangle = v_1\mathbf{i} + v_2\mathbf{j}$$

This is called the **standard representation** of the vector $\mathbf{v}$, and it can be shown that the representation is unique in the sense that if $\mathbf{v} = a\mathbf{i} + b\mathbf{j}$, then $a = v_1$ and $b = v_2$ as shown in Figure 9.11**b**. In this context, the scalars v_1 and v_2 are called the **horizontal** and **vertical components** of $\mathbf{v}$, respectively.



a. The standard basis vectors **b.** $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j}$

Figure 9.11 Standard representation of vectors in the plane

Example 6 Finding the standard representation of a vector

If $\mathbf{u} = 3\mathbf{i} + 2\mathbf{j}$, $\mathbf{v} = -2\mathbf{i} + 5\mathbf{j}$, and $\mathbf{w} = \mathbf{i} - 4\mathbf{j}$, what is the standard representation of the vector $2\mathbf{u} + 5\mathbf{v} - \mathbf{w}$?

Solution Using properties of vectors, we find that

$$\begin{aligned} 2\mathbf{u} + 5\mathbf{v} - \mathbf{w} &= 2(3\mathbf{i} + 2\mathbf{j}) + 5(-2\mathbf{i} + 5\mathbf{j}) - (\mathbf{i} - 4\mathbf{j}) \\ &= [2(3) + 5(-2) - 1]\mathbf{i} + [2(2) + 5(5) - (-4)]\mathbf{j} \\ &= -5\mathbf{i} + 33\mathbf{j} \end{aligned}$$

Example 7 A vector connecting two points

Find the standard representation of the vector **PQ**, for the points $P(3, -4)$ and $Q(-2, 6)$.

Solution The component form of **PQ** is

$$\mathbf{PQ} = \langle(-2) - 3, 6 - (-4)\rangle = \langle -5, 10\rangle$$

This means that **PQ** has the standard representation $\mathbf{PQ} = -5\mathbf{i} + 10\mathbf{j}$.

Example 8 Computing a resultant force

Two forces $\mathbf{F}_1$ and $\mathbf{F}_2$ act on the same body. Suppose it is known that $\mathbf{F}_1$ and $\mathbf{F}_2$ act on the same body. It is known that $\mathbf{F}_1$ has magnitude 3 newtons and acts in the direction of $-\mathbf{i}$, whereas $\mathbf{F}_2$ has magnitude 2 newtons and acts in the direction of the unit vector

$$\mathbf{u} = \frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}$$

What additional force $\mathbf{F}_3$ must be applied in order to keep the body at rest? (See Figure 9.12.)

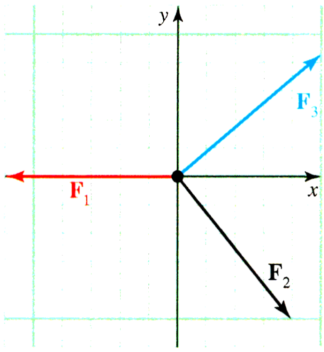


Figure 9.12 Keeping a body at rest

Solution We will calculate $\mathbf{F}_1$ and $\mathbf{F}_2$ and we want to find $\mathbf{F}_3 = a\mathbf{i} + b\mathbf{j}$ so that $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = 0$. According to the given information, we have

$$\mathbf{F}_1 = 3(-\mathbf{i}) = -3\mathbf{i} \quad \text{and} \quad \mathbf{F}_2 = 2\left(\frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}\right) = \frac{6}{5}\mathbf{i} - \frac{8}{5}\mathbf{j}$$

and we want to find $\mathbf{F}_3 = a\mathbf{i} + b\mathbf{j}$ so that $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \mathbf{0}$. Substituting into this vector we obtain

$$(-3\mathbf{i}) + \left(\frac{6}{5}\mathbf{i} - \frac{8}{5}\mathbf{j}\right) + (a\mathbf{i} + b\mathbf{j}) = 0\mathbf{i} + 0\mathbf{j}$$

By combining terms on the left, we find that

$$\left(-3 + \frac{6}{5} + a\right)\mathbf{i} + \left(-\frac{8}{5} + b\right)\mathbf{j} = 0\mathbf{i} + 0\mathbf{j}$$

Because the standard representation is unique, we must have

$$-3 + \frac{6}{5} + a = 0 \quad \text{and} \quad -\frac{8}{5} + b = 0$$
$$a = \frac{9}{5} \qquad\qquad b = \frac{8}{5}$$

The required force is $\mathbf{F}_3 = \frac{9}{5}\mathbf{i} + \frac{8}{5}\mathbf{j}$. This is a force of magnitude

$$\|\mathbf{F}_3\| = \sqrt{\left(\frac{9}{5}\right)^2 + \left(\frac{8}{5}\right)^2} = \frac{1}{5}\sqrt{145} \text{ newtons}$$

which acts in the direction of the unit vector

$$\mathbf{v} = \frac{\mathbf{F}_3}{\|\mathbf{F}_3\|} = \frac{5}{\sqrt{145}}\left(\frac{9}{5}\mathbf{i} + \frac{8}{5}\mathbf{j}\right) = \frac{9}{\sqrt{145}}\mathbf{i} + \frac{8}{\sqrt{145}}\mathbf{j}$$

PROBLEM SET 9.1

Level 1

Sketch each vector given in Problems 1-4 assuming that its initial point is at the origin.

1. $3\mathbf{i} - 4\mathbf{j}$
2. $-2\mathbf{i} - 3\mathbf{j}$
3. $-\frac{1}{2}\mathbf{i} + \frac{5}{2}\mathbf{j}$
4. $-2(-\mathbf{i} + 2\mathbf{j})$

The initial point P and terminal point Q of a vector are given in Problems 5-8. Sketch each vector and then write it in component form and find $\|PQ\|$.

5. $P(3,-1), Q(7,2)$
6. $P(5,-2), Q(5,8)$
7. $P(3,4), Q(-2,4)$
8. $P\left(\frac{1}{2},6\right), Q(-3,-2)$

Find the standard representation and length of each vector **PQ** *in Problems 9-12.*

9. $P(-1,-2), Q(1,-2)$
10. $P(5,7), Q(6,8)$
11. $P(-4,-3), Q(0,-1)$
12. $P(3,-5), Q(2,8)$

Find a unit vector that points in the direction of each of the vectors given in Problems 13-16.

13. $\mathbf{i} + \mathbf{j}$
14. $\frac{1}{2}\mathbf{i} + \frac{1}{4}\mathbf{j}$
15. $3\mathbf{i} - 4\mathbf{j}$
16. $-4\mathbf{i} + 7\mathbf{j}$

Let $\mathbf{u} = \langle -3, 4\rangle$ *and* $\mathbf{v} = \langle 1, -1\rangle$. *Find s and t so that* $s\mathbf{u} + t\mathbf{v} = \mathbf{w}$ *for the given vector in Problems 17-20.*

17. $\mathbf{w} = \langle 6, 0\rangle$
18. $\mathbf{w} = \langle 0, -3\rangle$
19. $\mathbf{w} = \langle -2, 1\rangle$
20. $\mathbf{w} = \langle 8, 11\rangle$

Suppose $\mathbf{u} = 3i - 4\mathbf{j}$, $\mathbf{v} = 4\mathbf{i} - 3\mathbf{j}$, *and* $\mathbf{w} = \mathbf{i} + \mathbf{j}$. *Express each of the expressions in Problems 21-24 in standard form.*

21. $2\mathbf{u} + 3\mathbf{v} - \mathbf{w}$
22. $\frac{1}{2}(\mathbf{u} + \mathbf{v}) - \frac{1}{4}\mathbf{w}$
23. $\|\mathbf{v}\|\,\mathbf{u} + \|\mathbf{u}\|\,\mathbf{v}$
24. $\|\mathbf{u}\|\,\|\mathbf{v}\|\,\mathbf{w}$

Find all real numbers x and y that satisfy the vector equations given in Problems 25-28.

25. $(x - y - 1)\mathbf{i} + (2x + 3y - 12)\mathbf{j} = \mathbf{0}$
26. $x\mathbf{i} - 4y^2\mathbf{j} = (5 - 3y)\mathbf{i} + (10 - 7x)\mathbf{j}$
27. $(x^2 + y^2)\mathbf{i} + y\mathbf{j} = 20\mathbf{i} + (x + 2)\mathbf{j}$
28. $(y - 1)\mathbf{i} + y\mathbf{j} = (\log x)\mathbf{i} + [\log 2 + \log(x + 4)]\mathbf{j}$

In Problems 29-32, find a unit vector **u** *with the given characteristics.*

29. **u** makes an angle of 30° with the positive x-axis.
30. **u** has the same direction as the vector $2\mathbf{i} - 3\mathbf{j}$.
31. **u** has the direction opposite that of $-4\mathbf{i} + \mathbf{j}$.
32. **u** has the direction of the vector from $P(-1,5)$ to $Q(7,-3)$.

In Problems 33-34, let $\mathbf{u} = 4\mathbf{i} - \mathbf{j}$, $\mathbf{v} = \mathbf{i} + 2\mathbf{j}$, *and* $\mathbf{w} = -3\mathbf{i} + 4\mathbf{j}$.

33. Find a unit vector in the same direction as $\mathbf{u} + \mathbf{v}$.
34. Find a vector of length 3 with the same direction as $\mathbf{u} - 2\mathbf{v} + 2\mathbf{w}$.
35. Find the terminal point of the vector $5\mathbf{i} + 7\mathbf{j}$ if the initial point is $(-2,3)$.
36. Find the initial point of the vector $-\mathbf{i} + 2\mathbf{j}$ if the terminal point is $(-1,-2)$.
37. **a.** Find the vector from the point P to the midpoint of the line segment joining the points $P(-3,-8)$ and $Q(9,-2)$.
    b. What is the vector from P to the point which is located $\frac{5}{6}$ of the distance from P to Q?
38. If $\|\mathbf{v}\| = 3$ and $-3 \le r \le 1$, what are the possible values of $r\,\|\mathbf{v}\|$?
39. Show that $\mathbf{v} = (\cos\theta)\mathbf{i} + (\sin\theta)\mathbf{j}$ is a unit vector for any angle θ.

Level 2

40. If **u** and **v** are nonzero vectors and
$$|r| = \frac{\|\mathbf{u}\|}{\|\mathbf{v}\|}$$
what is $\|r\mathbf{v}\|$?
41. If **u** and **v** are nonzero vectors with $\|\mathbf{u}\| = \|\mathbf{v}\|$, does it follow that $\mathbf{u} = \mathbf{v}$? Explain.
42. If $\mathbf{u} = 2\mathbf{i} - 3\mathbf{j}$ and $\mathbf{v} = x\mathbf{i} + y\mathbf{j}$, describe the set of points in the plane whose coordinates (x, y) satisfy $\|\mathbf{v} - \mathbf{u}\| \le 2$.
43. Let $\mathbf{u}_0 = x_0\mathbf{i} + y_0\mathbf{j}$ for constants x_0 and y_0, and let $\mathbf{u} = x\mathbf{i} + y\mathbf{j}$. Describe the set of all points in the plane whose coordinates satisfy:
    a. $\|\mathbf{u} - \mathbf{u}_0\| = 1$ **b.** $\|\mathbf{u} - \mathbf{u}_0\| \le r$
44. Let $\mathbf{u} = 3\mathbf{i} - \mathbf{j}$ and $\mathbf{v} = -6\mathbf{i} + 2\mathbf{j}$. Show that there are no numbers a, b for which $a\mathbf{u} + b\mathbf{v} = 2\mathbf{i} + 5\mathbf{j}$.
45. Suppose **u** and **v** are a pair of nonzero, nonparallel vectors. Find numbers a, b, c such that
$$a\mathbf{u} + b(\mathbf{u} - \mathbf{v}) + c(\mathbf{u} + \mathbf{v}) = \mathbf{0}$$
46. Let $\mathbf{u} = \langle 2, 1\rangle$ and $\mathbf{v} = \langle -3, 4\rangle$.
    a. Sketch the vector $c\mathbf{u}+(1 - c))\mathbf{v}$ for the cases where $c = 0, c = \frac{1}{4}, c = \frac{1}{2}, c = \frac{3}{4}$, and $c = 1$.
    b. In general, if the initial point of $c\mathbf{u} + (1 - c)\mathbf{v}$ for $0 \le c \le 1$ is at the origin, where is its terminal point (x, y)?
47. Two forces $\mathbf{F}_1 = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{F}_2 = 3\mathbf{i} - 7\mathbf{j}$ act on an object. What additional force should be applied to keep the body at rest?
48. Three forces $\mathbf{F}_1 = \mathbf{i} - 2\mathbf{j}$, $\mathbf{F}_2 = 3\mathbf{i} - 7\mathbf{j}$, and $\mathbf{F}_3 = \mathbf{i} + \mathbf{j}$ act on an object. What additional force $\mathbf{F}_4$ should be applied to keep the body at rest?

49. A river 2.1 mi wide flows south with a current of 3.1 mi/h. What speed and heading should a motorboat assume to travel directly across the river from east to west in 30 min?

50. Four forces act on an object: $\mathbf{F}_1$ has magnitude 10 lb and acts at an angle of $\frac{\pi}{6}$ measured counterclockwise from the positive x-axis; $\mathbf{F}_2$ has magnitude 8 lb and acts in the direction of the vector $\mathbf{j}$; $\mathbf{F}_3$ has magnitude 5 lb and acts at an angle of $4\pi/3$ measured counterclockwise from the positive x-axis. What must the fourth force $\mathbf{F}_4$ be to keep the object at rest?

Level 3

51. Use vector methods to show that the diagonals of a parallelogram bisect each other.

52. In a triangle, let **u**, **v**, and **w** be the vectors from each vertex to the midpoint of the opposite side. Use vector methods to show that $\mathbf{u}+\mathbf{v}+\mathbf{w}=\mathbf{0}$.

53. Prove that the medians of a triangle intersect at a single point by completing the following argument (see Figure 9.13).

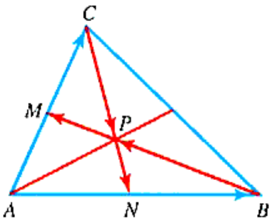


Figure 9.13 Median intersection

a. Let M and N be the midpoints of sides $\overline{AC}$ and $\overline{AB}$, respectively. Show that

$$\mathbf{CN} = \tfrac{1}{2}\mathbf{AB} - \mathbf{AC} \text{ and}$$
$$\mathbf{BM} = \tfrac{1}{2}\mathbf{AC} - \mathbf{AB}$$

b. Let P be the point where medians $\overline{BM}$ and $\overline{CN}$ intersect. Show that there are constants r and s such that

$$\mathbf{CP} = r\left(\tfrac{1}{2}\mathbf{AB} - \mathbf{AC}\right) \text{ and}$$
$$\mathbf{BP} = s\left(\tfrac{1}{2}\mathbf{AC} - \mathbf{AB}\right)$$

Note that $\mathbf{CP}+\mathbf{PB}=\mathbf{CB}$. Use this relationship to prove that $r=s=\frac{2}{3}$. Explain why this shows that any pair of medians meet at a point located $\frac{2}{3}$ the distance from each vertex to the midpoint of the opposite side. Why does this show that *all three* medians meet at a single point?

c. The *centroid* of a triangle is the point where the medians meet. Show that if a triangle has vertices $A(x_1,y_1)$, $B(x_2,y_2)$, $C(x_3,y_3)$, then the centroid has coordinates

$$\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$

54. Show that the vector with initial point $P(c,d)$ and terminal point $Q(a,b)$ has the component form $\mathbf{PQ} = \langle a-c, b-d\rangle$. See Figure 9.5.

55. Two nonzero vectors **u** and **v** are said to be *linearly independent* in the plane if they are not parallel.

a. If **u** and **v** have this property and $a\mathbf{u}=b\mathbf{v}$ for constants a,b, show that $a=b=0$.

b. Show that the standard representation of a vector is unique. That is, if the vector **u** has the representation $\mathbf{u}=a_1\mathbf{i}+b_1\mathbf{j}$ and $\mathbf{u}=a_2\mathbf{i}+b_2\mathbf{j}$ is another such representation, then $a_1=a_2$ and $b_1=b_2$.

56. a. Prove the commutativity property for vectors.

b. Prove the associative property for scalar multiplication.

57. a. Prove the identity property for vectors.

b. Prove the inverse property of addition for vectors.

58. a. Prove vector distributivity.

b. Prove scalar distributivity.

59. Let A,B,C, and D be any four points in the plane. If M and N are midpoints of $\overline{AC}$ and $\overline{BD}$, show that

$$\mathbf{MN} = \frac{1}{4}(\mathbf{AB}+\mathbf{AD}+\mathbf{CB}+\mathbf{CD})$$

60. Let A,B,C and D be vertices of a quadrilateral, and let M,N,P, and Q, be the midpoints of the sides $\overline{AB}$, $\overline{BC}$, $\overline{CD}$, and $\overline{AD}$, respectively, as shown in Figure 9.14. Use vector methods to show that $MNPQ$ is a parallelogram.

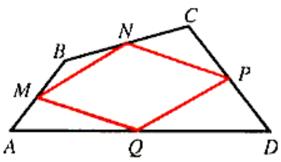


Figure 9.14 Quadrilateral *ABCD*

9.2 Coordinates and Vectors in $\mathbb{R}^3$

IN THIS SECTION: *Three-dimensional coordinate system, graphs in $\mathbb{R}^3$, vectors in $\mathbb{R}^3$*
Because we live in a three-dimensional world, it is essential that we be able to visualize surfaces in three dimensions. To that end, we consider ordered triples and their representation in three-dimensional space, as well as develop some knowledge of simple three-dimensional surfaces, namely, the quadric surfaces. In the next section, we will consider vectors in three dimensions to describe motion in space efficiently.

Three-Dimensional Coordinate System

We have already considered ordered pairs in $\mathbb{R}^2$, and because we exist in a three-dimensional world, it is also important to consider a three-dimensional system. We call this *three-space* and denote it by $\mathbb{R}^3$. We introduce a coordinate system to three-space by choosing three mutually perpendicular axes to serve as a frame of reference. The orientation of our reference system will be *right-handed* in the sense that if you stand at the origin with your right arm along the positive x-axis and your left arm along the positive y-axis, as shown in Figure 9.15, your head will then point in the direction of the positive z-axis.

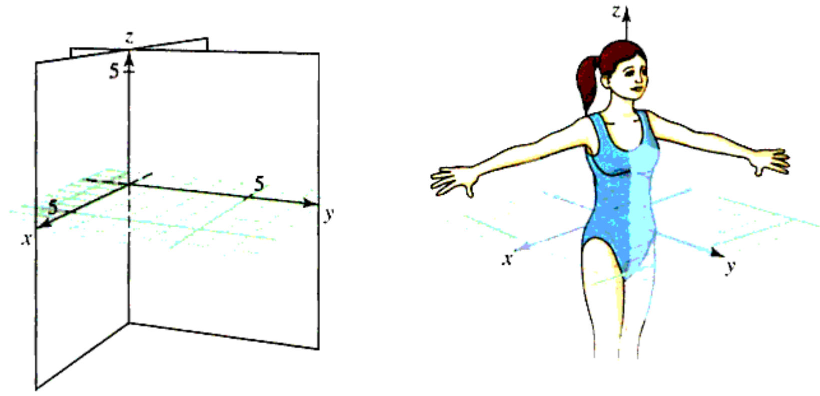


Figure 9.15 A "right-handed" rectangular coordinate system for $\mathbb{R}^3$

To orient yourself to a three-dimensional coordinate system, think of the x-axis and y-axis as lying in the plane of the floor and the z-axis as a line perpendicular to the floor. All the graphs that we have drawn in the first eight chapters of this book would now be drawn on the floor. If you orient yourself in a room as shown in Figure 9.16, you may notice some important planes. Assume that the room is 25 ft × 30 ft × 8 ft and fix the origin at a front corner (where the board hangs).

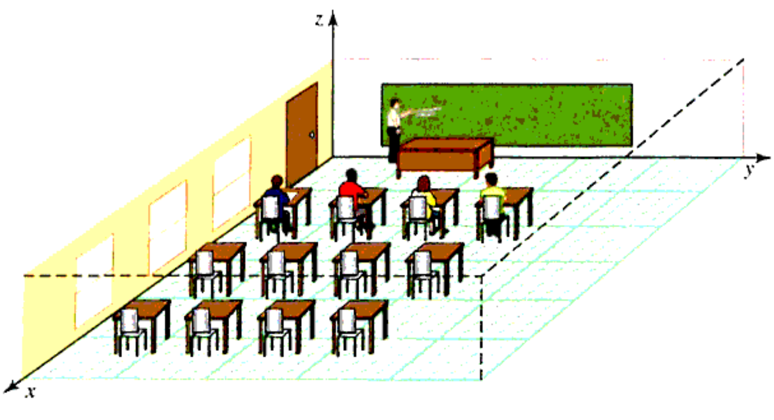
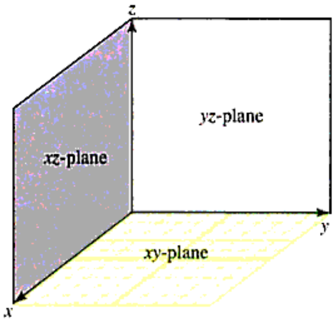


Figure 9.16 A typical classroom

Floor:	***xy*-plane**; equation is $z = 0$.
Ceiling:	Plane parallel to the xy-plane; equation is $z = 8$.
Front wall:	***yz*-plane**; equation is $x = 0$.
Back wall:	Plane parallel to the yz-plane; equation is $x = 30$.
Left wall:	***xz*-plane**; equation is $y = 0$.
Right wall:	Plane parallel to the xz-plane; equation is $y = 25$.



The xy-, xz-, and yz-planes are called the **coordinate planes**. We will obtain equations for planes in space after we discuss vectors in $\mathbb{R}^3$. However, in beginning to visualize objects in $\mathbb{R}^3$, we do not want to ignore planes, because they are so common (for example, the walls, ceiling, and floor in Figure 9.16). We will show in Section 9.5 that the graph of $Ax + By + Cz = D$ is a **plane** if A, B, C, and D are real numbers (not all zero). Points in $\mathbb{R}^3$ are located by their position in relation to the three coordinate planes and are given appropriate coordinates. Specifically, the point P is assigned coordinates (a, b, c) to indicate that it is a, b, and c units, respectively, from the yz-, xz-, and xy-planes.

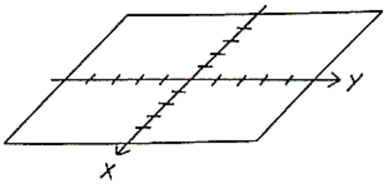
Example 1 **Drawing lesson** **Points in three dimensions**

Graph the following ordered triples without using technology:

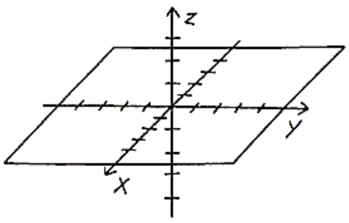
a. $(3, 4, -5)$ **b.** $(10, 20, 10)$ **c.** $(-12, 6, 12)$ **d.** $(-12, -18, 6)$ **e.** $(20, -10, 18)$

Solution

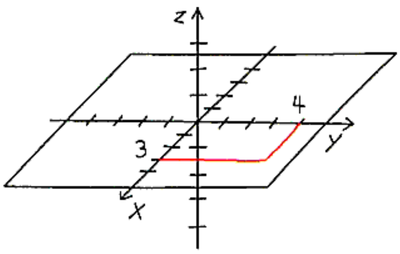
a. Step 1: Sketch the x- and y-axes, adding tick marks. Outline the xy-plane.



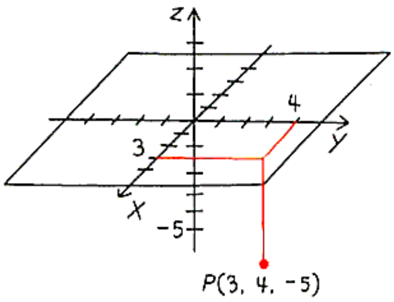
Step 2: Sketch the z-axis, adding tick marks. Use dashed segments for hidden parts.



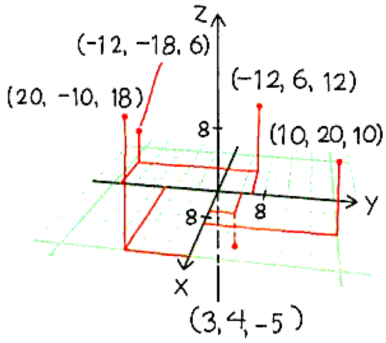
Step 3: Plot x-distance and y-distance on the xy-plane. Darken segments from each along grid lines. Colored pencil or highlighter may help you visualize the figure.



Step 4: Plot z-distance, using the unit size from the z-axis. Lightly sketch a grid on the xy-plane, using tick marks as guides.



The other points are plotted similarly as shown.



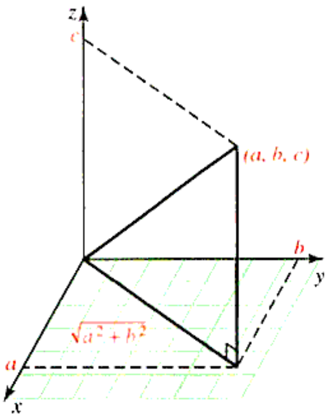


Figure 9.17 Distance from the origin (0,0,0) to the point (a, b, c)

In $\mathbb{R}^2$, the distance from the origin $(0, 0)$ to the point (a, b) is $d = \sqrt{a^2 + b^2}$, and in $\mathbb{R}^3$, the distance from $(0, 0, 0)$ to (a, b, c) is $d = \sqrt{a^2 + b^2 + c^2}$, as shown in Figure 9.17. We can now summarize.

DISTANCE FORMULA The distance $|P_1P_2|$ between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Example 2 Finding the distance between points in $\mathbb{R}^3$

Find the distance between $(10, 20, 10)$ and $(-12, 6, 12)$.

Solution
$$\begin{aligned} d &= \sqrt{(-12 - 10)^2 + (6 - 20)^2 + (12 - 10)^2} \\ &= \sqrt{684} \\ &= 6\sqrt{19} \end{aligned}$$

Graphs in $\mathbb{R}^3$

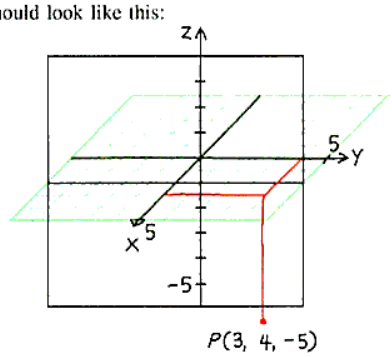
The **graph of an equation** in $\mathbb{R}^3$ is the collection of all points (x, y, z) whose coordinates satisfy a given equation. This graph is called a **surface**. You are not expected to spend a great deal of time graphing three-dimensional surfaces, but the drawing lessons in this section should help. You may also have access to a computer program to help you look at graphs in three dimensions.

Planes

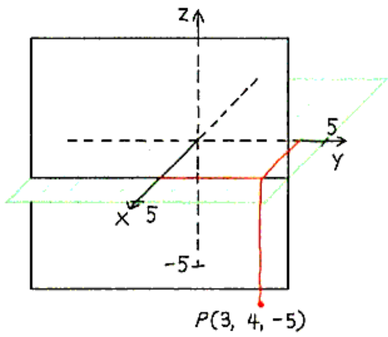
We shall obtain equations for planes in space after we discuss vectors in space. As we said above, we need to be able to draw planes. The following drawing lesson will help you sketch planes in three dimensions. The task described begins with the xy-plane and point $P(3, 4, -5)$ from Example 1. Then you are asked to draw (on your paper) the planes $x = 2$ and $y = 0$.

Drawing Lesson: Drawing Vertical and Horizontal Planes

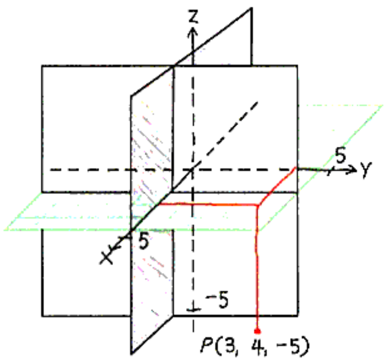
Begin with the xy-plane and the z-axis from Example 1, and then draw the line $x = 2$ on the xy-plane. Now, through each endpoint of the line segment you have drawn, draw a segment parallel to the z-axis. Then connect the endpoints. When you are finished, your drawing should look like this:



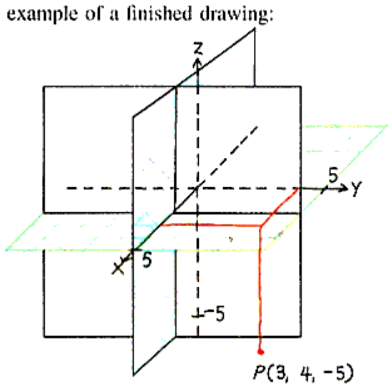
Shade the plane $x = 2$ where it is not hidden by the xy-plane. Erase hidden parts of both planes, and use your eraser to dash hidden parts of the axis. When you are finished, your drawing should look like this:



Follow the same procedure to draw and shade the plane $y = 0$. Draw the intersection of the two planes. Here is what it should look like:



Use colored pencils or highlighters to distinguish individual planes. Here is an example of a finished drawing:



The following example will also help you sketch planes.

Example 3 Graphing planes

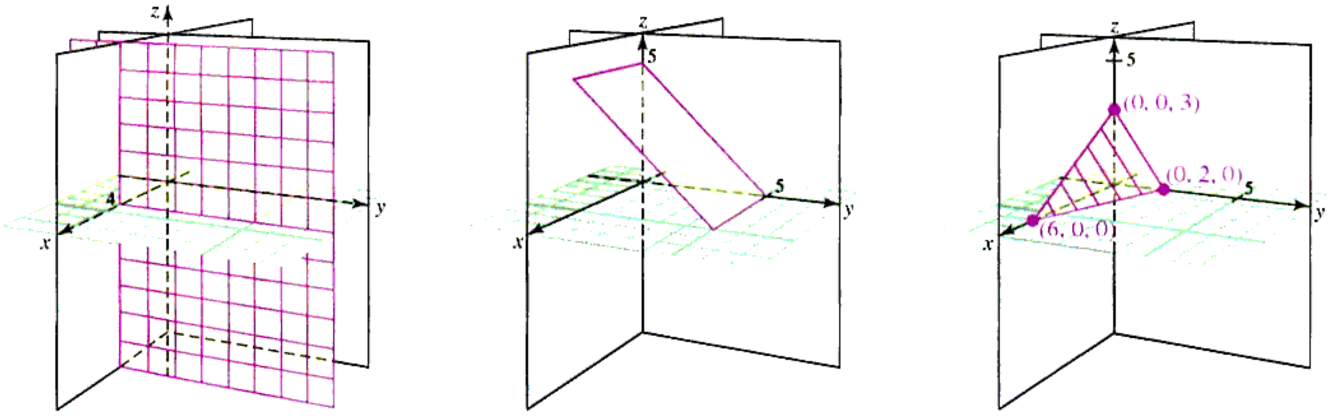
Graph the planes defined by the given equations.

a. $x = 4$ **b.** $y + z = 5$ **c.** $x + 3y + 2z = 6$

Solution To graph a plane, find some ordered triples satisfying the equation. The best ones to use are often those that fall on a coordinate axis (the intercepts).

a. When two variables are missing, then the plane is parallel to one of the coordinate planes, as shown in Figure 9.18**a**.

b. When one of the variables is missing from an equation of a plane, then that plane is parallel to the axis corresponding the missing variable; in this case it is parallel to the x-axis. Draw the line $y + z = 5$ on the yz-plane, and then complete the plane, as shown in Figure 9.18**b**.



a. $x = 4$ **b.** $y + z = 5$ **c.** $x + 3y + 2z = 6$

Figure 9.18 Graphs of planes

c. Let $x = 0$ and $y = 0$; then $z = 3$; plot the point $(0, 0, 3)$.
Let $x = 0$ and $z = 0$; then $y = 2$; plot the point $(0, 2, 0)$.
Let $y = 0$ and $z = 0$; then $x = 6$; plot the point $(6, 0, 0)$.
Use these points to draw the intersection lines (called **trace lines**) of the plane you are graphing with each of the coordinate planes. The result is shown in Figure 9.18**c**.

Spheres

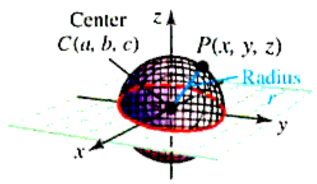


Figure 9.19 Graph of a sphere with center (a, b, c) and radius r

A **sphere** (see Figure 9.19) is defined as the collection of all points located a fixed distance (the **radius**) from a fixed point (the **center**).

In particular, if $P(x, y, z)$ is a point on the sphere with radius r and center $C(a, b, c)$, then the distance from C to P is r. Thus,

$$r = \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2}$$

If you square both sides of this equation, you can see that it is equivalent to the equation of a sphere displayed in the following box. Conversely, if the point (x, y, z) satisfies an equation of this form, it must lie on a sphere with center (a, b, c) and radius r.

EQUATION OF A SPHERE The graph of the equation

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$$

is a sphere with center (a, b, c) and radius r, and any sphere has an equation of this form. This is called the **standard form of the equation of a sphere** (*or simply standard-form sphere*).

Example 4 Center and radius of a sphere from a given equation

Show that the graph of the equation $x^2 + y^2 + z^2 + 4x - 6y - 3 = 0$ is a sphere, and find its center and radius.

Solution By completing the square in both variables x and y, we have

$$\begin{aligned}(x^2 + 4x) + (y^2 - 6y) + z^2 &= 3\\ (x^2 + 4x + 2^2) + (y^2 - 6y + (-3)^2) + z^2 &= 3 + 4 + 9\\ (x+2)^2 + (y-3)^2 + z^2 &= 16\end{aligned}$$

Comparing this equation with the standard form, we see that it is the equation of a sphere with center $(-2, 3, 0)$ and radius 4. The graph is shown in Figure 9.20. Note that you cannot see the bottom half of the sphere because your view is obstructed by the xy-plane.

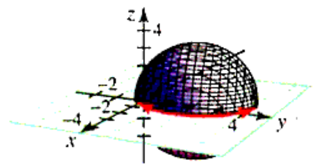
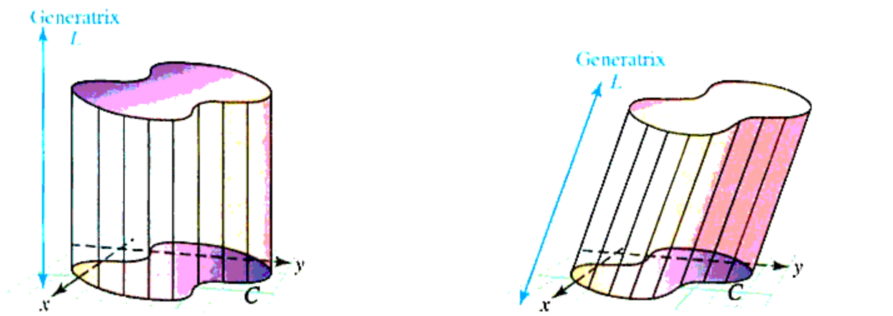


Figure 9.20 Sphere with center $(-2, 3, 0)$ and radius 4

Cylinders

A **cross-section** of a surface in space is a curve obtained by intersecting the surface with a plane. If parallel planes intersect a given surface in congruent cross-sectional curves, the surface is called a **cylinder**. We define a cylinder with **principal cross sections C and generating line L** to be the surface obtained by moving lines parallel to L along the boundary of the curve C, as shown in Figure 9.21. In this context, the curve C is called a **directrix** of the cylinder, and L is the **generatrix**.



The generatrix is perpendicular to the plane containing C

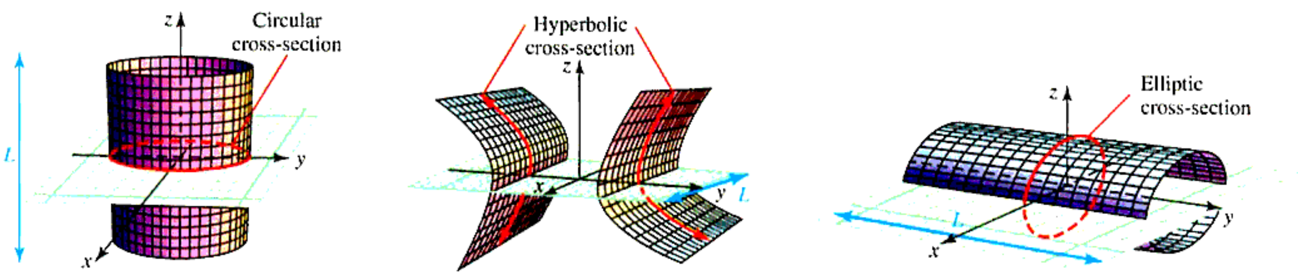
The line L does not need to be perpendicular to the plane containing the curve C

Figure 9.21 Cylinders

We shall deal primarily with cylinders in which the directrix is a conic section and the generatrix L is one of the coordinate axes. Such a cylinder is often named for the type of conic section in its principal cross-sections and is described by an equation involving only two of the variables x, y, z. In this case, the generating line L is parallel to the coordinate axis of the missing variable. Thus,

$x^2 + y^2 = 5$	is a **circular cylinder** with L parallel to the z-axis.
$y^2 - z^2 = 9$	is a **hyperbolic cylinder** with L parallel to the x-axis.
$x^2 + 2z^2 = 25$	is an **elliptic cylinder** with L parallel to the y-axis.

The graphs of these cylinders are shown in Figure 9.22.



a. Circular cylinder **b. Hyperbolic cylinder** **c. Elliptic cylinder**

Figure 9.22 Cylinders

Vectors in $\mathbb{R}^3$

A vector in $\mathbb{R}^3$ is a directed line segment (an "arrow") in space. The vector $\mathbf{P_1P_2}$ with initial point $P_1(x_1, y_1, z_1)$ and terminal point $P_2(x_2, y_2, z_2)$ has the component form analogous to the $\mathbb{R}^2$ case

$$\mathbf{P_1P_2} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Vector addition and multiplication of a vector by a scalar are defined for vectors in $\mathbb{R}^3$ in the same way as these operations were defined for vectors in $\mathbb{R}^2$. In addition, the properties of vector algebra listed in Theorem 9.1 apply to vectors in $\mathbb{R}^3$ as well as to those in $\mathbb{R}^2$.

For example, we observed that each vector in $\mathbb{R}^2$ can be expressed as a unique linear combination of the standard basis vectors $\mathbf{i}$ and $\mathbf{j}$. This representation can be extended to vectors in $\mathbb{R}^3$ by adding a vector $\mathbf{k}$ defined to be the unit vector in the direction of the positive z-axis. In component form, we have in $\mathbb{R}^3$,

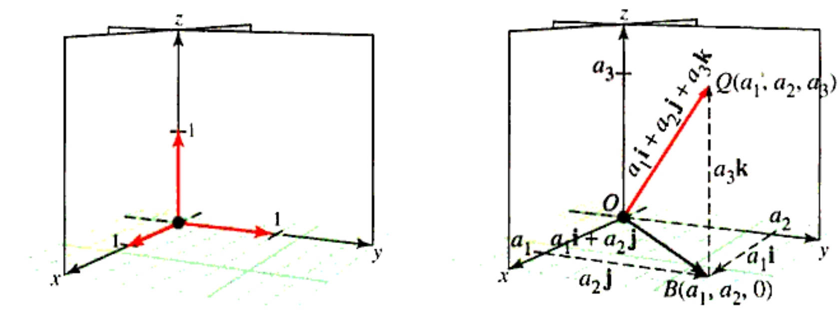
$$\mathbf{i} = \langle 1, 0, 0 \rangle \qquad \mathbf{j} = \langle 0, 1, 0 \rangle \qquad \mathbf{k} = \langle 0, 0, 1 \rangle$$

We call these **the standard basis vectors in** $\mathbb{R}^3$ (shown in Figure 9.23**a**). **The standard representation** of the vector with initial point at the origin O and terminal point $Q(a_1, a_2, a_3)$ is $\mathbf{OQ} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, as shown in Figure 9.23**b**. More generally, the vector $\mathbf{P_1P_2}$ with initial point $P_1(x_1, y_1, z_1)$ and terminal point $P_2(x_2, y_2, z_2)$ has the standard representation

$$\mathbf{P_1P_2} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$$

with magnitude

$$\|\mathbf{P_1P_2}\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



a. The standard basis vectors in $\mathbb{R}^3$ **b.** The vector from O to Q

Figure 9.23 Standard representation of vectors in $\mathbb{R}^3$

Example 5 Standard representation of a vector in $\mathbb{R}^3$

Find the standard representation of the vector $\mathbf{PQ}$ with initial point $P(-1, 2, 2)$ and terminal point $Q(3, -2, 4)$.

Solution We have

$$\mathbf{PQ} = [3 - (-1)]\mathbf{i} + [-2 \ - 2]\mathbf{j} + [4 - 2]\mathbf{k} = 4\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$$

Example 6 Magnitude of a vector and distance between points in $\mathbb{R}^3$

Find the magnitude of the vector $\mathbf{v} = 2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ and the distance between the points $A(1, -1, -4)$ and $B(-2, 3, 8)$.

Solution $\|\mathbf{v}\| = \sqrt{2^2 + (-3)^2 + 5^2} = \sqrt{38}$ and

$$\begin{aligned}\|\overline{AB}\| &= \sqrt{(-2-1)^2 + [3-(-1)]^2 + [8-(-4)^2]} \\ &= 13\end{aligned}$$

As in $\mathbb{R}^2$, if $\mathbf{v}$ is a given nonzero vector in $\mathbb{R}^3$, then the unit vector $\mathbf{u}$ that points in the same direction as $\mathbf{v}$ is

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

Example 7 Finding a direction vector

Find the unit vector that points in the direction of the vector $\mathbf{PQ}$ from $P(-1, 2, 5)$ to $Q(0, -3, 7)$.

Solution $\mathbf{PQ} = [0-(-1)]\,\mathbf{i} + [-3-2]\mathbf{j} + [7-5]\,\mathbf{k} = \mathbf{i} - 5\mathbf{j} + 2\mathbf{k}$

$$\|\mathbf{PQ}\| = \sqrt{1^2 + (-5)^2 + 2^2} = \sqrt{30}$$

Thus,

$$\begin{aligned}\mathbf{u} &= \frac{\mathbf{PQ}}{\|\mathbf{PQ}\|} \\ &= \frac{\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}}{\sqrt{30}} \\ &= \frac{1}{\sqrt{30}}\mathbf{i} - \frac{5}{\sqrt{30}}\mathbf{j} + \frac{2}{\sqrt{30}}\,\mathbf{k}\end{aligned}$$

As in $\mathbb{R}^2$, two vectors in $\mathbb{R}^3$ are **parallel** if they are scalar multiples of one another. Similarly, nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel if and only if $\mathbf{u} = s\mathbf{v}$ for some nonzero scalar s.

Example 8 Parallel vectors

A vector $\mathbf{PQ}$ has initial point $P(1, 0, -3)$ and length 3. Find Q so that $\mathbf{PQ}$ is parallel to $\mathbf{v} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$.

Solution Let Q have coordinates (a_1, a_2, a_3). Then

$$\begin{aligned}\mathbf{PQ} &= [a_1 - 1]\mathbf{i} + [a_2 - 0]\mathbf{j} + [a_3 - (-3)]\mathbf{k} \\ &= (a_1 - 1)\mathbf{i} + a_2\mathbf{j} + (a_3 + 3)\mathbf{k}\end{aligned}$$

Because $\mathbf{PQ}$ is parallel to $\mathbf{v}$, we have $\mathbf{PQ} = s\mathbf{v}$ for some scalar s, that is,

$$(a_1 - 1)\mathbf{i} + a_2\mathbf{j} + (a_3 + 3)\mathbf{k} = s(2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k})$$

Since the standard representation is unique, this implies that

$$\begin{array}{lll} a_1 - 1 = 2s & a_2 = -3s & a_3 + 3 = 6s \\ a_1 = 2s + 1 & & a_3 = 6s - 3 \end{array}$$

Because **PQ** has length 3, we have

$$\begin{aligned}
3 &= \sqrt{(a_1-1)^2+a_2^2+(a_3+3)^2}\\
&= \sqrt{[(2s+1)-1]^2+(-3s)^2+[(6s-3)+3]^2}\\
&= \sqrt{4s^2+9s^2+36s^2}\\
&= \sqrt{49s^2}\\
&= 7\,|s|
\end{aligned}$$

Thus, $s=\pm\frac{3}{7}$ and for the two cases, we have

$$s=\frac{3}{7}:\qquad a_1=2\left(\frac{3}{7}\right)+1=\frac{13}{7}\qquad a_2=-3\left(\frac{3}{7}\right)=-\frac{9}{7}\qquad a_3=6\left(\frac{3}{7}\right)-3=-\frac{3}{7}$$

$$s=-\frac{3}{7}:\qquad a_1=2\left(-\frac{3}{7}\right)+1=\frac{1}{7}\qquad a_2=-3\left(-\frac{3}{7}\right)=\frac{9}{7}\qquad a_3=6\left(-\frac{3}{7}\right)-3=-\frac{39}{7}$$

There are two points that satisfy the conditions for the required terminal point Q: $\left(\frac{13}{7},-\frac{9}{7},-\frac{3}{7}\right)$ and $\left(\frac{1}{7},\frac{9}{7},-\frac{39}{7}\right)$.

PROBLEM SET 9.2

Level 1

For the given vectors in Problems 1-4, find

a. $\mathbf{u}+\mathbf{v}$ **b.** $\mathbf{u}-\mathbf{v}$ **c.** $\frac{5}{2}\mathbf{u}$ **d.** $2\mathbf{u}+3\mathbf{v}$

1. $\mathbf{u}=\langle 4,-3,1\rangle,\ \mathbf{v}=\langle -2,5,3\rangle$
2. $\mathbf{u}=\langle 2,-1,0\rangle,\ \mathbf{v}=\langle 5,-3,4\rangle$
3. $\mathbf{u}=\langle 1,-2,5\rangle,\ \mathbf{v}=\langle 0,-1,3\rangle$
4. $\mathbf{u}=\langle 0,3,-1\rangle,\ \mathbf{v}=\langle 4,-2,0\rangle$

In Problems 5-8 plot the points P and Q in $\mathbb{R}^3$, and find the distance $\|\mathbf{PQ}\|$.

5. $P(3,-4,5), Q(1,5,-3)$
6. $P(3,0,0), Q(-2,5,7)$
7. $P(-3,-5,8), Q(3,6,-7)$
8. $P(0,5,-3), Q(2,-1,0)$

In Problems 9-12 find the standard form equation of the sphere with the given center C and radius r.

9. $C(0,0,0), r=1$
10. $C(-3,5,7), r=2$
11. $C(0,4,-5), r=3$
12. $C(-2,3,-1), r=\sqrt{5}$

Find the center and radius of each sphere whose equations are given in Problems 13-16.

13. $x^2+y^2+z^2-2y+2z-2=0$
14. $x^2+y^2+z^2+4x-2z-8-0$
15. $x^2+y^2+z^2-6x+2y-2z+10=0$
16. $x^2+y^2+z^2-2x-4y+8z+17=0$

Follow the steps shown in the drawing lesson to plot the planes in Problems 17-20.

17. Sketch the planes $x=0, y=0$, and $z=0$.
18. Sketch the planes $x=4, y=0$, and $z=0$.
19. Sketch the planes $x=0, y=2$, and $z=0$.
20. Sketch the planes $x=0, y=3$, and $z=0$.

Find the standard representation of the vector **PQ**, *and then find* $\|\mathbf{PQ}\|$ *in Problems 21-24.*

21. $P(1,-1,3), Q(-1,1,4)$
22. $P(0,2,3), Q(2,3,0)$
23. $P(1,1,1), Q(-3,-3,-3)$
24. $P(3,0,-4), Q(0,-4,3)$

In Problems 25-28, perform the indicated operations.

$\mathbf{u}=2\mathbf{i}-\mathbf{j}+3\mathbf{k},\ \mathbf{v}=\mathbf{i}+\mathbf{j}-5\mathbf{k},\ \mathbf{w}=5\mathbf{i}+7\mathbf{k}$

25. $\mathbf{u}+\mathbf{v}-2\mathbf{w}$
26. $2\mathbf{u}-\mathbf{v}+3\mathbf{w}$
27. $4\mathbf{u}+\mathbf{w}$
28. $3\mathbf{u}-\mathbf{v}-5\mathbf{w}$

In Problems 29-32, find a unit vector that points in the same direction as the given vector $\mathbf{v}$.

29. $\mathbf{v}=\langle 3,-2,1\rangle$
30. $\mathbf{v}=\langle -1,1,\sqrt{2}\rangle$
31. $\mathbf{v}=\langle -5,3,4\rangle$
32. $\mathbf{v}=\langle 1,\sqrt{2},7\rangle$

Sketch the cylindrical surfaces given in Problems 33-36.

33. $x^2+y^2=4$
34. $z=9-x^2$
35. $y=3z^2$
36. $z=\sin x$

37. Find an equation for a sphere, given that the endpoints of a diameter of the sphere are $(1, 2, -3)$ and$(-2, 3, 3)$.

38. Find an equation for a sphere with center $C(2, -1, 3)$ and radius $r = 4$.

Evaluate the expressions given in Problems 39-42.

39. $\|\mathbf{i} - \mathbf{j} + \mathbf{k}\|$

40. $\|\mathbf{i} + \mathbf{j} + \mathbf{k}\|$

41. $\|2(\mathbf{i} - \mathbf{j} + \mathbf{k}) - 3(2\mathbf{i} + \mathbf{j} - \mathbf{k})\|^2$

42. $\|2\mathbf{i} + \mathbf{j} - 3\mathbf{k}\|^2$

Let $\mathbf{v} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ *and* $\mathbf{w} = 2\mathbf{i} + 4\mathbf{j} - \mathbf{k}$; *find the vector or scalar requested in Problems* 43-46.

43. $\|\mathbf{v}\|\mathbf{w}$

44. $2\,\|\mathbf{v}\| - 3\,\|\mathbf{w}\|$

45. $\|\mathbf{v} - \mathbf{w}\|\,(\mathbf{v}+\mathbf{w})$

46. $\|2\mathbf{v} - 3\mathbf{w}\|$

Determine whether each vector in Problems 47-50 *is parallel to* $\mathbf{u} = 2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$.

47. $\mathbf{v} = \langle -2, 6, -10 \rangle$

48. $\mathbf{v} = \langle 4, -6, 10 \rangle$

49. $\mathbf{v} = \langle -1, \frac{3}{2}, -\frac{5}{2} \rangle$

50. $\mathbf{v} = \langle 1, -\frac{3}{2}, 2 \rangle$

Level 2

The vertices $A, B,$ *and* C *of a triangle in* $\mathbb{R}^3$ *are given in Problems* 51-54. *Find the lengths of the sides of the triangle and determine whether it is a right triangle, an isosceles triangle, both, or neither.*

51. $A(1, 1, 1), B(3, 3, 2), C(3, -3, 5)$

52. $A(3, -1, 0), B(7, 1, 4), C(1, 3, 4)$

53. $A(2, 4, 3),\ B(-3, 2, -4),\ C(-6, 8, -10)$

54. $A(1, 2, 3), B(-3, 2, 4), C(1, -4, 3)$

In Problems 55 *and* 56, *determine whether the given points are collinear (that is, lie on the same line). Note: for three points* $A, B,$ *and* C *to be collinear,* $\mathbf{AC}$ *must be a multiple of* $\mathbf{AB}$.

55. $(2, 3, 2), (-1, 4, 0), (-4, 5, -2)$

56. $(3, 0, 3), (2, -3, 5), (4, 3, 1)$

57. A 150-lb professional television camera is mounted on a three-legged tripod. Suppose a coordinate system is chosen with the camera at point $P(0, 0, 6)$ and the feet of the tripod at $A(0, -2, 0)$, $B(-\sqrt{3}, 1, 0)$, and $C(\sqrt{3}, 1, 0)$, as shown in Figure 9.24. Find the force exerted on each of the three legs.

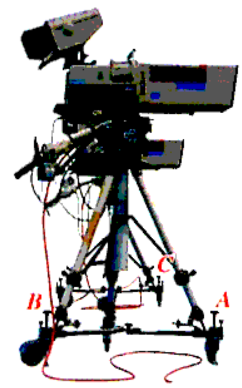


Figure 9.24 T.V. camera

58. Let $\mathbf{u} = \langle -1, 1, 2 \rangle$, $\mathbf{v} = \langle 0, 2, -3 \rangle$, and $\mathbf{w} = \langle 5, -1, 0 \rangle$; find a vector $\mathbf{q}$ so that $2\mathbf{u} - \mathbf{v} + 5\mathbf{q} = 3\mathbf{w}$.

Level 3

59. Find the point P that lies $\frac{2}{3}$ of the distance from the point $A(-1, 3, 9)$ to the midpoint of the line segment joining points $B(-2, 3, 7)$ and $C(4, 1, -3)$.

60. Let $P(3, 2, -1)$, $Q(-2, 1, c)$, and $R(c, 1, 0)$ be points in $\mathbb{R}^3$. For what values of c (if any) is PQR a right triangle?

9.3 The Dot Product

IN THIS SECTION: *Definition of dot product, angle between two vectors, direction cosines, projections, work as a dot product*

In this section, we introduce a product of vectors that yields a scalar and use it to find the angle between three-dimensional vectors.

Definition of Dot Product

In the first two sections of this chapter, you have learned how to add vectors and multiply a vector by a scalar. We now turn our attention to the product of two vectors. The *dot (scalar) product* and the *cross (vector) product* are two important vector operations. We shall examine the cross product in the next section. The dot product is also known as a scalar product because it is a product of vectors that gives a scalar (that is, real number) as a result. Sometimes the dot product is called the **inner product**.

DOT PRODUCT The **dot product** of vectors $\mathbf{v} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{w} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ is the scalar denoted by $\mathbf{v} \cdot \mathbf{w}$ and given by

$$\mathbf{v} \cdot \mathbf{w} = a_1b_1 + a_2b_2 + a_3b_3$$

The dot product of two vectors $\mathbf{v} = a_1\mathbf{i} + a_2\mathbf{j}$ and $\mathbf{w} = b_1\mathbf{i} + b_2\mathbf{j}$ in a plane is given by a similar formula with $a_3 = b_3 = 0$: $\mathbf{v} \cdot \mathbf{w} = a_1b_1 + a_2b_2$.

Example 1 Dot product

Find the dot product of $\mathbf{v} = -3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and $\mathbf{w} = 4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

Solution $\mathbf{v} \cdot \mathbf{w} = -3(4) + 2(-1) + 1(2) = -12$

Example 2 Dot product in component form

If $\mathbf{v} = \langle 4, -1, 3\rangle$ and $\mathbf{w} = \langle -1, -2, 5\rangle$ find the dot product, $\mathbf{v} \cdot \mathbf{w}$.

Solution $\mathbf{v} \cdot \mathbf{w} = 4(-1) + (-1)(-2) + 3(5) = 13$

Before we can apply the dot product to geometric and physical problems, we need to know how it behaves algebraically. A number of important general properties of the dot product are listed in the following theorem.

Theorem 9.2 Properties of dot product

If **u**, **v**, and **w** are vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$ and c is a scalar, then:

Magnitude of a vector	$\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$
Zero product	$\mathbf{0} \cdot \mathbf{v} = 0$
Commutativity	$\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$
Scalar multiple	$c(\mathbf{v} \cdot \mathbf{w}) = (c\mathbf{v}) \cdot \mathbf{w} = \mathbf{v} \cdot (c\mathbf{w})$
Distributivity	$\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$

Proof: Let $\mathbf{u} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{v} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, and $\mathbf{w} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$. We begin with the magnitude of a vector property.

$$\begin{aligned}\|\mathbf{v}\|^2 &= \left(\sqrt{b_1^2 + b_2^2 + b_3^2}\right)^2 \\ &= b_1^2 + b_2^2 + b_3^2 \\ &= \mathbf{v} \cdot \mathbf{v}\end{aligned}$$

Zero product, commutativity, and **scalar multiple** properties can be established in a similar fashion. For the distributive property, we have:

$$\begin{aligned}\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) &= (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) \cdot [(b_1 + c_1)\mathbf{i} + (b_2 + c_2)\mathbf{j} + (b_3 + c_3)\mathbf{k}] \\ &= a_1(b_1 + c_1) + a_2(b_2 + c_2) + a_3(b_3 + c_3) \\ &= a_1b_1 + a_1c_1 + a_2b_2 + a_2c_2 + a_3b_3 + a_3c_3\end{aligned}$$

Also,

$$\begin{aligned}\mathbf{u}\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{w} &= (a_1b_1+a_2b_2+a_3b_3)+(a_1c_1+a_2c_2+a_3c_3)\\ &= a_1b_1+a_1c_1+a_2b_2+a_2c_2+a_3b_3+a_3c_3\end{aligned}$$

Since these are the same, we see

$$\mathbf{u}\cdot(\mathbf{v}+\mathbf{w})=\mathbf{u}\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{w}$$

Angle Between Two Vectors

The angle between two nonzero vectors $\mathbf{v}$ and $\mathbf{w}$ is defined to be the angle θ with $0 \le \theta \le \pi$ that is formed when the vectors are in standard position (initial points at the origin), as shown in Figure 9.25.

The angle between two vectors plays an important role in certain applications and may be computed by using the following formula involving the dot product.

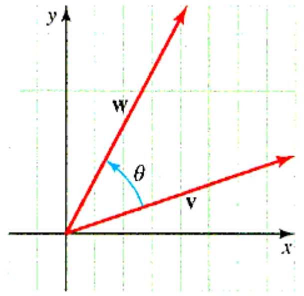


Figure 9.25 The angle between two vectors

Theorem 9.3 Angle between two vectors

If θ is the angle between the nonzero vectors $\mathbf{v}$ and $\mathbf{w}$, then

$$\cos\theta=\frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{v}\|\ \|\mathbf{w}\|}$$

Proof: We can derive this formula as follows. Suppose $\mathbf{v}=a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}$ and $\mathbf{w}=b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k}$, and consider $\triangle AOB$ with vertices at the origin O and the points $A(a_1,a_2,a_3)$ and $B(b_1,b_2,b_3)$, as shown in Figure 9.26.

Note that sides $\overline{OA}$ and $\overline{OB}$ have lengths $\|\mathbf{v}\|=\sqrt{a_1^2+a_2^2+a_3^3}$ and $\|\mathbf{w}\|=\sqrt{b_1^2+b_2^2+b_3^2}$, respectively, and that side $\overline{AB}$ has length $\|\mathbf{v}-\mathbf{w}\|=\sqrt{(a_1-b_1)^2+(a_2-b_2)^2+(a_3-b_3)^2}$. Next, we use the law of cosines.

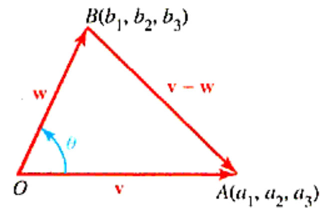


Figure 9.26 Angle between two vectors

$$a^2=b^2+c^2-2bc\cos\theta \qquad \textit{Law of cosines}$$

$$\|\mathbf{v}-\mathbf{w}\|^2=\|\mathbf{v}\|^2+\|\mathbf{w}\|^2-2\,\|\mathbf{v}\|\,\|\mathbf{w}\|\cos\theta \qquad \textit{See Figure 9.26.}$$

$$\begin{aligned}\cos\theta &= \frac{\|\mathbf{v}\|^2+\|\mathbf{w}\|^2-\|\mathbf{v}-\mathbf{w}\|^2}{2\,\|\mathbf{v}\|\,\|\mathbf{w}\|} \qquad \textit{Solve for } \cos\theta.\\ &= \frac{a_1^2+a_2^2+a_3^2+b_1^2+b_2^2+b_3^2-[(a_1-b_1)^2+(a_2-b_2)^2+(a_3-b_3)^2]}{2\,\|\mathbf{v}\|\,\|\mathbf{w}\|}\\ &= \frac{2a_1b_1+2a_2b_2+2a_3b_3}{2\,\|\mathbf{v}\|\,\|\mathbf{w}\|}\\ &= \frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{v}\|\,\|\mathbf{w}\|}\end{aligned}$$

Example 3 Angle between two given vectors

Let $\triangle ABC$ be the triangle with vertices $A(1,1,8)$, $B(4,-3,-4)$, and $C(-3,1,5)$. Find the angle formed at A.

Solution Draw $\triangle\ ABC$ and label the angle formed at A as α as shown in Figure 9.27.

The angle α is the angle between vectors $\mathbf{AB}$ and $\mathbf{AC}$, where

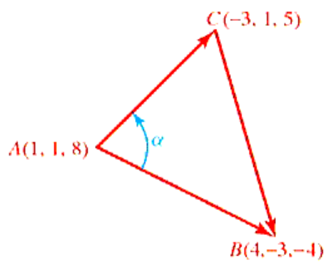


Figure 9.27 Angle in a triangle

$$\begin{aligned}\mathbf{AB} &= (4-1)\mathbf{i}+(-3-1)\mathbf{j}+(-4-8)\mathbf{k}\\ &= 3\mathbf{i}-4\mathbf{j}-12\mathbf{k}\end{aligned} \qquad \begin{aligned}\mathbf{AC} &= (-3-1)\mathbf{i}+(1-1)\mathbf{j}+(5-8)\mathbf{k}\\ &= -4\mathbf{i}-3\mathbf{k}\end{aligned}$$

Thus,

$$\begin{aligned}\cos\alpha &= \frac{\mathbf{AB}\cdot\mathbf{AC}}{\|\mathbf{AB}\|\,\|\mathbf{AC}\|}\\ &= \frac{3(-4)+(-4)(0)+(-12)(-3)}{\sqrt{3^2+(-4)^2+(-12)^2}\,\sqrt{(-4)^2+(-3)^2}}\\ &= \frac{24}{\sqrt{169}\,\sqrt{25}}\\ &= \frac{24}{65}\end{aligned}$$

and the required angle is $\alpha = \cos^{-1}\left(\frac{24}{65}\right) \approx 1.19$ (or about 68°).

The formula for the angle between two vectors is often used in conjunction with the dot product. If we multiply both sides of the formula by $\|\mathbf{v}\|\ \|\mathbf{w}\|$, we obtain the following alternate form for the dot product formula.

GEOMETRICAL FORMULA FOR THE DOT PRODUCT

$$\mathbf{v}\cdot\mathbf{w} = \|\mathbf{v}\|\,\|\mathbf{w}\|\cos\theta$$

where θ is the angle ($0 \le \theta \le \pi$) between the vectors $\mathbf{v}$ and $\mathbf{w}$.

Two vectors are said to be **perpendicular**, or **orthogonal**, if the angle between them is $\theta = \frac{\pi}{2}$. The following theorem provides a useful condition for orthogonality.

Theorem 9.4 The orthogonal vector theorem

Nonzero vectors $\mathbf{v}$ and $\mathbf{w}$ are orthogonal if and only if

$$\mathbf{v}\cdot\mathbf{w} = 0$$

What this says The orthogonal vector theorem allows you to show that two vectors are orthogonal by finding the dot product of the vectors and showing this dot product is 0. On the other hand, if you know the vectors are orthogonal, then it follows that the dot product of the vectors is 0. In particular, $\mathbf{i}\cdot\mathbf{j} = \mathbf{j}\cdot\mathbf{k} = \mathbf{k}\cdot\mathbf{i} = 0$.

Proof: If the vectors are orthogonal, then the angle between them is $\frac{\pi}{2}$; therefore,

$$\mathbf{v}\cdot\mathbf{w} = \|\mathbf{v}\|\,\|\mathbf{w}\|\cos\frac{\pi}{2} = 0$$

Conversely, if $\mathbf{v}\cdot\mathbf{w} = 0$ and $\mathbf{v}$ and $\mathbf{w}$ are nonzero vectors, then $\cos\theta = 0$, so that $\theta = \frac{\pi}{2}$ and the vectors are orthogonal. ♦

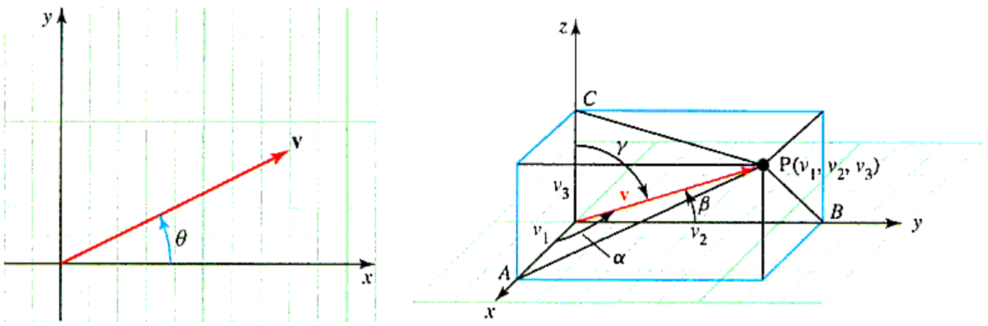
Example 4 Orthogonal vectors

Determine which (if any) of the following vectors are orthogonal:
$\mathbf{u} = 3\mathbf{i} + 7\mathbf{j} - 2\mathbf{k}$ $\quad$ $\mathbf{v} = 5\mathbf{i} - 3\mathbf{j} - 3\mathbf{k}$ $\quad$ $\mathbf{w} = \mathbf{j} - \mathbf{k}$

Solution $\mathbf{u} \cdot \mathbf{v} = 3(5) + 7(-3) + (-2)(-3) = 0$; orthogonal vectors.
$\mathbf{u} \cdot \mathbf{w} = 3(0) + 7(1) + (-2)(-1) = 9$; not orthogonal vectors.
$\mathbf{v} \cdot \mathbf{w} = 5(0) + (-3)(1) + (-3)(-1) = 0$; orthogonal vectors.

Direction Cosines

The direction of a vector in $\mathbb{R}^2$ positioned with its initial point at the origin may be specified by determining its angle of inclination; that is, the angle ϕ is the smallest positive angle the vector makes with the positive x-axis, as shown in Figure 9.28**a**. In $\mathbb{R}^3$, it is common practice to measure the direction of a nonzero vector $\mathbf{v}$ in terms of angles α, β, and γ between $\mathbf{v}$ and the positive coordinate axes as shown in Figure 9.28**b**. These angles are called the **direction angles** of $\mathbf{v}$, and their cosines are the **direction cosines** of $\mathbf{v}$.



a. In $\mathbb{R}^2$, the direction of $\mathbf{v}$ is given by the angle of inclination θ

b. Interactive In $\mathbb{R}^3$, the direction of $\mathbf{v}$ is given by the angles α, β, and γ

Figure 9.28 The direction angles of a vector $\mathbf{v}$

The direction cosines of $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ can be computed by taking the dot product of $\mathbf{v}$ with the three standard basis vectors, $\mathbf{i} = \langle 1, 0, 0 \rangle$, $\mathbf{j} = \langle 0, 1, 0 \rangle$, and $\mathbf{k} = \langle 0, 0, 1 \rangle$. For example, to find $\cos \alpha$, note that

$$v_1 = \mathbf{v} \cdot \mathbf{i} = \|\mathbf{v}\| \, \|\mathbf{i}\| \cos \alpha, \text{ so that } \cos \alpha = \frac{v_1}{\|\mathbf{v}\|}$$

and, similarly,

$$\cos \beta = \frac{v_2}{\|\mathbf{v}\|} \qquad \text{and} \qquad \cos \gamma = \frac{v_3}{\|\mathbf{v}\|}$$

It can be shown (Problem 59) that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

This identity can be used to check whether you have computed the direction cosines correctly.

Example 5 Direction cosines and direction angles

Find the direction cosines and direction angles (to the nearest degree) of the vector $\mathbf{v} = -2\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$, and verify the formula

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

Solution We find $\|\mathbf{v}\| = \sqrt{(-2)^2 + 3^2 + 5^2} = \sqrt{38}$. Therefore,

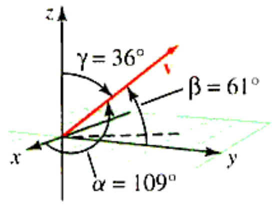


Figure 9.29 The vector **v** with direction angles α, β, and γ

$$\cos\alpha = \frac{v_1}{\|\mathbf{v}\|} = \frac{-2}{\sqrt{38}} \approx -0.324428 \quad \alpha \approx \cos^{-1}(-0.324428) \approx 109^\circ$$
$$\cos\beta = \frac{v_2}{\|\mathbf{v}\|} = \frac{3}{\sqrt{38}} \approx 0.4866642 \quad \beta \approx \cos^{-1}(0.4866642) \approx 61^\circ$$
$$\cos\gamma = \frac{v_3}{\|\mathbf{v}\|} = \frac{5}{\sqrt{38}} \approx 0.811107 \quad \gamma \approx \cos^{-1}(0.811107) \approx 36^\circ$$

and $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \left(\frac{-2}{\sqrt{38}}\right)^2 + \left(\frac{3}{\sqrt{38}}\right)^2 + \left(\frac{5}{\sqrt{38}}\right)^2 = 1$. These direction angles are shown in Figure 9.29.

Projections

Let **v** and **w** be two vectors in $\mathbb{R}^2$ drawn so that they have a common initial point, as shown in Figure 9.30.*

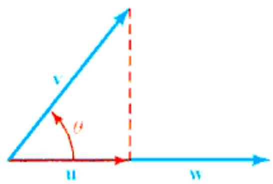


Figure 9.30 Projection of **v** onto **w**

If we drop a perpendicular from the head of **v** to the line determined by **w**, we determine a vector called the **vector projection of v onto w**, which we have labeled **u** in Figure 9.30. The **scalar projection of v onto w** (also called the **component of v along w**) is the length of the vector projection, so it is denoted by $\|\mathbf{u}\|$. Let θ be the acute angle between **v** and **w**. Then, by definition of cosine we have

$$\begin{aligned}\|\mathbf{u}\| &= \|\mathbf{v}\|\cos\theta \\ &= \|\mathbf{v}\|\left(\frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{v}\|\,\|\mathbf{w}\|}\right) \\ &= \frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{w}\|} \qquad \textit{Cosine is positive because } \theta \textit{ is acute.}\end{aligned}$$

Note that $\left(\frac{\mathbf{v}\cdot\mathbf{w}}{\mathbf{w}\cdot\mathbf{w}}\right)\mathbf{w}$ *is not the same as* $\left(\frac{\mathbf{v}}{\mathbf{w}}\right)\mathbf{w}$*; you cannot "cancel" the vector* **w**. *Remember,* $\mathbf{v}\cdot\mathbf{w}$ *and* $\mathbf{w}\cdot\mathbf{w}$ *are real numbers whereas* $\mathbf{v}/\mathbf{w}$ *is not defined.*

If θ is obtuse, its cosine is negative and $\|\mathbf{u}\| = -\|\mathbf{v}\|\cos\theta$. To find a formula for the vector projection, we note that it is the scalar component of **v** in the direction of **w**. If θ is acute, then the vector projection has length $\|\mathbf{v}\|\cos\theta$ and has direction $\mathbf{w}/\|\mathbf{w}\|$ (the unit vector of **w**). If the angle is obtuse, the vector projection has length $-\|\mathbf{v}\|\cos\theta$ and has direction $-\mathbf{w}/\|\mathbf{w}\|$. In either case,

$$\begin{aligned}\mathbf{u} &= (\|\mathbf{v}\|\cos\theta)\frac{\mathbf{w}}{\|\mathbf{w}\|} \\ &= \frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{w}\|}\left(\frac{\mathbf{w}}{\|\mathbf{w}\|}\right) \\ &= \frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{w}\|^2}\,\mathbf{w} \\ &= \left(\frac{\mathbf{v}\cdot\mathbf{w}}{\mathbf{w}\cdot\mathbf{w}}\right)\mathbf{w}\end{aligned}$$

*Even though Figure 9.30 is drawn in $\mathbb{R}^2$, the projection formula applies to $\mathbb{R}^3$ as well.

PROJECTIONS If $\mathbf{v}$ and $\mathbf{w}$ are nonzero vectors, then the **vector projection** of $\mathbf{v}$ in the direction of $\mathbf{w}$ (a vector), denoted by $\text{proj}_{\mathbf{w}}\mathbf{v}$ is

$$\text{proj}_{\mathbf{w}}\mathbf{v} = \left(\frac{\mathbf{v}\cdot\mathbf{w}}{\mathbf{w}\cdot\mathbf{w}}\right)\mathbf{w}$$

scalar projection of $\mathbf{v}$ onto $\mathbf{w}$ (a number), denoted by $\text{comp}_{\mathbf{w}}\mathbf{v}$ is

$$\text{comp}_{\mathbf{w}}\mathbf{v} = \frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{w}\|}$$

Note $comp_{w}v > 0$ *when* $0 \le \theta < \frac{\pi}{2}$ *and* $comp_{w}v < 0$ *when* $\frac{\pi}{2} < \theta \le \pi$.

In $\mathbb{R}^3$, the $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ components of any vector $\mathbf{v}$ are the scalar projections of $\mathbf{v}$ onto the appropriate basis vector, as shown in Figure 9.31.

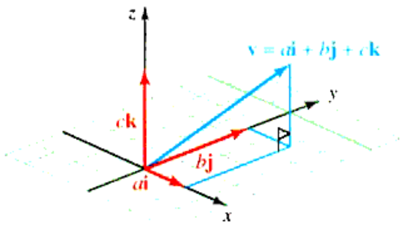


Figure 9.31 Basis vectors are vector projections

Example 6 Vector and scalar projections

Find the scalar and vector projections of $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ onto $\mathbf{w} = 2\mathbf{i} - 2\mathbf{j} - \mathbf{k}$.

Solution We first find the vector projection:

$$\begin{aligned}
\text{proj}_{\mathbf{w}}\mathbf{v} &= \left(\frac{\mathbf{v}\cdot\mathbf{w}}{\mathbf{w}\cdot\mathbf{w}}\right)\mathbf{w}\\
&= \left(\frac{2(2)+3(-2)+5(-1)}{2^2+(-2)^2+(-1)^2}\right)(2\mathbf{i}-2\mathbf{j}-\mathbf{k})\\
&= -\frac{7}{9}(2\mathbf{i}-2\mathbf{j}-\mathbf{k})\\
&= -\frac{14}{9}\mathbf{i}+\frac{14}{9}\mathbf{j}+\frac{7}{9}\mathbf{k}
\end{aligned}$$

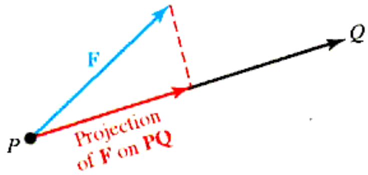
To find the scalar projection, we can find the length of the vector projection, or we can use the scalar projection formula (which is usually easier than finding the length directly):

$$\begin{aligned}
\text{comp}_{\mathbf{w}}\mathbf{v} &= \frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{w}\|}\\
&= \frac{2(2)+3(-2)+5(-1)}{\sqrt{2^2+(-2)^2+(-1)^2}}\\
&= \frac{-7}{3}
\end{aligned}$$

Work as a Dot Product

One important application of the dot product and projections occurs in physics when calculating the amount of work done by a constant force. When a constant force of magnitude F is applied to an object through a distance d, the **work** performed is defined to be the product of force and distance. This means that the work, W, done by a constant force directed along the line of the motion of the object to be $W = Fd$ (Figure 9.32**a**). We now consider a force acting in some other direction (Figure 9.32**b**).

a. If the force **F** acts along the line of motion, then $W = \|\mathbf{F}\|\,\|\mathbf{PQ}\|$



b. If **F** acts at a nonzero angle with the line of motion, then $W = \|\text{proj. of } \mathbf{F} \text{ onto } \mathbf{PQ}\|\,\|\mathbf{PQ}\|$

Figure 9.32 Work W as a dot product

Physics experiments indicate that when a force F moves an object along the line from point P to point Q, the work performed is given by the product

$$W = [\text{SCALAR COMPONENT OF } \mathbf{F} \text{ ALONG } \mathbf{PQ}]\ [\text{DISTANCE MOVED BY THE OBJECT}]$$

The distance moved by the object is the length of the *displacement vector* **PQ**, and we find that

$$W = \overbrace{\frac{\mathbf{F}\cdot\mathbf{PQ}}{\|\mathbf{PQ}\|}}^{\text{Scalar component of } F \text{ along } PQ}\ \underbrace{\|\mathbf{PQ}\|}_{\text{Distance moved by the object}} = \mathbf{F}\cdot\mathbf{PQ}$$

WORK AS A DOT PRODUCT An object that moves along a line with displacement **PQ** against a constant force **F** performs

$$W = \mathbf{F}\cdot\mathbf{PQ}$$

units of work.

Example 7 Work performed by a constant force

Suppose that the wind is blowing with a force **F** of magnitude 500 lb in the direction of N30°E over a boat's sail. How much work does the wind perform in moving the boat in a northerly direction a distance of 100 ft? Give your answer in foot-pounds.

Solution We see that $\|\mathbf{F}\| = 500$ lb and is in the direction of N30°E, as shown in Figure 9.33.

The displacement direction is $\mathbf{PQ} = 100\mathbf{j}$, so $\|\mathbf{PQ}\| = 100$ ft. Thus,

$$\begin{aligned}\mathbf{F} &= 500\cos 60^\circ\mathbf{i} + 500\sin 60^\circ\mathbf{j}\\ &= 250\mathbf{i} + 250\sqrt{3}\mathbf{j}\end{aligned}$$

Thus, the work performed is

$$W = \mathbf{F}\cdot\mathbf{PQ} = 100(250\sqrt{3}) = 25{,}000\sqrt{3}$$

Thus, the work is approximately 43,300 ft-lb.

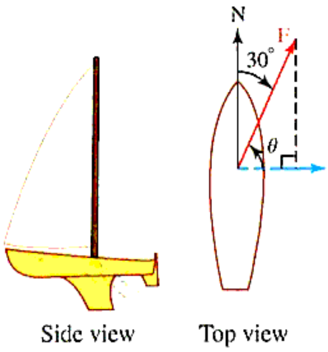


Figure 9.33 Work performed

In the U.S. system of measurements, work is typically expressed in *foot-pounds, inch-pounds*, or *foot-tons*. In the International System (SI), work is expressed in newton-meters (called *joules*), and in the centimeter-gram-second (CGS) system, the basic unit of work is the dyne-centimeter (called an *erg*). These units are shown in Table 9.1.

Table 9.1 Common Units of Work and Force

Mass	Distance	Unit of force	Work
kg	m	newton (N)	joule
g	cm	dyne (dyn)	erg
slug	ft	pound	ft-lb

PROBLEM SET 9.3

Level 1

1. ■ What does this say? Discuss how to find a dot product, and describe an application of the dot product.
2. ■ What does this say? What does it mean for vectors to be orthogonal? What is meant by the vector and scalar projections of a vector $\mathbf{v}$ onto another vector $\mathbf{w}$?

Find the dot product $\mathbf{v}\cdot\mathbf{w}$ *in Problems* 3-6.

3. $\mathbf{v} = \langle 3, -2, 4\rangle$; $\mathbf{w} = \langle 2, -1, -6\rangle$
4. $\mathbf{v} = \langle 2, -6, 0\rangle$; $\mathbf{w} = \langle 0, -3, 7\rangle$
5. $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$; $\mathbf{w} = -3\mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$
6. $\mathbf{v} = 3\mathbf{i} - \mathbf{j}$; $\mathbf{w} = 2\mathbf{i} + 5\mathbf{j}$

State whether the given pairs of vectors in Problems 7-10 *are orthogonal.*

7. $\mathbf{v} = \mathbf{i}$; $\mathbf{w} = \mathbf{k}$
8. $\mathbf{v} = \mathbf{j}$; $\mathbf{w} = -\mathbf{k}$
9. $\mathbf{v} = 3\mathbf{i} - 2\mathbf{j}$; $\mathbf{w} = 6\mathbf{i} + 9\mathbf{j}$
10. $\mathbf{v} = 4\mathbf{i} - 5\mathbf{j} + \mathbf{k}$; $\mathbf{w} = 8\mathbf{i} + 10\mathbf{j} - 2\mathbf{k}$

Let $\mathbf{v} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ *and* $\mathbf{w} = \mathbf{i} + \mathbf{j} - \mathbf{k}$. *Evaluate the expressions in Problems* 11-14.

11. $(\mathbf{v}+\mathbf{w})\cdot(\mathbf{v}-\mathbf{w})$
12. $(\mathbf{v}\cdot\mathbf{w})\mathbf{w}$
13. $(\|\mathbf{v}\|\mathbf{w})\cdot(\|\mathbf{w}\|\,\mathbf{v})$
14. $\dfrac{2\mathbf{v} + 3\mathbf{w}}{\|3\mathbf{v} + 2\mathbf{w}\|}$

Let $\mathbf{v} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ *and* $\mathbf{w} = 2\mathbf{i} + 4\mathbf{j} - \mathbf{k}$; *and find the vector or scalar requested in Problems* 15-18.

15. $2\mathbf{v} - 3\mathbf{w}$
16. $\|\mathbf{v}\|\mathbf{w}$
17. $\|2\mathbf{v} - 3\mathbf{w}\|$
18. $\|\mathbf{v} + \mathbf{w}\|(\mathbf{v}+\mathbf{w})$

Evaluate the expressions given in Problems 19-22.

19. $\|\mathbf{i} + \mathbf{j} + \mathbf{k}\|$
20. $\|\mathbf{i} - \mathbf{j} + \mathbf{k}\|$
21. $\|2\mathbf{i} + \mathbf{j} - 3\mathbf{k}\|^2$
22. $\|2(\mathbf{i} - \mathbf{j} + \mathbf{k}) - 3(2\mathbf{i} + \mathbf{j} - \mathbf{k})\|^2$

Find the angle between the vectors given in Problems 23-26. *Round to the nearest degree.*

23. $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{k}$; $\mathbf{w} = \mathbf{i} - \mathbf{j} + \mathbf{k}$
24. $\mathbf{v} = 2\mathbf{i} + \mathbf{k}$; $\mathbf{w} = \mathbf{j} - 3\mathbf{k}$
25. $\mathbf{v} = 2\mathbf{j} + \mathbf{k}$; $\mathbf{w} = \mathbf{i} - 2\mathbf{k}$
26. $\mathbf{v} = 4\mathbf{i} - \mathbf{j} + \mathbf{k}$; $\mathbf{w} = 2\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$

Find the scalar and vector projections of $\mathbf{v}$ *onto* $\mathbf{w}$ *in Problems* 27-30.

27. $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{k}$; $\mathbf{w} = 2\mathbf{k}$
28. $\mathbf{v} = 2\mathbf{i} - 3\mathbf{j}$; $\mathbf{w} = 2\mathbf{j} - 3\mathbf{k}$
29. $\mathbf{v} = \mathbf{i} + 2\mathbf{k}$; $\mathbf{w} = -3\mathbf{j}$
30. $\mathbf{v} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$; $\mathbf{w} = \mathbf{i} + \mathbf{j} + \mathbf{k}$
31. Find two distinct unit vectors orthogonal to both $\mathbf{v} = \mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{w} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$.
32. Find two distinct unit vectors orthogonal to both $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{w} = -\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.
33. Find a unit vector that points in the direction opposite to $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$.
34. Find a vector that points in the same direction as $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and has length one-third.
35. Find x, y, and z that solve
$$x(\mathbf{i} + \mathbf{j} + \mathbf{k}) + y(\mathbf{i} - \mathbf{j} + 2\mathbf{k}) + z(\mathbf{i} + \mathbf{k}) = 2\mathbf{i} + \mathbf{k}$$
36. Find x, y, and z that solve
$$x(\mathbf{i} - \mathbf{k}) + y(\mathbf{j} + \mathbf{k}) + z(\mathbf{i} - \mathbf{j}) = 5\mathbf{i} - \mathbf{k}$$

*Some of the measurements and units in this table may not be familiar to you. For example, a *slug* is a unit of measurement in physics defined to be "the unit of mass that is accelerated at the rate of one foot per second per second when acted on by a force of one pound weight." If you have not studied physics, this may be meaningless to you right now, but remember that our focus here is not on the units of measurement, but on how to calculate *work* using scalar multiplication of vectors. However, if you are interested, a *dyne* is defined to be "the force required to accelerate a mass of one centimeter per second per second to a mass of one gram" and a *newton* is defined to be "the unit of force required to accelerate a mass of one kilogram one meter per second per second." Thus, one newton is equal to 100,000 dynes.

37. Find a number a that guarantees that the vectors $3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $2\mathbf{i} + a\mathbf{j} - 2a\mathbf{k}$ will be orthogonal.
38. Find x if the vectors $\mathbf{v} = 3\mathbf{i} - x\mathbf{j} + 2\mathbf{k}$ and $\mathbf{w} = x\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ are to be orthogonal.

Find the direction cosines and the direction angles for the vectors given in Problems 39-42.

39. $\mathbf{v} = 2\mathbf{i} - 3\mathbf{j} - 5\mathbf{k}$
40. $\mathbf{v} = 3\mathbf{i} - 2\mathbf{k}$
41. $\mathbf{v} = 5\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}$
42. $\mathbf{v} = \mathbf{j} - 5\mathbf{k}$
43. Let $\mathbf{v} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$ and $\mathbf{w} = 4\mathbf{i} + 3\mathbf{k}$.
    Find
    a. $\mathbf{v} \cdot \mathbf{w}$
    b. $\cos\theta$, where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$
    c. a scalar s such that $\mathbf{v}$ is orthogonal to $\mathbf{v} - s\mathbf{w}$
    d. a scalar t such that $\mathbf{v} + t\mathbf{w}$ is orthogonal to $\mathbf{w}$
44. Let $\mathbf{v} = 4\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{w} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$.
    Find
    a. $\mathbf{v} \cdot \mathbf{w}$
    b. $\cos\theta$, where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$
    c. a scalar s such that $\mathbf{v}$ is orthogonal to $\mathbf{v} - s\mathbf{w}$
    d. a scalar t such that $t\mathbf{v} + \mathbf{w}$ is orthogonal to $\mathbf{w}$
45. Find the cosine of the angle between the vectors $\mathbf{v} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{w} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$. Then find the vector projection of $\mathbf{v}$ onto $\mathbf{w}$.
46. Find the scalar projection of the force $\mathbf{F} = 4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ in the direction of the vector $\mathbf{v} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$.
47. Find the work done by the constant force $\mathbf{F} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ when it moves a particle along the line from $P(1, 0, -1)$ to $Q(3, 1, 2)$.
48. Find the work performed when a force $\mathbf{F} = \frac{6}{7}\mathbf{i} - \frac{2}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$ is applied to an object moving along the line from $P(-3, -5, 4)$ to $Q(4, 9, 11)$.

Level 2

49. Fred and his friend Sam are pulling a heavy log along flat horizontal ground by ropes attached to the front of the log. The ropes are 8 ft long. Fred holds his rope 2 ft above the log and 1 ft to the side, and Sam holds his end 1 ft above the log and 1 ft to the opposite side, as shown in Figure 9.34.

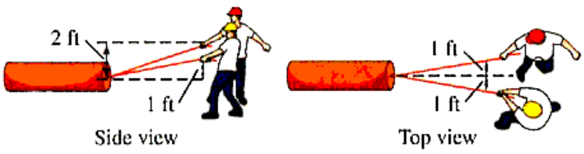


Figure 9.34 Problem 49

If Fred exerts a force of 30 lb and Sam exerts a force of 20 lb, what is the resultant force on the log?

50. Find the force required to keep a 5,000-lb van from rolling downhill if it is parked on a 10° slope (see Figure 9.35).

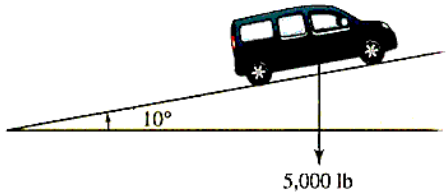


Figure 9.35 Problem 50

51. A trunk is dragged 20 ft across a floor, using a force of 50 lb, as shown in Figure 9.36.

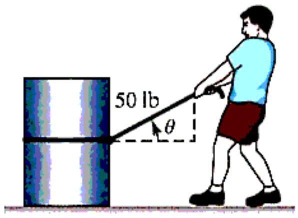


Figure 9.36 Problem 51

Find the work done if the direction of the force is inclined θ to the horizontal, where
a. $\theta = \frac{\pi}{3}$
b. $\theta = \frac{\pi}{4}$

52. Suppose that the wind is blowing with a 1,000-lb magnitude force $\mathbf{F}$ in the direction of N60°W behind a boat's sail. How much work does the wind perform in moving the boat in a northerly direction a distance of 50 ft? Give your answer in foot-pounds.

Orthogonality of functions *One of the big discoveries in modern mathematics was that it is very fruitful to extend some of the geometric notions of this chapter to functions—for example, taking the dot product of two functions or asking whether two functions are orthogonal, or attempting to project a function onto a second function (or onto a "plane" of functions). You are introduced to these important concepts in Problems* 53 *and* 54.

53. We say that two functions f and g are **orthogonal** on $[a, b]$ if
$$\int_a^b f(x)g(x)\,dx = 0$$
    a. Show that the two functions x^2 and $x^3 - 5x$ are orthogonal on $[-b, b]$ for any positive b.
    b. For distinct positive integers k and n, show that $\sin kx$ and $\sin nx$ are orthogonal on interval $[-\pi, \pi]$.

That is, the *family* of functions $\sin x$, $\sin 2x, \cdots$ are *mutually* orthogonal on $[-\pi, \pi]$. You may need the product-to-sum identity (trigonometric identity 28, Appendix E):

$$2\sin\alpha\sin\beta = \cos(\alpha-\beta) - \cos(\alpha+\beta)$$

54. *Historical Quest*

Jean Fourier (1768-1830)

Jean Baptiste Joseph Fourier began his studies by training for the priesthood. In spite of this first calling, his interest in mathematics remained intense, and in 1789 he wrote, "Yesterday was my 21st birthday, and at that age Newton and Pascal had already acquired many claims to immortality." He did not take his religious vows but continued his mathematical research while teaching. Fourier was renowned as an outstanding teacher, but it was not until around 1804-1807 that he did his important mathematical work on the theory of heat. During that time, he expanded on an idea of Daniel Bernoulli by showing how many functions, both continuous and discontinuous, can be represented by a trigonometric series of the general form

$$\frac{1}{2}a_0 + a_1\cos x + a_2\cos 2x + a_3\cos 3x + \cdots + b_1\sin x + b_2\sin 2x + b_3\sin 3x + \cdots$$

The study of such series, called **Fourier series**, is of fundamental importance in both theoretical and applied mathematics, and they are used for modeling phenomena in engineering and physics as well as in areas such as computer science and medical research.

The issue of whether a given function f can be represented by a Fourier series is a matter for a more advanced course. For this *Quest*, assume that $f(x)$ is such a function and that

$$f(x) = \frac{1}{2}a_0 + \sum_{k=1}^{\infty} a_k\cos kx + \sum_{k=1}^{\infty} b_k\sin kx$$

on the interval $[-\pi, \pi]$. In particular, if f is a continuous odd function (that is, $f(-x) = -f(x)$ on $[-\pi, \pi]$), it can be shown that $a_k = 0$ for $k = 0, 1, 2, \cdots$, and that

$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin kx\,dx = \frac{2}{\pi}\int_0^{\pi} f(x)\sin kx\,dx$$

for $k = 1, 2, \cdots$.

a. Using this result, show that the function $f(x) = x$ on $[-\pi, \pi]$ has the Fourier series representation

$$f(x) = 2\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k}\sin kx$$

b. Use the Fourier series in part **a** to evaluate the convergent alternating series

$$\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{2k-1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots$$

Level 3

55. Show that the vector $\mathbf{B} = \|\mathbf{v}\|\,\mathbf{u} + \|\mathbf{u}\|\,\mathbf{v}$ bisects the angle θ between the two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$.

56. Use vector methods to prove that an angle inscribed in a semicircle must be a right angle.

57. a. Show that
$$(\mathbf{v}+\mathbf{w})\cdot(\mathbf{v}+\mathbf{w}) = \|\mathbf{v}\|^2 + \|\mathbf{w}\|^2 + 2(\mathbf{v}\cdot\mathbf{w})$$

b. Use part **a** to prove the **triangle inequality:**

$$\|\mathbf{v}+\mathbf{w}\| \le \|\mathbf{v}\| + \|\mathbf{w}\|$$

Hint: Note that

$$\|\mathbf{v}+\mathbf{w}\|^2 = (\mathbf{v}+\mathbf{w})\cdot(\mathbf{v}+\mathbf{w})$$

58. The **Cauchy-Schwarz** inequality in $\mathbb{R}^3$ states that for any vectors $\mathbf{v}$ and $\mathbf{w}$

$$|\mathbf{v}\cdot\mathbf{w}| \le \|\mathbf{v}\|\,\|\mathbf{w}\|$$

a. Prove the Cauchy-Schwarz inequality. *Hint:* Use the formula for the angle between two vectors.

b. Show that equality $|\mathbf{v}\cdot\mathbf{w}| = \|\mathbf{v}\|\,\|\mathbf{w}\|$ occurs if and only if $\mathbf{v} = t\mathbf{w}$ for some scalar t.

c. Use the Cauchy-Schwarz inequality to prove the **triangle inequality:**

$$\|\mathbf{v}+\mathbf{w}\| \le \|\mathbf{v}\| + \|\mathbf{w}\|$$

59. Show that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$, where α, β, and γ are the direction angles of a triangle $\triangle ABC$.

60. Find the angle (to the nearest degree) between the diagonal of a cube and a diagonal of one of its faces. Assume both diagonals have the same initial point.

9.4 The Cross Product

IN THIS SECTION: *Definition of the cross product, geometric interpretation of the cross product, properties of the cross product, the scalar triple product and volume, torque*

In this section, we introduce a second type of vector multiplication called *cross product*, which turns out to be a vector. We shall find that the cross product is a vector orthogonal to each of the given vectors. This geometric property will be particularly useful in the next section.

We have considered two types of products involving vectors. The first was *scalar multiplication*, which is the multiplication of a vector by a number (which is called a *scalar*). In the last section we defined a product of vectors that produces a number (i.e., scalar); this product is known as *scalar product* (because the answer is a scalar) or *dot product* since the notation we use for this product is a dot. We are now ready to consider a third product of vectors that produces a vector. It is called the *vector product* or *cross product* because we use the multiplication cross ($\times$) to denote this type of product.

Definition of the Cross Product*

The cross product is sometimes called **vector product** because the result is a vector. In other applications it is called the **outer product**. The definition requires a basis of **i**, **j**, and **k** and therefore is a definition that makes sense only in $\mathbb{R}^3$.

CROSS PRODUCT If $\mathbf{v} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{w} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, the **cross product**, written $\mathbf{v} \times \mathbf{w}$, is the vector

$$\mathbf{v} \times \mathbf{w} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$$

This definition seems very strange, indeed, until we show that these terms can be obtained by using a determinant

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

We can verify this is the formula for cross product by expanding this determinant about the first row.

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix}\mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix}\mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}\mathbf{k}$$

j is in row 1, column 2, so do not forget the negative sign here.

$$= (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$$
$$= (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$$

*Determinants (see Appendix F) are necessary for this section.

Example 1 Cross product

Find $\mathbf{v} \times \mathbf{w}$ where $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{w} = 7\mathbf{j} - 4\mathbf{k}$.

Solution $\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 3 \\ 0 & 7 & -4 \end{vmatrix}$

Do not forget minus here (j is negative position).

$$= [(-1)(-4) - 3(7)]\mathbf{i} - [2(-4) - 0(3)]\mathbf{j} + [2(7) - 0(-1)]\mathbf{k}$$
$$= -17\mathbf{i} + 8\mathbf{j} + 14\mathbf{k}$$

Properties of determinants can also be used to establish properties of the cross product. For instance, the following computation shows that the cross product is *not* commutative. This property is sometimes called **anticommutativity**.

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = -\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \end{vmatrix} = -(\mathbf{w} \times \mathbf{v})$$

The cross product has interesting properties. We have already seen that $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$. This and other properties are listed in the following box.

Theorem 9.5 Properties of the cross product

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in $\mathbb{R}^3$ and s and t are scalars, then several properties can be derived:

Scalar distributivity	$(s\mathbf{v}) \times (t\mathbf{w}) = st(\mathbf{v} \times \mathbf{w})$
Vector distributivity*	$\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$
	$(\mathbf{u} + \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \times \mathbf{w}) + (\mathbf{v} \times \mathbf{w})$
Anticommutativity	$\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$
Product of a multiple	$\mathbf{v} \times \mathbf{v} = 0$; if $\mathbf{w} = s\mathbf{v}$, then $\mathbf{v} \times \mathbf{w} = 0$
Zero product	$\mathbf{v} \times \mathbf{0} = \mathbf{0} \times \mathbf{v} = \mathbf{0}$
Lagrange's identity	$\lVert\mathbf{v} \times \mathbf{w}\rVert^2 = \lVert\mathbf{v}\rVert^2 \lVert\mathbf{w}\rVert^2 - (\mathbf{v} \cdot \mathbf{w})^2$
***cab-bac* formula**	$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{c} \cdot \mathbf{a})\mathbf{b} - (\mathbf{b} \cdot \mathbf{a})\mathbf{c}$

Proof: Let

$$\mathbf{u} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}, \qquad \mathbf{v} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}, \qquad \text{and} \qquad \mathbf{w} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$$

Scalar distributivity and *vector distributivity* are proved by using the definition of the cross product and the corresponding properties of real numbers. For *product of a multiple* we find:

$$\mathbf{v} \times s\mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ sb_1 & sb_2 & sb_3 \end{vmatrix} = s\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \mathbf{0}$$

Property of determinants (two rows the same)

*Properly, it is the distributive property of vectors for cross product over addition, which, for convenience, we shorten to vector distributivity.

The *zero product* property is obvious. You are asked to prove *Lagrange's identity* in Problem 52 and the *cab-bac* formula in Problem 55. ♦

Geometric Interpretation of the Cross Product

We will show with the following property that the vector $(\mathbf{v} \times \mathbf{w})$ is orthogonal to both the vectors $\mathbf{v}$ and $\mathbf{w}$. The only way this can occur is in a three-dimensional setting. Any two distinct nonzero vectors in $\mathbb{R}^3$ that are not parallel (that is, not scalar multiples of one another) can be arranged to determine a plane. Then, according to the geometric property of the cross product the vector product *must* be orthogonal to this plane, as shown in Figure 9.37.

This property is one of the most important properties when working in three dimensions. We will use this property extensively in Section 9.6.

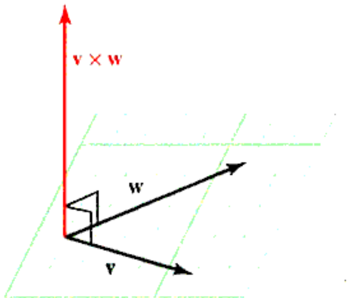


Figure 9.37 Vector product

Theorem 9.6 Geometric property of the cross product

If $\mathbf{v}$ and $\mathbf{w}$ are nonzero vectors in $\mathbb{R}^3$ that are not multiples of one another, then $\mathbf{v} \times \mathbf{w}$ is orthogonal to both $\mathbf{v}$ and $\mathbf{w}$.

Proof: We will show $(\mathbf{v} \times \mathbf{w})$ is orthogonal to $\mathbf{v}$ and leave the proof that $\mathbf{v} \times \mathbf{w}$ is orthogonal to $\mathbf{w}$ as an exercise. Let $\mathbf{v} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{w} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$. Then,

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$$

To show this vector is orthogonal to $\mathbf{v}$, we find $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w})$:

$$\begin{aligned} \mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) &= a_1(a_2b_3 - a_3b_2) - a_2(a_1b_3 - a_3b_1) + a_3(a_1b_2 - a_2b_1) \\ &= a_1a_2b_3 - a_1a_3b_2 - a_1a_2b_3 + a_2a_3b_1 + a_1a_3b_2 - a_2a_3b_1 \\ &= 0 \end{aligned}$$

Example 2 A vector orthogonal to two given vectors

Find a nonzero vector that is orthogonal to both $\mathbf{v} = -2\mathbf{i} + 3\mathbf{j} - 7\mathbf{k}$ and $\mathbf{w} = 5\mathbf{i} + 9\mathbf{k}$.

Solution The cross product $\mathbf{v} \times \mathbf{w}$ is orthogonal to both $\mathbf{v}$ and $\mathbf{w}$.

$$\begin{aligned} \mathbf{v} \times \mathbf{w} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 3 & -7 \\ 5 & 0 & 9 \end{vmatrix} \\ &= (27 + 0)\mathbf{i} - (-18 + 35)\mathbf{j} + (0 - 15)\mathbf{k} \\ &= 27\mathbf{i} - 17\mathbf{j} - 15\mathbf{k} \end{aligned}$$

Because both $\mathbf{v} \times \mathbf{w}$ and $\mathbf{w} \times \mathbf{v}$ are orthogonal to the plane determined by $\mathbf{v}$ and $\mathbf{w}$, and because $(\mathbf{v} \times \mathbf{w}) = -(\mathbf{w} \times \mathbf{v})$, we see that one points up from the given plane and the other points down. In order to see which is which, we state the **right-hand rule**, which is described in Figure 9.38.

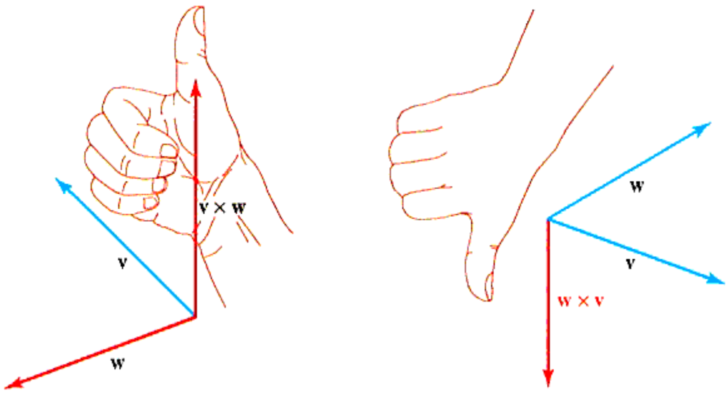


Figure 9.38 Right-hand rule

If you place the palm of your right hand along $\mathbf{v}$ in the positive direction and curl your fingers toward $\mathbf{w}$, then your fingers are pointing in the direction of positive $\mathbf{v}$, and your thumb points in the positive direction of $\mathbf{v} \times \mathbf{w}$.

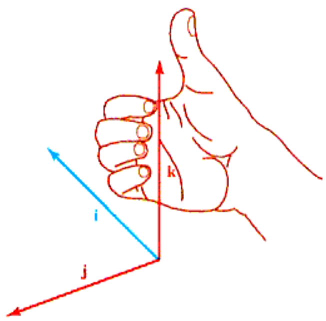
Example 3 Right-hand rule

Use the right hand rule to verify each of the following cross products.

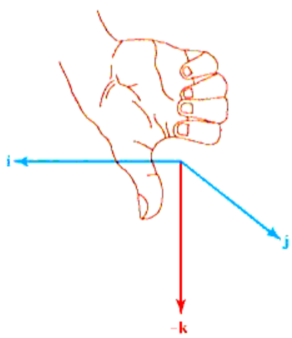
$$\begin{array}{lll} \mathbf{i} \times \mathbf{j} = \mathbf{k} & \mathbf{j} \times \mathbf{i} = -\mathbf{k} & \mathbf{k} \times \mathbf{i} = \mathbf{j} \\ \mathbf{i} \times \mathbf{k} = -\mathbf{j} & \mathbf{j} \times \mathbf{k} = \mathbf{i} & \mathbf{k} \times \mathbf{j} = -\mathbf{i} \\ \mathbf{i} \times \mathbf{i} = \mathbf{0} & \mathbf{j} \times \mathbf{j} = \mathbf{0} & \mathbf{k} \times \mathbf{k} = \mathbf{0} \end{array}$$

Solution

$\mathbf{i} \times \mathbf{j}$ Place the palm of your right hand along $\mathbf{i}$ and curl your fingers toward $\mathbf{j}$. The answer is $\mathbf{k}$.



$\mathbf{j} \times \mathbf{i}$ Place the palm of your right hand along $\mathbf{j}$ and curl your fingers toward $\mathbf{i}$. The answer is $-\mathbf{k}$.



$\mathbf{i} \times \mathbf{i}$ Can't use right-hand rule. However, we know

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{vmatrix} = \mathbf{0}$$

The other parts are left for you to verify.

Properties of the Cross Product

Next, we will find the magnitude of the cross product of two vectors.

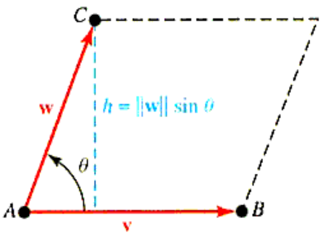


Figure 9.39 Area of a parallelogram

Theorem 9.7 Magnitude of a cross product

If $\mathbf{v}$ and $\mathbf{w}$ are nonzero vectors in $\mathbb{R}^3$ with θ the angle between $\mathbf{v}$ and $\mathbf{w}$ $(0 \le \theta \le \pi)$, then

$$\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\|\,\|\mathbf{w}\| \sin\theta$$

What this says The magnitude of the cross product of two vectors $\|\mathbf{v} \times \mathbf{w}\|$ is equal the area of the parallelogram having $\mathbf{v}$ and $\mathbf{w}$ as adjacent sides, as shown in Figure 9.39.

Proof: Let $\mathbf{v} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{w} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$.

$$\begin{aligned}
\|\mathbf{v}\|\,(\|\mathbf{w}\| \sin\theta) &= \|\mathbf{v}\|\|\mathbf{w}\| \sqrt{1 - \cos^2\theta} && \textit{From } \sin^2\theta + \cos^2\theta = 1. \\
&= \|\mathbf{v}\|\|\mathbf{w}\| \sqrt{1 - \left[\frac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{v}\|\,\|\mathbf{w}\|}\right]^2} && \textit{Angle between two vectors: } \cos\theta = \tfrac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} \\
&= \|\mathbf{v}\|\|\mathbf{w}\| \frac{\sqrt{\|\mathbf{v}\|^2\,\|\mathbf{w}\|^2 - (\mathbf{v}\cdot\mathbf{w})^2}}{\|\mathbf{v}\|\,\|\mathbf{w}\|} && \textit{Common denominator} \\
&= \sqrt{\|\mathbf{v}\|^2\,\|\mathbf{w}\|^2 - (\mathbf{v}\cdot\mathbf{w})^2} \\
&= \sqrt{(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2} \\
&= \sqrt{(a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2} \\
&= \|\mathbf{v} \times \mathbf{w}\|
\end{aligned}$$

Consider the parallelogram shown in Figure 9.39. Then the area of this parallelogram is

$$\begin{aligned}
\text{AREA} &= (\text{BASE})(\text{HEIGHT}) \\
&= \|\mathbf{v}\|\,(\|\mathbf{w}\| \sin\theta) \\
&= \|\mathbf{v} \times \mathbf{w}\|
\end{aligned}$$

Here is an example that uses this formula. You might also note that even though $\mathbf{v} \times \mathbf{w} \neq \mathbf{w} \times \mathbf{v}$, it is true that $\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{w} \times \mathbf{v}\|$.

Example 4 Area of a triangle

Find the area of the triangle with vertices $P(-2,4,5), Q(0,7,-4)$, and $R(-1,5,0)$.

Solution Draw this triangle as shown in Figure 9.40.

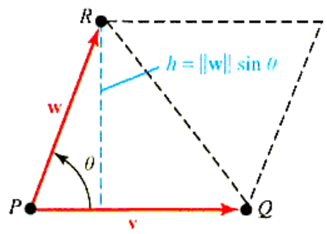


Figure 9.40 Area of a triangle

Then $\triangle PQR$ has half the area of the parallelogram determined by the vectors **PQ** and **PR**; that is, the triangle has area

$$A = \frac{1}{2}\|\mathbf{PQ} \times \mathbf{PR}\|$$

First find

$$\mathbf{PQ} = (0+2)\mathbf{i} + (7-4)\mathbf{j} + (-4-5)\mathbf{k} = 2\mathbf{i} + 3\mathbf{j} - 9\mathbf{k}$$
$$\mathbf{PR} = (-1+2)\mathbf{i} + (5-4)\mathbf{j} + (0-5)\mathbf{k} = \mathbf{i} + \mathbf{j} - 5\mathbf{k}$$

and compute the cross product:

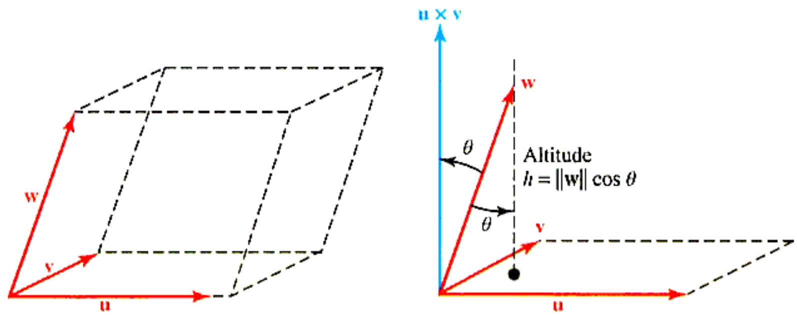
$$\begin{aligned}\mathbf{PQ} \times \mathbf{PR} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & -9 \\ 1 & 1 & -5 \end{vmatrix} \\ &= (-15+9)\mathbf{i} - (-10+9)\mathbf{j} + (2-3)\mathbf{k} \\ &= -6\mathbf{i} + \mathbf{j} - \mathbf{k}\end{aligned}$$

Thus, the triangle has area

$$\begin{aligned}A &= \frac{1}{2}\|\mathbf{PQ} \times \mathbf{PR}\| \\ &= \frac{1}{2}\sqrt{(-6)^2 + 1^2 + (-1)^2} \\ &= \frac{1}{2}\sqrt{38}\end{aligned}$$

The Scalar Triple Product and Volume

The cross product can also be used to compute the volume of a parallelepiped in $\mathbb{R}^3$. Consider the parallelepiped determined by three nonzero vectors **u**, **v**, and **w** that do not all lie in the same plane, as shown in Figure 9.41.



a. Parallelepiped **b.** $V = Ah$

Figure 9.41 Computing the volume of a parallelepiped

It is known from solid geometry that this parallelogram has volume $V = Ah$, where A is the area of the face determined by **u** and **v** and h is the altitude from the tip of **w** to this face.

The face determined by **u** and **v** is a parallelogram with area $A = \|\mathbf{u} \times \mathbf{v}\|$, and we know that the cross-product vector $\mathbf{u} \times \mathbf{v}$ is perpendicular to both **u** and **v** and hence to the face determined by **u** and **v**. From the alternative form of the dot product, we have

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \|\mathbf{u} \times \mathbf{v}\| \, \|\mathbf{w}\| \cos\,\theta$$

where θ is the angle between $\mathbf{u} \times \mathbf{v}$ and **w**. Thus, the parallelepiped has altitude

$$h = |\|\mathbf{w}\| \cos\,\theta| = \left| \frac{(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}}{\|\mathbf{u} \times \mathbf{v}\|} \right|$$

and the volume is given as

$$V = Ah = \|\mathbf{u} \times \mathbf{v}\| \left| \frac{(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}}{\|\mathbf{u} \times \mathbf{v}\|} \right| = |(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|$$

Remark: Note the absolute value in the above formula. Volume can never be negative. The combined operation $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ is called the **scalar triple product** of **u, v,** and **w**.

Example 5 Volume of a parallelepiped

Find the volume of the parallelepiped determined by the vectors $\mathbf{u} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$, $\mathbf{v} = -4\mathbf{i} + 7\mathbf{j} - 11\mathbf{k}$, and $\mathbf{w} = 5\mathbf{i} + 9\mathbf{j} - \mathbf{k}$.

Solution We first find the cross product.

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 3 \\ -4 & 7 & -11 \end{vmatrix} \\ &= (22 - 21)\mathbf{i} - (-11 + 12)\mathbf{j} + (7 - 8)\mathbf{k} \\ &= \mathbf{i} - \mathbf{j} - \mathbf{k} \end{aligned}$$

Thus,

$$\begin{aligned} V &= |(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}| \\ &= |(\mathbf{i} - \mathbf{j} - \mathbf{k}) \cdot (5\mathbf{i} + 9\mathbf{j} - \mathbf{k})| \\ &= |5 - 9 + 1| \\ &= 3 \end{aligned}$$

The volume of the parallelepiped is 3 cubic units.

Computationally, there is an easier way to work Example 5.

SCALAR TRIPLE PRODUCT If $\mathbf{u} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{v} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, and $\mathbf{w} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$, then the volume determined by these vectors has volume using the scalar triple product

$$V = |(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}| = \left| \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \right|$$

Theorem 9.8 Determinant form for a scalar triple product

If $\mathbf{u} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}, \mathbf{v} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, and $\mathbf{w} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$, then the scalar product can be found by evaluating the determinant

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{array}{l} \leftarrow \textit{components of } u \\ \leftarrow \textit{components of } v \\ \leftarrow \textit{components of } w \end{array}$$

Proof: The proof follows by expanding the determinant (see Problem 49).

We can use this formula to rework Example 5.

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \begin{vmatrix} 1 & -2 & 3 \\ -4 & 7 & -11 \\ 5 & 9 & -1 \end{vmatrix} = -3$$

Thus, the volume is $|-3| = 3$, as computed directly in Example 5.
We summarize the area and volume vector formulas in the following box.

AREAS AND VOLUMES Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be nonzero vectors that do not all lie in the same plane. Then,

Area of a parallelogram: $A = \|\mathbf{u} \times \mathbf{v}\|$

Area of a triangle: $A = \dfrac{1}{2}\|\mathbf{u} \times \mathbf{v}\|$

Volume of a parallelepiped: $V = |(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|$

In the problem set, we have included several exercises involving the scalar triple product and the vector triple product $\mathbf{u} \times \mathbf{v} \times \mathbf{w}$. In case you wonder why we neglect the product $(\mathbf{u} \cdot \mathbf{v}) \times \mathbf{w}$, notice that such a product makes no sense because $\mathbf{u} \cdot \mathbf{v}$ is a *scalar* and the cross product is an operation involving only vectors. Thus, the product

$$\mathbf{u} \cdot \mathbf{v} \times \mathbf{w} \text{ must mean } \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$$

Torque

A useful physical application of the cross product involves **torque**. Suppose the force $\mathbf{F}$ is applied to the point Q. Then the torque of $\mathbf{F}$ around P is defined as the cross product of the "arm" vector $\mathbf{PQ}$ with the force $\mathbf{F}$, as shown in Figure 9.42.

The torque, $\mathbf{T}$, of $\mathbf{F}$ at Q about P is

$$\mathbf{T} = \mathbf{PQ} \times \mathbf{F}$$

The magnitude of the torque, $\|\mathbf{T}\|$, provides a measure of the tendency of the vector arm $\mathbf{PQ}$ to rotate counterclockwise about an axis perpendicular to the plane determined by $\mathbf{PQ}$ and $\mathbf{F}$.

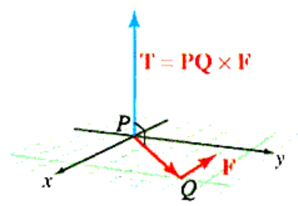


Figure 9.42 Torque as a cross product

Example 6 Torque on the hinge of a door

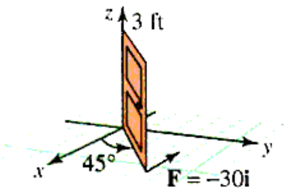


Figure 9.43 Swinging door

Figure 9.43 shows a half-open door that is 3 ft wide. A horizontal force of 30 lb is applied at the edge of the door. Find the torque of the force about the hinge on the door.

Solution We represent the force by $\mathbf{F} = -30\mathbf{i}$. Because the door is half open, it makes an angle of $\frac{\pi}{4}$ with the horizontal, and we can represent the door (i.e., the "arm" **PQ**) by the vector

$$\mathbf{PQ} = 3\left(\cos\frac{\pi}{4}\mathbf{i} + \sin\frac{\pi}{4}\mathbf{j}\right) = 3\left(\frac{\sqrt{2}}{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j}\right) = \frac{3\sqrt{2}}{2}\mathbf{i} + \frac{3\sqrt{2}}{2}\mathbf{j}$$

The torque can now be found:

$$\mathbf{T} = \mathbf{PQ} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{3\sqrt{2}}{2} & \frac{3\sqrt{2}}{2} & 0 \\ -30 & 0 & 0 \end{vmatrix} = 45\sqrt{2}\,\mathbf{k}$$

The magnitude of the torque ($45\sqrt{2}$ ft-lb) is a measure of the tendency of the door to rotate about its hinges.

PROBLEM SET 9.4

Level 1

Find $\mathbf{v} \times \mathbf{w}$ *for the vectors given in Problems* 1-10.

1. $\mathbf{v} = \mathbf{i}$; $\mathbf{w} = \mathbf{j}$
2. $\mathbf{v} = \mathbf{k}$; $\mathbf{w} = \mathbf{k}$
3. $\mathbf{v} = 3\mathbf{i} + 2\mathbf{k}$; $\mathbf{w} = 2\mathbf{i} + \mathbf{j}$
4. $\mathbf{v} = \mathbf{i} - 3\mathbf{j}$; $\mathbf{w} = \mathbf{i} + 5\mathbf{k}$
5. $\mathbf{v} = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$; $\mathbf{w} = \mathbf{i} + 4\mathbf{j} - 7\mathbf{k}$
6. $\mathbf{v} = 5\mathbf{i} - \mathbf{j} + 2\mathbf{k}$; $\mathbf{w} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$
7. $\mathbf{v} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$; $\mathbf{w} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$
8. $\mathbf{v} = -\mathbf{j} + 4\mathbf{k}$; $\mathbf{w} = 5\mathbf{i} + 6\mathbf{k}$
9. $\mathbf{v} = \mathbf{i} - 6\mathbf{j} + 10\mathbf{k}$; $\mathbf{w} = -\mathbf{i} + 5\mathbf{j} - 6\mathbf{k}$
10. $\mathbf{v} = \cos\theta\mathbf{i} + \sin\theta\mathbf{j}$; $\mathbf{w} = -\sin\theta\mathbf{i} + \cos\theta\mathbf{j}$ for any angle θ

Use the cross product to find $\sin\theta$ *where* θ *is the angle between* $\mathbf{v}$ *and* $\mathbf{w}$ *in Problems* 11-14.

11. $\mathbf{v} = \mathbf{i} + \mathbf{k}$; $\mathbf{w} = \mathbf{i} + \mathbf{j}$
12. $\mathbf{v} = \mathbf{i} + \mathbf{j}$; $\mathbf{w} = \mathbf{i} + \mathbf{j} + \mathbf{k}$
13. $\mathbf{v} = \mathbf{j} + \mathbf{k}$; $\mathbf{w} = \mathbf{i} + \mathbf{k}$
14. $\mathbf{v} = \mathbf{i} + \mathbf{j}$; $\mathbf{w} = \mathbf{j} + \mathbf{k}$

Find a unit vector that is orthogonal to both $\mathbf{v}$ *and* $\mathbf{w}$ *in Problems* 15-18.

15. $\mathbf{v} = 2\mathbf{i} + \mathbf{k}$; $\mathbf{w} = \mathbf{i} - \mathbf{j} - \mathbf{k}$
16. $\mathbf{v} = \mathbf{j} - 3\mathbf{k}$; $\mathbf{w} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$
17. $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{k}$; $\mathbf{w} = 3\mathbf{i} + 12\mathbf{j} - 4\mathbf{k}$
18. $\mathbf{v} = 2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$; $\mathbf{w} = 4\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$

Find the area of the parallelogram determined by the vectors in Problems 19-22.

19. $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{w} = \mathbf{i} + \mathbf{j} - \mathbf{k}$
20. $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{w} = 4\mathbf{i} - 3\mathbf{j}$
21. $\mathbf{v} = 4\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{w} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$
22. $\mathbf{v} = 2\mathbf{i} + 3\mathbf{k}$ and $\mathbf{w} = 2\mathbf{j} - 3\mathbf{k}$

Find the area of $\triangle PQR$ *in Problems* 23-26.

23. $P(0,1,1)$, $Q(1,1,0)$, $R(1,0,1)$
24. $P(1,0,0)$, $Q(2,1,-1)$, $R(0,1,-2)$
25. $P(1,2,3)$, $Q(2,3,1)$, $R(3,1,2)$
26. $P(-1,-1,-1)$, $Q(1,-1,-1)$, $R(-1,1,-1)$

Determine whether each product in Problems 27-30 *is a scalar or a vector or does not exist. Explain your reasoning.*

27. a. $\mathbf{u} \times (\mathbf{v} \cdot \mathbf{w})$ **b.** $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$
28. a. $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$ **b.** $\mathbf{u} \cdot (\mathbf{v} \cdot \mathbf{w})$
29. a. $(\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{u} \times \mathbf{w})$ **b.** $(\mathbf{u} \times \mathbf{v}) \times (\mathbf{u} \times \mathbf{w})$
30. a. $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ **b.** $|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|$

In Problems 31-34, *find the volume of the parallelepiped determined by vectors* $\mathbf{u}$, $\mathbf{v}$, *and* $\mathbf{w}$.

31. $\mathbf{u} = \mathbf{j} + \mathbf{k}$; $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$; $\mathbf{w} = 5\mathbf{i}$
32. $\mathbf{u} = \mathbf{i} + \mathbf{j}$; $\mathbf{v} = \mathbf{j} + 2\mathbf{k}$; $\mathbf{w} = 3\mathbf{k}$
33. $\mathbf{u} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$; $\mathbf{v} = 3\mathbf{i} + \mathbf{k}$; $\mathbf{w} = \mathbf{j} + \mathbf{k}$
34. $\mathbf{u} = \mathbf{i} + \mathbf{j} + \mathbf{k}$; $\mathbf{v} = \mathbf{i} - \mathbf{j} - \mathbf{k}$; $\mathbf{w} = 2\mathbf{i} + 3\mathbf{k}$

Level 2

Explain your reasoning.

35. ■ What does this say? Contrast dot and cross products of vectors, including a discussion of some of their properties.

36. ■ What does this say? What is the right-hand rule?

37. Find a number s that guarantees that the vectors $\mathbf{i}$, $\mathbf{i}+\mathbf{j}+\mathbf{k}$, and $\mathbf{i}+2\mathbf{j}+s\mathbf{k}$ will all be coplanar (in the same plane).

38. Find a number t that guarantees the vectors $\mathbf{i}+\mathbf{j}$, $2\mathbf{i}-\mathbf{j}+\mathbf{k}$, and $\mathbf{i}+\mathbf{j}+t\mathbf{k}$ will all be coplanar (in the same plane).

39. Let $\mathbf{u}=\mathbf{i}+\mathbf{j}$, $\mathbf{v}=2\mathbf{i}-\mathbf{j}+\mathbf{k}$, and $\mathbf{w}=3\mathbf{i}$. Compute $(\mathbf{u}\times\mathbf{v})\times\mathbf{w}$ and $\mathbf{u}\times(\mathbf{v}\times\mathbf{w})$. What does this say about the associativity of cross product?

40. Show that $(a\mathbf{u})\times(b\mathbf{v})=ab(\mathbf{u}\times\mathbf{v})$ for any scalars a and b.

41. For a given vector $\mathbf{v}$ in $\mathbb{R}^3$, find all vectors $\mathbf{w}$ such that $\mathbf{v}\times\mathbf{w}=\mathbf{w}$.

42. What can be said about nonzero vectors $\mathbf{v}$ and $\mathbf{w}$ if $\mathbf{v}\cdot\mathbf{w}=0$? What can be said if $\mathbf{v}\times\mathbf{w}=\mathbf{0}$?

43. One end of a 2-ft lever pivots about the origin in the yz-plane, as shown in Figure 9.44.

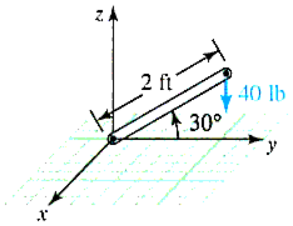


Figure 9.44 Finding the torque

If a vertical force of 40 lb is applied at the end of the lever, what is the torque of the lever about the pivot point (the origin) when the lever makes an angle of 30° with the xy-plane?

44. A 3-lb weight hangs from a rope at the end of Q of a 5-ft stick PQ that is held at an angle of 60° to the horizontal, as shown in Figure 9.45. What is the torque about the point P due to the weight?

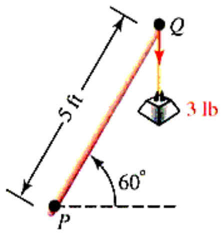


Figure 9.45 Finding the torque

45. Find the angle between the vector $2\mathbf{i}-\mathbf{j}+\mathbf{k}$ and the plane determined by the points $P(1,-2,3)$, $Q(-1,2,3)$, and $R(1,2,-3)$.

46. a. Show that the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are coplanar (all in the same plane) if

$$\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})=0 \text{ or } (\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}=0$$

b. Are the vectors $\mathbf{u}=\mathbf{i}+3\mathbf{j}+\mathbf{k}$, $\mathbf{v}=2\mathbf{i}-\mathbf{j}-\mathbf{k}$, and $\mathbf{w}=7\mathbf{j}+3\mathbf{k}$ coplanar?

47. Show that the triangle with vertices (x_1,y_1), (x_2,y_2), (x_3,y_3) has area $A=\frac{1}{2}|D|$, where

$$D=\begin{vmatrix} x_1 & y_1 & 1\\ x_2 & y_2 & 1\\ x_3 & y_3 & 1\end{vmatrix}$$

48. Using the properties of determinants, show that

$$\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})=(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}$$

for any vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.

Level 3

49. Prove the determinant formula for evaluating a scalar triple product.

50. Let A, B, C, and D be four points that do not lie in the same plane; see Figure 9.46, for example.

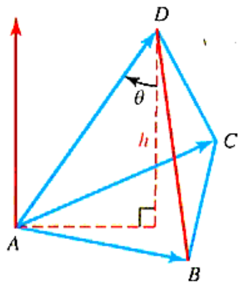


Figure 9.46 Problem 50

It can be shown that the volume of the tetrahedron with vertices A, B, C and D satisfies

$$[\text{VOLUME OF TETRAHEDRON } ABCD] = \frac{1}{3}\begin{pmatrix}\text{AREA OF}\\ \triangle ABC\end{pmatrix}\begin{pmatrix}\text{ALTITUDE FROM}\\ D \text{ TO } \triangle ABC\end{pmatrix}$$

Show that the volume is given by

$$V=\frac{1}{6}|(\mathbf{AB}\times\mathbf{AC})\cdot\mathbf{AD}|$$

51. Show that if **u**, **v**, and **w** are vectors in $\mathbb{R}^3$ with $\mathbf{u}+\mathbf{v}+\mathbf{w}=\mathbf{0}$, then

$$\mathbf{u}\times\mathbf{v}=\mathbf{v}\times\mathbf{w}=\mathbf{w}\times\mathbf{u}$$

52. Prove Lagrange's identity

$$\|\mathbf{v}\times\mathbf{w}\|^2=\|\mathbf{v}\|^2\,\|\mathbf{w}\|^2-(\mathbf{v}\cdot\mathbf{w})^2$$

53. Show that

$$\tan\theta=\frac{\|\mathbf{v}\times\mathbf{w}\|}{\mathbf{v}\cdot\mathbf{w}}$$

where θ $(0\le\theta<\frac{\pi}{2})$ is the angle between **v** and **w**.

54. Let **u**, **v**, and **w** be nonzero vectors in $\mathbb{R}^3$ that do not all lie in the same plane. Show that

$$|\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})|=|\mathbf{v}\cdot(\mathbf{u}\times\mathbf{w})|$$

What other scalar triple products involving **u**, **v**, and **w** have the same absolute values?

55. Let **a**, **b**, and **c** be vectors in space. Prove the *cab-bac* formula

$$\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=(\mathbf{c}\cdot\mathbf{a})\mathbf{b}-(\mathbf{b}\cdot\mathbf{a})\mathbf{c}$$

Establish the validity of the equations in Problems 56-59 for arbitrary vectors **u**, **v**, **w**, *and* **z** *in* $\mathbb{R}^3$. *You may use the result of Problem 55.*

56. $(\mathbf{u}\times\mathbf{v})\times(\mathbf{w}\times\mathbf{z})=(\mathbf{u}\cdot\mathbf{w}\times\mathbf{z})\mathbf{v}-(\mathbf{v}\cdot\mathbf{w}\times\mathbf{z})\mathbf{u}$

57. $\mathbf{u}\times(\mathbf{v}\times\mathbf{w})+\mathbf{v}\times(\mathbf{w}\times\mathbf{u})+\mathbf{w}\times(\mathbf{u}\times\mathbf{v})=\mathbf{0}$

58. $\mathbf{u}\times\mathbf{v}=(\mathbf{u}\cdot\mathbf{v}\times\mathbf{i})\mathbf{i}+(\mathbf{u}\cdot\mathbf{v}\times\mathbf{j})\mathbf{j}+(\mathbf{u}\cdot\mathbf{v}\times\mathbf{k})\mathbf{k}$

59. $\mathbf{u}\times[\mathbf{u}\times(\mathbf{u}\times\mathbf{v})]\cdot\mathbf{w}=-\|\mathbf{u}\|^2\,\mathbf{u}\cdot\mathbf{v}\times\mathbf{w}$

60. Show that if vectors **OP**, **OQ**, **OR**, and **OS** lie in the same plane, then

$$(\mathbf{OP}\times\mathbf{OQ})\times(\mathbf{OR}\times\mathbf{OS})=\mathbf{0}$$

9.5 LINES IN $\mathbb{R}^3$

IN THIS SECTION: *Equations of lines in* $\mathbb{R}^3$, *parametric equations, parameterizing a curve*

As we move from $\mathbb{R}^2$ to $\mathbb{R}^3$, we look for patterns. As we have seen, the number of components of points corresponds to the dimension. In $\mathbb{R}$, we locate a point with one component, in $\mathbb{R}^2$ we use ordered pairs, and in $\mathbb{R}^3$ we use triples. Circles in $\mathbb{R}^2$ are two points in $\mathbb{R}$ and spheres in $\mathbb{R}^3$. However, lines in $\mathbb{R}^2$ generalize to planes in $\mathbb{R}^3$, so we need to pay special attention to developing lines in $\mathbb{R}^3$.

Equations of Lines in $\mathbb{R}^3$

In precalculus, and in Section 1.3, we graphed lines in $\mathbb{R}^2$ using a variety of forms, and one of those forms was the parametric form of a line passing through (x_1, y_1) with slope $m=\frac{b}{a}$, $a\neq 0$, namely $x=x_1+at$ and $y=y_1+bt$. In order to graph a line in $\mathbb{R}^3$, we need to write the slope $m=\frac{b}{a}$ in a form which involves changes in three components instead of two. That is, instead of talking about the slope $m=\frac{b}{a}$, we talk about the direction $[a,b]$ where a is the change in x and b is the change in y, and we note that this line has the same direction as a vector $\mathbf{v}=a\mathbf{i}+b\mathbf{j}$. Using this representation, we now generalize to $\mathbb{R}^3$.

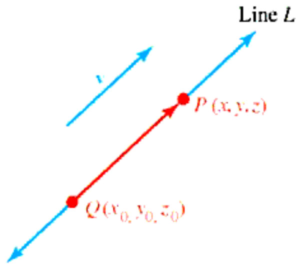


Figure 9.47 L is aligned with **v**

Suppose L is a line in space that contains $Q(x_0, y_0, z_0)$ whose location is determined by the vector $\mathbf{v}=A\mathbf{i}+B\mathbf{j}+C\mathbf{k}$. We say that L is **aligned with v**, as shown in Figure 9.47.

We also say that the line has **direction numbers** A, B, and C and denote these direction numbers by $[A,B,C]$. If $P(x,y,z)$ is any point on L, then the vector **QP** is parallel to **v** and must satisfy the vector equation $\mathbf{QP}=t\mathbf{v}$ for some number t. If we introduce coordinates and use the standard representation, we can rewrite this vector equation as

$$(x-x_0)\mathbf{i}+(y-y_0)\mathbf{j}+(z-z_0)\mathbf{k}=t[A\mathbf{i}+B\mathbf{j}+C\mathbf{k}]$$

By equating components on both sides of this equation, we find that the coordinates of P must satisfy the linear system

$$x - x_0 = tA \qquad y - y_0 = tB \qquad z - z_0 = tC$$

where t is a real number.

PARAMETRIC FORM OF A LINE IN $\mathbb{R}^3$ If L is a line that contains the point (x_0, y_0, z_0) and is aligned with the vector $\mathbf{v} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$, then the point (x, y, z) is on L if and only if its coordinates satisfy

$$x - x_0 = tA \qquad y - y_0 = tB \qquad z - z_0 = tC$$

for some number t.

The parameter t will be a different value for each point (x, y, z)

Turning things around, if we are given the equation of a line with direction numbers $[A, B, C]$, then $\mathbf{v} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ is the **vector aligned with** L.

Example 1 Parametric equations of a line in space

Find the parametric equations for the line that contains the point $(3, 1, 4)$ and is aligned with the vector $\mathbf{v} = -\mathbf{i} + \mathbf{j} - 2\mathbf{k}$. Find where this line passes through the coordinate planes and sketch the line.

Solution The direction numbers are $[-1, 1, -2]$ and $x_0 = 3$, $y_0 = 1$, $z_0 = 4$, so the line has the parametric form

$$\begin{aligned} x - 3 &= -t & y - 1 &= t & z - 4 &= -2t \\ x &= 3 - t & y &= 1 + t & z &= 4 - 2t \end{aligned}$$

This line will intersect the xy-plane when $z = 0$; solve

$$0 = 4 - 2t \qquad \text{implies} \qquad t = 2$$

If $t = 2$, then $x = 3 - 2 = 1$ and $y = 1 + 2 = 3$. This is the point $(1, 3, 0)$. Similarly, the line intersects the xz-plane at $(4, 0, 6)$ and the yz-plane at $(0, 4, -2)$. Plot these points and draw the line, as shown in Figure 9.48.

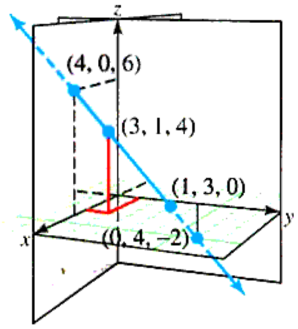


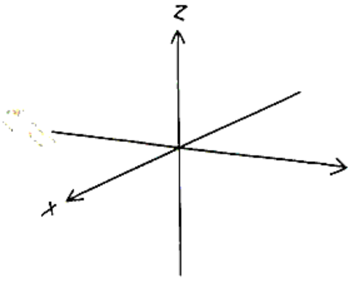
Figure 9.48 Graph of a line in space

Example 1 shows the graph of a line in $\mathbb{R}^3$, but it does not show how you will draw a line on your own paper. The following drawing lesson shows steps you can use to graph the line

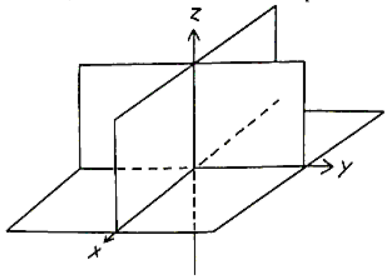
$$x = 2 + 2t \qquad y = 10t \qquad z = 3 - 3t$$

Drawing Lesson: Sketching a line in $\mathbb{R}^3$

Draw the three coordinate axes.

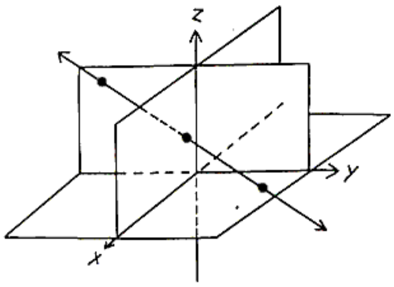


Next, draw the three coordinate planes.



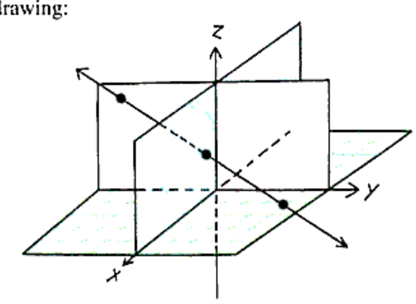
We focus, instead, on the points where the graph passes through each of the coordinate planes.

We see, by inspection, that the line passes through (2,0,3); plot this point.



If $x = 0$, then $t = -1$, so $y = -10$ and $z = 6$; plot the point $(0, -10, 6)$ in the yz-plane. Finally, if $z = 0$, $t = 1$, so $x = 4, y = 10$; plot $(4, 10, 0)$ in the xy-plane.

Use colored pencils or highlighters to distinguish individual planes. Here is an example of a finished drawing:



In the special case where none of the direction numbers A, B, or C is 0, we can solve each of the parametric-form equations for t to obtain the following **symmetric equations** for a line.

SYMMETRIC FORM OF A LINE IN $\mathbb{R}^3$ If L is a line that contains the point (x_0, y_0, z_0) and is aligned with the vector $\mathbf{v} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$, ($A, B$, and C nonzero numbers) then the point (x, y, z) is L if and only if its coordinates satisfy

$$\frac{x - x_0}{A} = \frac{y - y_0}{B} = \frac{z - z_0}{C}$$

Example 2 Symmetric form of the equation of a line in space

Find symmetric equations for the line L through the points $P(-1, 3, 7)$ and $Q(4, 2, -1)$. Find the points of intersection with the coordinate planes and sketch the line.

Solution The required line passes through P and is aligned with the vector

$$\mathbf{PQ} = [4 - (-1)]\mathbf{i} + [2 - 3]\mathbf{j} + [-1 - 7]\mathbf{k} = 5\mathbf{i} - \mathbf{j} - 8\mathbf{k}$$

Thus, the direction numbers of the line are $[5, -1, -8]$, and we can choose either P or Q as (x_0, y_0, z_0). Choosing P, we obtain:

$$\frac{x + 1}{5} = \frac{y - 3}{-1} = \frac{z - 7}{-8}$$

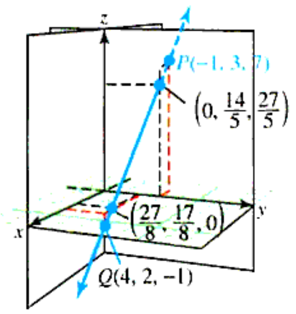


Figure 9.49 Interactive Graph of $\dfrac{x+1}{5} = \dfrac{y-3}{-1} = \dfrac{z-7}{-8}$

Next, we find points of intersection with the coordinate planes:

xy-plane: $z = 0$, so $\dfrac{x+1}{5} = \dfrac{7}{8}$ implies $x = \dfrac{27}{8}$ and $\dfrac{y-3}{-1} = \dfrac{7}{8}$ implies $y = \dfrac{17}{8}$.

The point of intersection of the line with the xy-plane is $(\frac{27}{8}, \frac{17}{8}, 0)$. Similarly, the other intersections are xz-plane: $(14, 0, -17)$; yz-plane: $(0, \frac{14}{5}, \frac{27}{5})$. The graph is shown in Figure 9.49.

If one or more of the direction numbers $[A, B, C]$ of a line L in $\mathbb{R}^3$ is zero, then L is parallel to at least one coordinate plane. For example, the line

$$x = 3 + 2t, \ y = 5, \ z = 4 - t$$

has direction numbers $[2, 0, -1]$ and lies in the plane $y = 5$, parallel to the xz-plane. The symmetric form for this line is

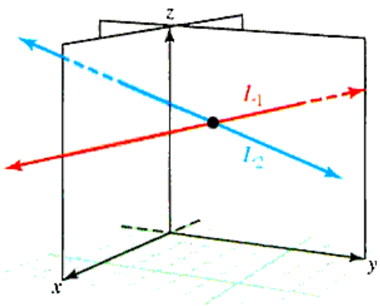
$$\frac{x-3}{2} = \frac{z-4}{-1};\ y = 5$$

On the other hand, consider a line with symmetric form

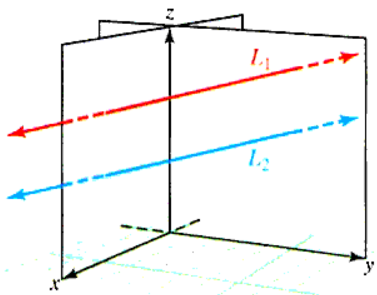
$$\frac{y+1}{4} = \frac{z-3}{-2};\ x = 7$$

lies in the plane $x = 7$, parallel to the yz-plane and has direction numbers $[0, 4, -2]$.

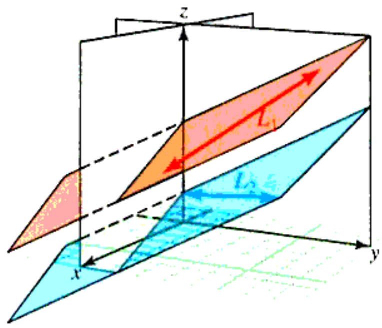
You might remember that two lines in $\mathbb{R}^2$ must intersect if their slopes are different (because they cannot be parallel), but two lines in space may have different direction numbers and still not intersect. In this case, the lines are said to be **skew**. The reason why the situation in $\mathbb{R}^3$ is different from that in $\mathbb{R}^2$ is that even though lines with different direction numbers cannot be parallel, there is still enough "room" in space for the lines to lie in parallel planes and be aligned with vectors that are not parallel. This situation is shown in Figure 9.50.



a. Intersecting lines



b. Parallel lines



c. Skew lines

Figure 9.50 Lines in space may intersect, be parallel, or skew

Example 3 Skew lines in space

Determine whether the following pair of lines intersect, are parallel, or are skew.

$$L_1\colon \frac{x-1}{2} = \frac{y+1}{1} = \frac{z-2}{4} \quad \text{and} \quad L_2\colon \frac{x+2}{4} = \frac{y}{-3} = \frac{z+1}{1}$$

Solution Note that L_1 has direction numbers $[2, 1, 4]$ (that is, L_1 is aligned with $2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$) and L_2 has direction numbers $[4, -3, 1]$. If we solve $\langle 2, 1, 4\rangle = t\langle 4, -3, 1\rangle$ for t, we find no possible solution for any value of t. This implies that the lines are not parallel. (See Figure 9.51.)

Next, we determine whether the lines intersect or are skew. Note that $S(1, -1, 2)$ lies on L_1 and $T(-2, 0, -1)$ lies on L_2. The lines intersect if and only if there is a point P that lies on both lines. To determine this, we write the equations of the lines in parametric form. We use a different parameter for each line so that points on one line do not depend on the value of the parameter of the other line:

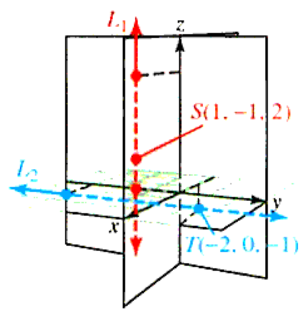


Figure 9.51 Interactive Graphs of L_1 and L_2

L_1:	$x = 1 + 2s$	$y = -1 + s$	$z = 2 + 4s$
L_2:	$x = -2 + 4t$	$y = -3t$	$z = -1 + t$
Thus:	$x = 1 + 2s = -2 + 4t$	$y = -1 + s = -3t$	$z = 2 + 4s = -1 + t$
or:	$2s - 4t = -3$	$s + 3t = 1$	$4s - t = -3$

This is equivalent to the system of linear equations

$$\begin{cases} 2s - 4t = -3 \\ s + 3t = 1 \\ 4s - t = -3 \end{cases}$$

Any solution of this system must correspond to a point of intersection of L_1 and L_2, and if no solution exists, then L_1 and L_2 are skew. Because this is a system of three equations with two unknowns, we first solve the first two equations simultaneously to find $s = -\frac{1}{2}, t = \frac{1}{2}$. Because $s = -\frac{1}{2}$ and $t = \frac{1}{2}$ do not satisfy the third equation, it follows that L_1 and L_2 do not intersect, so they must be skew.

Example 4 Intersecting lines

Show that the lines

$$L_1\colon \frac{x-1}{2} = \frac{y+1}{1} = \frac{z-2}{4} \quad \text{and} \quad L_2\colon \frac{x+2}{4} = \frac{y}{-3} = \frac{z-\frac{1}{2}}{-1}$$

intersect and find the point of intersection. (See Figure 9.52.)

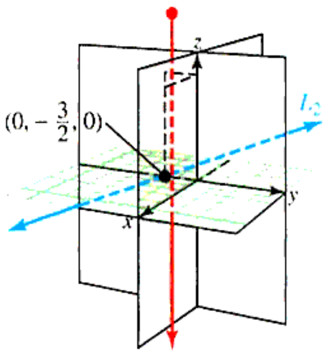


Figure 9.52 Interactive
Graph of L_1 and L_2

Notice that we used two parameters (s and t) for the lines in Example 4. This is because it is quite possible for two lines to intersect at "different times" (that is, different values of the parameter). In Example 4, we have $s = -\frac{1}{2}$ and $t = \frac{1}{2}$ at the point of intersection, rather than $s = t$.

Solution L_1 has direction numbers $[2, 1, 4]$ and L_2 has direction numbers $[4, -3, -1]$. Because there is no t for which $[2, 1, 4] = t[4, -3, -1]$, the lines are not parallel. Express the lines in parametric form:

$$\begin{array}{llll} L_1\colon & x = 1 + 2s & y = -1 + s & z = 2 + 4s \\ L_2\colon & x = -2 + 4t & y = -3t & z = \frac{1}{2} - t \\ \text{Thus:} & x = 1 + 2s = -2 + 4t & y = -1 + s = -3t & z = 2 + 4s = \frac{1}{2} - t \\ \text{or:} & 2s - 4t = -3 & s + 3t = 1 & 4s + t = -\frac{3}{2} \end{array}$$

Solving the first two equations simultaneously, we find $s = -\frac{1}{2}, t = \frac{1}{2}$. This solution satisfies the third equation, namely,

$$4\left(-\frac{1}{2}\right) + \frac{1}{2} = -\frac{3}{2}$$

To find the coordinates of the point of intersection, substitute $s = -\frac{1}{2}$ into the parametric-form equations for L_1 (or substitute $t = \frac{1}{2}$ into L_2) to obtain

$$x_0 = 1 + 2\left(-\frac{1}{2}\right) = 0$$
$$y_0 = -1 + \left(-\frac{1}{2}\right) = -\frac{3}{2}$$
$$z_0 = 2 + 4\left(-\frac{1}{2}\right) = 0$$

Thus, the lines intersect at $P(0, -\frac{3}{2}, 0)$.

Parametric Equations

In this book we have used parameters to graph lines in $\mathbb{R}^2$ (Section 1.3), and in $\mathbb{R}^3$ in this section. However, parametric representation can be used to graph a wide variety of curves and surfaces in both two and three dimensions. We begin with a definition.

PARAMETRIC REPRESENTATION Let f_1, f_2, and f_3 be continuous functions of t on an interval I; then the equations

$$x = f_1(t), \quad y = f_2(t), \quad z = f_3(t)$$

are called **parametric equations** with **parameter** t. As t varies over the **parametric set** I, the points

$$(x, y, z) = (f_1(t), f_2(t), f_3(t))$$

trace out a **parametric curve** in $\mathbb{R}^3$. If $z = f_3(t) = 0$, then the curve is in the xy-plane and we say the parametric curve is in $\mathbb{R}^2$.

Note: The letter "t" used for the parameter does not necessarily denote time, although in many applications, time is a suitable parameter. Indeed, any letter or symbol may be used to denote a parameter. If $z = 0$ for all values of t, then the curve is called a **plane curve**, and if $z \neq 0$ for at least one value of t, then the curve is called a **space curve**. We will investigate space curves at length in the next chapter.

Example 5 Sketching the path of a parametric curve

Sketch the path of the curve $x = t^2 - 9$, $y = \frac{1}{3}t$ for $-3 \leq t \leq 2$.

Solution Values of x and y corresponding to various choices of the parameter t are shown in the following table:

t	x	y	
-3	0	-1	(Starting or **initial** point)
-2	-5	$-\frac{2}{3}$	
-1	-8	$-\frac{1}{3}$	
0	-9	0	
1	-8	$\frac{1}{3}$	
2	-5	$\frac{2}{3}$	(Ending or **terminal** point)

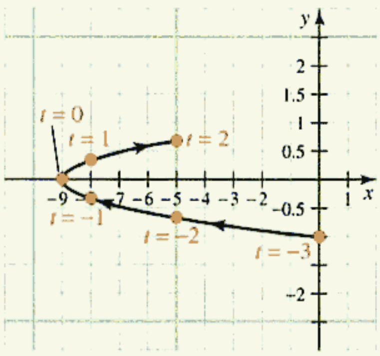


Figure 9.53 **Interactive** Graph of spiral with restricted domain

The graph is shown in Figure 9.53. Note how the arrows show the orientation as t increases from -3 to 2.

If you are using a computer or a graphing calculator, plotting points can be an effective way of sketching a parametric curve. Sometimes, however, we wish to eliminate the parameter to obtain a Cartesian equation. For instance, in Example 5, we have $y = \frac{1}{3}t$, so $t = 3y$, and by substituting into the equation $x = t^2 - 9$, we obtain

$$x = (3y)^2 - 9 = 9y^2 - 9$$

which is the Cartesian equation for a parabola that opens to the right. Because of the domain of the parameter t, we see that the parametric curve in Figure 9.53 is a subset of the set of points that satisfy the equation

$$x = 9y^2 - 9$$

Parameterizations are not unique. For example, the curve with parametric equations

$$x = 9(9s^2 - 1),\ y = 3s \text{ for } -\frac{1}{3} \le s \le \frac{2}{9}$$

is the same as the curve in Figure 9.53.

Example 6 Sketching the path by eliminating the parameter

Describe the path $x = \sin \pi t$, $y = \cos 2\pi t$ for $0 \le t \le 0.5$.

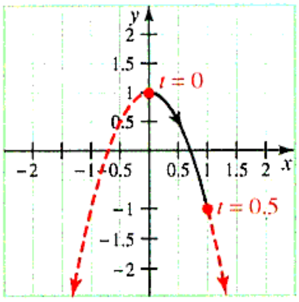


Figure 9.54 Interactive Graph of parabola with restricted domain

Solution Using a double angle identity, we find

$$\cos 2\pi t = 1 - 2\sin^2 \pi t$$

so that

$$y = 1 - 2x^2$$

We recognize this as a Cartesian equation for a parabola. Because $y' = -4x$, we can find the critical number $x = 0$, which locates the vertex of the parabola at $(0, 1)$. The parabola is the curve shown in color as the dashed curve in Figure 9.54.

Because t is restricted to the interval $0 \le t \le 0.5$, the parametric representation involves only part of the right side of the parabola $y = 1 - 2x^2$. The curve is oriented from the point $(0, 1)$, where $t = 0$, to the point $(1, -1)$, where $t = 0.5$, and is the portion of the parabola shown in black in Figure 9.54.

When it is difficult to eliminate the parameter from a given parametric representation, we can sometimes get a good picture of the parametric curve by plotting points.

Example 7 Describing a spiraling path

Discuss the path of the curve described by the parametric equations

$$x = e^{-t}\cos t, \quad y = e^{-t}\sin t \qquad \text{for } t \ge 0$$

Solution We have no convenient way of eliminating the parameter so we write out a table of values (x, y) that correspond to various values of t. The curve is obtained by plotting these points in a Cartesian plane and passing a smooth curve through the plotted points, as shown in Figure 9.55.

t	x	y
0	1	0
$\frac{\pi}{4}$	0.32	0.32
1	0.20	0.31
$\frac{\pi}{2}$	0	0.21
2	−0.06	0.12
π	−0.04	0
$\frac{3\pi}{2}$	0	−0.01
2π	0.00	0

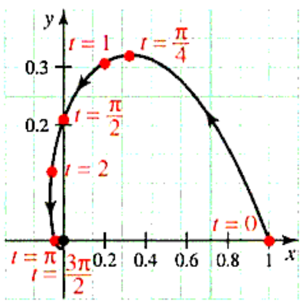


Figure 9.55 Interactive Graph of spiral with restricted domain

Note that for each value of t, the distance from $P(x, y)$ on the curve to the origin is

$$\sqrt{x^2 + y^2} = \sqrt{(e^{-t}\cos t)^2 + (e^{-t}\sin t)^2} = \sqrt{e^{-2t}(1)} = e^{-t}$$

Because e^{-t} decreases as t increases, it follows that P gets closer and closer to the origin as t increases. However, because $\cos t$ and $\sin t$ vary between -1 and $+1$, the approach is not direct but takes place along a spiral.

Parameterizing a Curve

So far, our examples have dealt with sketching a parametric curve given the parametric equations. In general, this process may be tedious, but generally can be done easily using technology. However, the reverse process, finding a suitable set of parametric equations for a given curve, is an art for which there is no simple procedure. Indeed, a given curve can have many different parameterizations and there are curves for which no simple parameterization be given. The following two examples illustrate various methods for parameterizating a given curve.

Example 8 Parameterizing two curves

In each of the following cases, parameterize the given curve:

a. $y = 9x^2$ **b.** $r = 5\cos^3\theta$ in polar coordinates

Solution

a. The usual parameterization for a parabola is to let the parameter t be the variable that is squared: $x = t$, $y = 9t^2$. However, another parameterization is to let $t = 3x$ so that $x = \frac{1}{3}t$ and $y = t^2$.

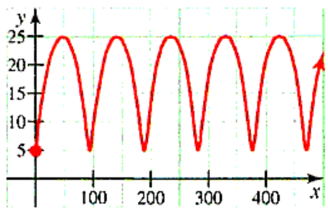
b. In polar coordinates we have $x = r\cos\theta$, $y = r\sin\theta$, so we can parameterize x and y in terms of the parameter θ:

$$\begin{aligned} x &= r\cos\theta & y &= r\sin\theta \\ &= \left(5\cos^3\theta\right)\cos\theta & &= \left(5\cos^3\theta\right)\sin\theta \\ &= 5\cos^4\theta & &= 5\cos^3\theta\sin\theta \end{aligned}$$

Example 9 Modeling Problem: Finding parametric equations for a trochoid

A bicycle wheel has radius a and a reflector is attached at a point P on a spoke of the bicycle wheel at a fixed distance d from the center. Find parametric equations for the curve described by P as the wheel rolls along a straight line without slipping. Such a curve is called a **trochoid**, and is shown in Figure 9.56.

Solution Assume that the wheel rolls along the x-axis and that the center C of the wheel begins at $(0, a)$ on the y-axis. Further assume that P also starts on the y-axis, d units below C. Figure 9.57 shows the initial position of the wheel and its position after turning through an angle θ (radians).



The path of a reflector placed 5 in. from the tire on the wheel of a bike with a 30-in wheel. This graph is an example of a trochoid.

Figure 9.56 Graph of a trochoid

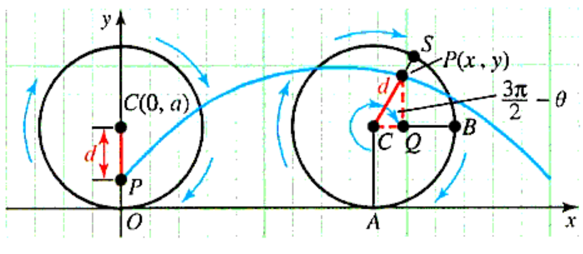


Figure 9.57 The path of a reflector on a bicycle

We begin by labeling some points: The point A is on the x-axis directly beneath C, whereas B is the point where the horizontal line through C meets the rim of the wheel. Finally, Q is the point on BC directly beneath P, and S is the point where the line through C and P intersects the rim. Let P have coordinates (x, y).

We need to find representations (in terms of a, d, and θ) for x and y.

$$\begin{aligned} x &= |\overline{OA}| + |\overline{CQ}| \\ &= a\theta + |\overline{CQ}| \end{aligned}$$

Because the wheel rolls along the x-axis without slipping, $|\overline{OA}|$ is the same as the arc length from A to S, so $|\overline{OA}| = a\theta$.

$$\begin{aligned} y &= |\overline{AC}| + |\overline{QP}| \\ &= a + |\overline{QP}| \end{aligned}$$

To complete our evaluation of x and y, we need to compute $|\overline{CQ}|$ and $|\overline{QP}|$. These are sides of $\triangle\ PCQ$. Note that $\angle PCQ = \frac{3\pi}{2} - \theta$; therefore, by the definition, of cosine and sine, we have

$$\cos\left(\frac{3\pi}{2} - \theta\right) = \frac{|\overline{CQ}|}{d} \quad \text{so} \quad |\overline{CQ}| = d\cos\left(\frac{3\pi}{2} - \theta\right) = -d\sin\theta$$

Similarly, $|\overline{QP}| = d\sin\left(\frac{3\pi}{2} - \theta\right) = -d\cos\theta$.

We can now substitute these values for $|\overline{CQ}|$ and $|\overline{QP}|$ into the equations we derived for x and y.

$$\begin{aligned} x &= a\theta + |\overline{CQ}| = a\theta - d\sin\theta \\ y &= a + |\overline{QP}| \;\;= a - d\cos\theta \end{aligned}$$

The special case where P is on the rim of the wheel in Example 9 (when $d = a$) is a curve called a **cycloid**. There are several problems involving these and similar curves in the problem set.

PROBLEM SET 9.5

Level 1

1. ■ What does this say? What is a parameter?
2. ■ What does this say? Contrast the parametric and symmetric forms of the equation of a line.

Find the parametric and symmetric equations for the line(s) passing through the given points with the properties described in Problems 3-10.

3. $(1, -1, -2)$ parallel to $3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$
4. $(1, 0, -1)$ parallel to $3\mathbf{i} + 4\mathbf{j}$
5. $(1, -1, 2)$ through $(2, 1, 3)$
6. $(2, 2, 3)$ through $(1, 3, -1)$
7. $(1, -3, 6)$ parallel to $\dfrac{x-5}{1} = \dfrac{y+2}{-3} = \dfrac{z}{-5}$
8. $(1, -1, 2)$ parallel to $\dfrac{x+3}{4} = \dfrac{y-2}{5} = \dfrac{z+5}{1}$
9. $(0, 4, -3)$ parallel to $\dfrac{x-1}{22} = \dfrac{y+2}{-6} = \dfrac{z-1}{10}$
10. $(1, 0, -4)$ parallel to $x = -2 + 3t$, $y = 4 + t$, $z = 2 + 2t$
11. Find the parametric form of the equation of the line passing through $(3, -1, 0)$ parallel to both the xy- and xz-planes.
12. Find the parametric form of the equation of the line passing through $(3, -1, 0)$ parallel to both the xy- and yz-planes.

Find the points of intersection of each line in Problems 13-16 *with each of the coordinate planes.*

13. $\dfrac{x-4}{4} = \dfrac{y+3}{3} = \dfrac{z+2}{1}$
14. $\dfrac{x+1}{1} = \dfrac{y+2}{2} = \dfrac{z-6}{3}$
15. $x = 6 - 2t$, $y = 1 + t$, $z = 3t$
16. $x = 6 + 3t$, $y = 2 - t$, $z = 2t$

In Problems 17-22, tell whether the two lines intersect, are parallel, are skew, or coincide. If they intersect, give the point of intersection.

17. $\dfrac{x-4}{2}=\dfrac{y-6}{-3}=\dfrac{z+2}{5}$

$\dfrac{x}{4}=\dfrac{y+2}{-6}=\dfrac{z-3}{10}$

18. $x=4-2t,\ y=6t,\ z=7-4t$

$x=5+t,\ y=1-3t,\ z=-3+2t$

19. $x=3+3t,\ y=1-4t,\ z=-4-7t$

$x=2+3t,\ y=5-4t,\ z=3-7t$

20. $x=2-4t,\ y=1+t,\ z=\frac{1}{2}+5t$

$x=3t,\ y=-2-t,\ z=4-2t$

21. $\dfrac{x-3}{2}=\dfrac{y-1}{-1}=\dfrac{z-4}{1}$

$\dfrac{x+2}{3}=\dfrac{y-3}{-1}=\dfrac{z-2}{1}$

22. $\dfrac{x+1}{2}=\dfrac{y-3}{-1}=\dfrac{z-2}{1}$

$\dfrac{x+1}{2}=\dfrac{y+1}{3}=\dfrac{z-3}{-4}$

Find an explicit relationship between x and y in Problems 23-36 by eliminating the parameter. In each case, sketch the path described by the parametric equations over the prescribed interval.

23. $x=t, y=t-1, 0\le t\le 3$

24. $x=-t,\ y=3-2t,\ 0\le t\le 1$

25. $x=60t,\ y=80t-16t^2,\ 0\le t\le 3$

26. $x=30t,\ y=60t-9t^2,\ -1\le t\le 2$

27. $x=t^3, y=t^2,\ t\ge 0$

28. $x=t^4,\ y=t^2,\ -1\le t\le\sqrt{2}$

29. $x=3\cos\theta, y=3\sin\theta, 0\le\theta\le 2\pi$

30. $x=2\sin\theta, y=2\cos\theta, 0\le\theta\le 2\pi$

31. $x=1+\sin t,\ y=-2+\cos t,\ 0\le t\le 2\pi$

32. $x=1+\sin^2 t,\ y=-2+\cos t,\ 0\le t\le\pi$

33. $x=4\tan 2t,\ y=3\sec 2t,\ 0\le t\le\pi$

34. $x=4\sec 2t,\ y=2\tan 2t,\ 0\le t\le\pi$

35. $x=t^3,\ y=3\ln t,\ t>0$

36. $x=e^t,\ y=e^{-t},\ (-\infty,\infty)$

37. Find two unit vectors parallel to the line

$$\frac{x-3}{4}=\frac{y-1}{2}=\frac{z+1}{1}$$

38. Find two unit vectors parallel to the line

$$\frac{x-1}{2}=\frac{y+2}{4}=\frac{z+5}{1}$$

Level 2

Find parametric equations for each of the curves in Problems 39-44.

39. A circle of radius 3, centered at the origin, oriented counterclockwise.

40. A circle of radius 2 centered at the origin, oriented clockwise.

41. The ellipse

$$\frac{x^2}{9}+\frac{y^2}{4}=1$$

oriented counterclockwise.

42. The parabola $y^2=4x+9$, oriented from $(4,-5)$ to $(0,3)$.

43. The hyperbola $\dfrac{x^2}{16}-\dfrac{y^2}{9}=1$.

44. The ellipse $\dfrac{(x-2)^2}{3}+\dfrac{(y+3)^2}{5}=1$.

45. Describe the path of the curve described by $x=\sin\pi t$, $y=\cos 2\pi t$ for $0\le t\le 1$.

46. Describe the path of the curve described by $x=\cos\pi t$, $y=\cos 2\pi t$ for $0\le t\le 1$.

47. Let $x=4a\sin t$, $y=b\cos^2 t$. Express y as a function of x.

48. Let $x=a\cos t$, $y=2b\sin^2 t$. Express y as a function of x.

49. Show that the vector $\mathbf{v}=3\mathbf{i}-4\mathbf{j}+\mathbf{k}$ is orthogonal to the line that passes through the points $P(0,0,1)$ and $Q(2,1,-1)$.

50. Show that the vector $\mathbf{v}=7\mathbf{i}+4\mathbf{j}+3\mathbf{k}$ is orthogonal to the line passing through the points $P(-2,2,7)$ and $Q(3,-3,2)$.

51. Find constants a and b so that the following lines coincide:

$$L_1:\ \frac{x-a}{2}=\frac{y-1}{4}=\frac{z+2}{1}$$

and

$$L_2:\ \frac{x-2}{-4}=\frac{y-b}{-8}=\frac{z+1}{-2}$$

52. Find constants a and b so that the following lines coincide:

$$L_1:\ \frac{x-1}{1}=\frac{y+a}{2}=\frac{z+2}{4}$$

and

$$L_2:\ \frac{x-b}{-2}=\frac{y-1}{-4}=\frac{z+1}{-8}$$

53. Consider the lines

$$L_1: x = -1 + 2t,\ y = 3 - t,\ z = 2 + 2t$$
$$L_2: x = -2 - t,\ y = 5 + 2t,\ z = -2t$$

Show that L_1 and L_2 intersect and find the point of intersection and the acute angle θ (to the nearest degree) between them.

54. Consider the lines

$$L_1: \frac{x+2}{3} = \frac{y-1}{2} = \frac{z}{-1}$$
$$L_2: \frac{x+1}{-2} = \frac{y}{3} = \frac{z-2}{1}$$

Show that L_1 and L_2 are skew.

55. Find an equation for the line L_1 that contains the point $P(-1, 3, 1)$ and meets the line L_2 orthogonally, where

$$L_2: x = 2 - t,\ y = 1 - 2t,\ z = 5 + t$$

56. Find an equation for the line L_1 that contains the point $P(2, 3, -1)$ and meets the line L_2 orthogonally, where

$$L_2: \frac{x-2}{2} = \frac{y+1}{-2} = \frac{z}{1}$$

Level 3

57. What can be said about the lines

$$\frac{x - x_0}{a_1} = \frac{y - y_0}{b_1} = \frac{z - z_0}{c_1}$$

and

$$\frac{x - x_0}{a_2} = \frac{y - y_0}{b_2} = \frac{z - z_0}{c_2}$$

in the case where $a_1a_2 + b_1b_2 + c_1c_2 = 0$?

58. **Modeling Problem** A circle of radius R rolls without slipping on the *outside* of a fixed circle of radius a. Assume the fixed circle is centered at the origin and that the moving point begins at $(a, 0)$. Use the angle t measured from the positive x-axis to the ray from the origin to the center of the rolling circle. Show that this *epicycloid* may be modeled by the parametric equations

$$x = (a + R)\cos t - R\cos\left(\frac{a+R}{R}\right)t$$
$$y = (a + R)\sin t - R\sin\left(\frac{a+R}{R}\right)t$$

59. **Modeling Problem** A circle of radius R rolls without slipping on the *inside* of a fixed circle of radius a. Find parametric equations to model the curve traced out by a point P on the circumference of the rolling circle of radius R. Let t be the angle measured from the positive x-axis to the ray that passes through the center of the rolling circle, and assume that the point P begins on the x-axis (that is, P has coordinates $(a, 0)$ when $t = 0$). Show that this *hypocycloid* has parametric equations

$$x = (a - R)\cos t + R\cos\left(\frac{a-R}{R}\right)t$$
$$y = (a - R)\sin t - R\sin\left(\frac{a-R}{R}\right)t$$

60. **Journal Problem** (*The MATYC Journal**) A rectangle is "rolled" along a straight line by rotations of 90° about each vertex in turn. One vertex traces out a series of arches where each arch is the union of three circular arcs. Show that the area of the region bounded by the straight line and one arch equals the area of the rectangle plus twice the area of the circumscribed circle.

9.6 PLANES IN $\mathbb{R}^3$

IN THIS SECTION: *Forms for the equation of a plane in $\mathbb{R}^3$, vector methods for measuring distances in $\mathbb{R}^3$*
Planes in three dimensions provide one of the foundational tools for working in $\mathbb{R}^3$.

Forms for the Equation of a Plane in $\mathbb{R}^3$

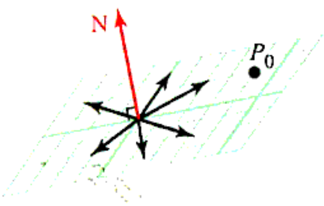


Figure 9.58 A plane through P with a normal vector **N**

Planes in space can also be characterized by vector methods. In particular, any plane is completely determined once we know one of its points and its orientation; that is, the "direction it faces." A common way to specify the direction of a plane is by means of a vector **N** that is orthogonal to every vector in the plane, as shown in Figure 9.58. Such a vector is called a **normal** to the plane.

*Problem 148 by Frank Kocher, *The MATYC Journal*, Vol. 14, 1980, p. 155.

Example 1 Obtain the equation for a plane

Find an equation for the plane that contains the point $Q(3, -7, 2)$ and is normal to the vector $\mathbf{N} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$.

Solution The normal vector $\mathbf{N}$ is orthogonal to every vector in the plane. In particular, if $P(x, y, z)$ is any point in the plane, then $\mathbf{N}$ must be orthogonal to the vector

$$\mathbf{QP} = (x - 3)\mathbf{i} + (y + 7)\mathbf{j} + (z - 2)\mathbf{k}$$

Because the dot (or scalar) product of two orthogonal vectors is 0, we have

$$\begin{aligned} \mathbf{N} \cdot \mathbf{QP} = 2(x - 3) + (1)(y + 7) + (-3)(z - 2) &= 0 \\ 2x - 6 + y + 7 - 3z + 6 &= 0 \\ 2x + y - 3z + 7 &= 0 \end{aligned}$$

Therefore, $2x + y - 3z + 7 = 0$ is the equation of the plane.

By generalizing the approach illustrated in Example 1, we can show that the plane that contains the point (x_0, y_0, z_0) and has normal vector $\mathbf{N} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ must have the Cartesian equation

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

This is called the **point-normal form** of the equation of a plane. By rearranging terms, we can rewrite this equation in the form $Ax + By + Cz + D = 0$. This is called the **standard form** of the equation of a plane. The numbers $[A, B, C]$ called **attitude numbers** of the plane.

Notice from Figure 9.59 that *attitude numbers of a plane are the same as direction numbers of a normal line.* This means that you can find normal vectors to a plane *by inspecting the equation of the plane.*

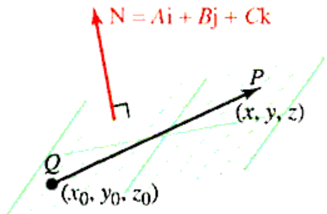


Figure 9.59 The graph of a plane with attitude numbers $[A, B, C]$

Example 2 Relationship between normal vectors and planes

Find normal vectors to the planes

a. $5x + 7y - 3z = 0$ **b.** $x - 5y + \sqrt{2}z = 6$ **c.** $3x - 7z = 10$

Solution

a. A normal to the plane $5x + 7y - 3z = 0$ is $\mathbf{N} = 5\mathbf{i} + 7\mathbf{j} - 3\mathbf{k}$.
b. For $x - 5y + \sqrt{2}z = 6$ the normal is $\mathrm{N} = \mathbf{i} - 5\mathbf{j} + \sqrt{2}\mathbf{k}$.
c. For $3x - 7z = 10$ it is $\mathbf{N} = 3\mathbf{i} - 7\mathbf{k}$.

EQUATIONS OF A PLANE A plane with normal $\mathbf{N} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ that contains the point (x_0, y_0, z_0) has the following equations:
Point-normal form: $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$
Standard form: $Ax + By + Cz + D = 0$ for some constants A, B, C, and D

Example 3 Equation of a line orthogonal to a given plane

Find an equation of the line that passes through the point $Q(2, -1, 3)$ and is orthogonal to the plane $3x - 7y + 5z + 55 = 0$. Where does the line intersect the plane? (See Figure 9.60.)

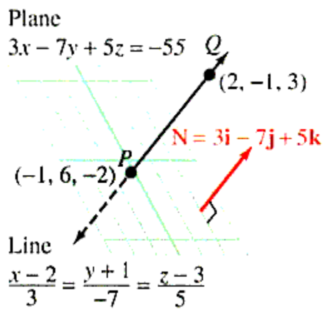


Figure 9.60 Graph of a plane

Solution By inspection of the equation of the plane, we see that $\mathbf{N} = 3\mathbf{i} - 7\mathbf{j} + 5\mathbf{k}$ is a normal vector to the plane. Because the required line is also orthogonal to the plane, it must be parallel to **N**. Thus, the line contains the point $Q(2,-1,3)$ and has direction numbers $[3,-7,5]$, so that its equation is

$$\frac{x-2}{3}=\frac{y+1}{-7}=\frac{z-3}{5}$$

To find the point where this line intersects the plane, we rewrite it in parametric form:

$$x=2+3t, y=-1-7t, \text{ and } z=3+5t$$

Now, substitute into the equation of the plane:

$$\begin{aligned} 3(2+3t)-7(-1-7t)+5(3+5t) &= -55 \\ 6+9t+7+49t+15+25t &= -55 \\ 83t &= -83 \\ t &= -1 \end{aligned}$$

Thus, the point of intersection is found by substituting $t=-1$ for x, y, and z:

$$\begin{aligned} x &= 2+3(-1)=-1 \\ y &= -1-7(-1)=6 \\ z &= 3+5(-1)=-2 \end{aligned}$$

The point of intersection is $(-1,6,-2)$.

Example 4 Equation of a plane containing three given points

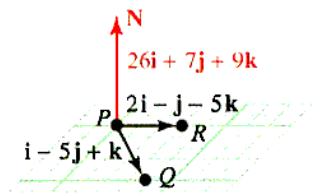


Figure 9.61 Plane passing through three points

Find the standard form equation of a plane containing $P(-1,2,1)$, $Q(0,-3,2)$, and $R(1,1,-4)$. (See Figure 9.61.)

Solution Because a normal **N** to the required plane is orthogonal to the vectors **PR** and **PQ**, we find **N** by computing the cross product $\mathbf{N}=\mathbf{PR}\times\mathbf{PQ}$.

$$\begin{aligned} \mathbf{PR} &= (1+1)\mathbf{i}+(1-2)\mathbf{j}+(-4-1)\mathbf{k}=2\mathbf{i}-\mathbf{j}-5\mathbf{k} \\ \mathbf{PQ} &= (0+1)\mathbf{i}+(-3-2)\mathbf{j}+(2-1)\mathbf{k}=\mathbf{i}-5\mathbf{j}+\mathbf{k} \end{aligned}$$

$$\begin{aligned} \mathbf{N}=\mathbf{PR}\times\mathbf{PQ} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & -5 \\ 1 & -5 & 1 \end{vmatrix} \\ &= (-1-25)\mathbf{i}-(2+5)\mathbf{j}+(-10+1)\mathbf{k} \\ &= -26\mathbf{i}-7\mathbf{j}-9\mathbf{k} \end{aligned}$$

We can now find the equation of the plane using this normal vector and any point in the plane. We will use the point P:

Attitude numbers of the plane: from $N=-26i-7j-9k$

$$-26(x+1)-7(y-2)-9(z-1)=0$$

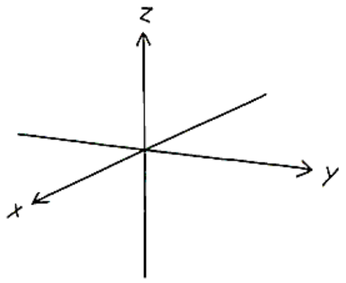
Point on the plane; we are using $P(-1,2,1)$.

Thus, the equation of the plane is

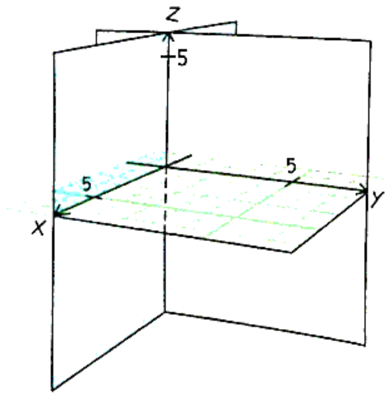
$$\begin{aligned} -26x-26-7y+14-9z+9 &= 0 \\ 26x+7y+9z+3 &= 0 \end{aligned}$$

Drawing Lesson: Sketching a plane in one octant of $\mathbb{R}^3$

Sketch the graph of
$4(x-1)+6y+2(z-4)=0$
Begin by drawing the three coordinate axes.



Next, draw the three coordinate planes, shaded for easy viewing.

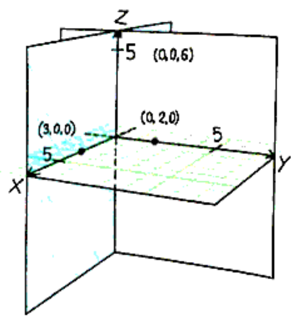


Plot the points where the plane passes through each of the coordinate axes.

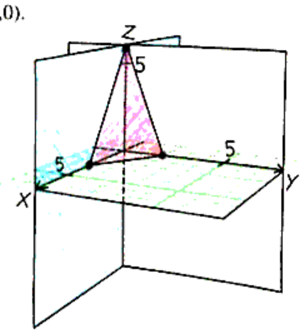
If $y=z=0$, then
$$4(x-1)+6(0)+2(0-4)=0$$
$$4x-4-8=0$$
$$x=3;\ \text{Plot } (3,0,0)$$

If $x=y=0$, then
$$4(0-1)+6(0)+2(z-4)=0$$
$$-4+0+2z-8=0$$
$$z=6;\ \text{Plot } (0,0,6)$$

If $x=z=0$, then
$$4(0-1)+6y+2(0-4)=0$$
$$-4+6y-8=0$$
$$y=2;\ \text{Plot } (0,2,0).$$



Connect the three points, (3,0,0), (0,0,6), and (0,2,0).



Shade the plane in order to add depth to the figure.

Example 5 Line parallel to the intersection of two planes

Find the equation of a line passing through $(-1,2,3)$ that is parallel to the line of intersection of the planes $3x-2y+z=4$ and $x+2y+3z=5$. (See Figure 9.62.)

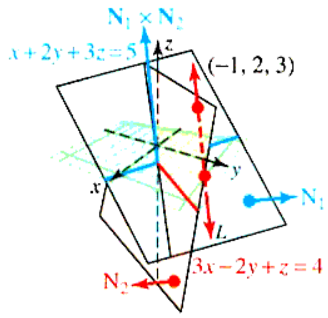


Figure 9.62 Given planes

Solution By inspection, we see that the normals to the given planes are $\mathbf{N}_1 = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\mathbf{N}_2 = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. The desired line is perpendicular to both of these normals, so the aligned vector is found by by computing with the cross computing with the cross product:

$$\begin{aligned}
\mathbf{N}_1 \times \mathbf{N}_2 &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -2 & 1 \\ 1 & 2 & 3 \end{vmatrix} \\
&= (-6-2)\mathbf{i} - (9-1)\mathbf{j} + (6+2)\mathbf{k} \\
&= -8\mathbf{i} - 8\mathbf{j} + 8\mathbf{k}
\end{aligned}$$

The direction of this vector is $\langle -8, -8, 8 \rangle = -8\langle 1, 1, -1 \rangle$. The equation of the desired line is

$$\frac{x+1}{1} = \frac{y-2}{1} = \frac{z-3}{-1}$$

Example 5 can also be used to find the equation of the line of intersection of the two planes. Instead of using the given point $(-1, 2, 3)$, you will first need to find a point in the intersection and then proceed, using the steps of Example 5. We conclude this discussion by finding the equation of a plane containing two given (nonparallel) lines.

Example 6 Equation of a plane containing two intersecting lines

Find the standard-form equation of the plane determined by the intersecting lines

$$\frac{x-2}{3} = \frac{y+5}{-2} = \frac{z+1}{4} \quad \text{and} \quad \frac{x+1}{2} = \frac{y}{-1} = \frac{z-16}{5}$$

Solution Proceeding as in Example 4 of Section 9.5, we find that the lines intersect at $(-19, 9, -29)$.

The aligned vectors for these two lines are $\mathbf{v}_1 = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\mathbf{v}_2 = 2\mathbf{i} - \mathbf{j} + 5\mathbf{k}$ and the normal to the desired plane is orthogonal to both $\mathbf{v}_1$ and $\mathbf{v}_2$. (See Figure 9.63.)

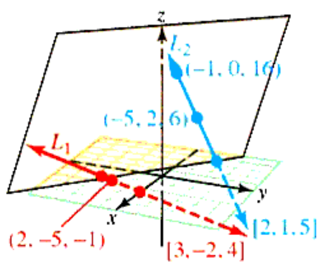


Figure 9.63 Plane determined by lines

Thus, we take the normal to be the cross product:

$$\begin{aligned}
\mathbf{N} &= \mathbf{v}_1 \times \mathbf{v}_2 \\
&= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -2 & 4 \\ 2 & -1 & 5 \end{vmatrix} \\
&= (-10+4)\mathbf{i} - (15-8)\mathbf{j} + (-3+4)\mathbf{k} \\
&= -6\mathbf{i} - 7\mathbf{j} + \mathbf{k}
\end{aligned}$$

The point of intersection $P(-19, 9, -29)$ is certainly in the plane, so we can use $(2, -5, -1)$, $(-1, 0, 16)$, or $(-19, 9, -29)$ to obtain

$$-6(x-2) - 7(y+5) + 1(z+1) = 0$$
$$-6x + 12 \ldots + z + 1 = 0$$
$$22 = 0$$
$$= 0$$

Vector Metho... $\mathbb{R}^3$

Vector met... in $\mathbb{R}^3$, between points, lines, and p... or the distance from a point to

Theorem 9.9 Distance from a point to a plane in $\mathbb{R}^3$

The distance from the point $P(x_0, y_0, z_0)$ to the plane $Ax + By + Cz + D = 0$ is given by

$$d = \frac{|\mathbf{QP} \cdot \mathbf{N}|}{\|\mathbf{N}\|} = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

where Q is any point in the given plane and $\mathbf{N}$ is a normal to the given plane.

Proof: The required distance is found by projecting the vector $\mathbf{QP}$ onto the normal $\mathbf{N}$. Thus, the distance from the point to the plane is given by

$$d = \|\mathbf{QP}\| \, |\cos\theta| = \frac{\|\mathbf{QP}\| \, \|\mathbf{N}\| \, |\cos\theta|}{\|\mathbf{N}\|} = \frac{|\mathbf{QP} \cdot \mathbf{N}|}{\|\mathbf{N}\|}$$

where θ is the angle between $\mathbf{QP}$ and $\mathbf{N}$ (see Figure 9.64.)

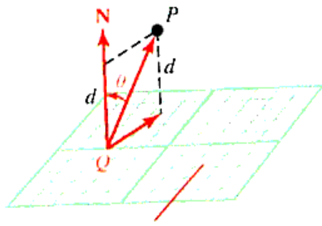


Figure 9.64 The distance from a point to a plane in $\mathbb{R}^3$

Suppose $Q(x_1, y_1, z_1)$ is any particular point in the plane, so that

$$\mathbf{QP} = (x_0 - x_1)\mathbf{i} + (y_0 - y_1)\mathbf{j} + (z_0 - z_1)\mathbf{k}$$

The dot product of $\mathbf{QP}$ with $\mathbf{N} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ is given by

$$\begin{aligned} \mathbf{QP} \cdot \mathbf{N} &= (x_0 - x_1)A + (y_0 - y_1)B + (z_0 - z_1)C \\ &= (Ax_0 + By_0 + Cz_0) - (Ax_1 + By_1 + Cz_1) \\ &= Ax_0 + By_0 + Cz_0 - (-D) \qquad \textit{Because } Ax_1 + By_1 + Cz_1 + D = 0. \end{aligned}$$

Because the normal vector has length $\|\mathbf{N}\| = \sqrt{A^2 + B^2 + C^2}$, we can substitute into the formula for d to obtain

$$d = \frac{|\mathbf{QP} \cdot \mathbf{N}|}{\|\mathbf{N}\|} = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}} \qquad \blacklozenge$$

Example 7 Finding an equation for a sphere tangent to a given plane

Find an equation for the sphere with center $C(-3, 1, 5)$ that is tangent to the plane $6x - 2y + 3z = 9$ is shown in Figure 9.65.

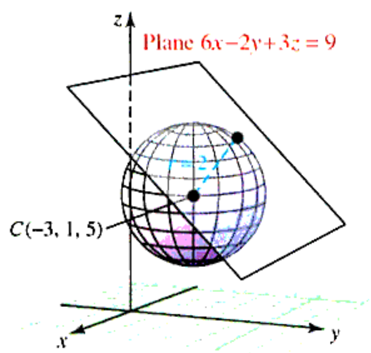


Figure 9.65 Sphere with a tangent plane

Solution The radius r of the sphere is the distance from the center C to the given plane, as shown in Figure 9.65; namely,

$$r = \left| \frac{6(-3) + (-2)(1) + 3(5) - 9}{\sqrt{6^2 + (-2)^2 + 3^2}} \right| = \left| \frac{-14}{7} \right| = 2$$

Therefore, an equation of the sphere is

$$(x+3)^2+(y-1)^2+(z-5)^2=2^2$$

Vector methods can also be used to derive a formula for the distance between two skew lines L_1 and L_2 in $\mathbb{R}^3$.

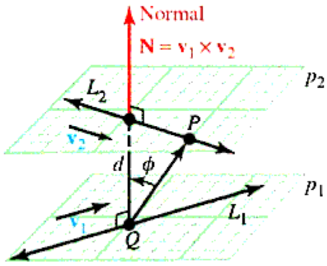


Figure 9.66 Distance between skew lines

Example 8 Computing the distance between two skew lines

Find the distance between the two skew lines

$$L_1\colon x=1+2s,\ y=-1+s,\ z=2+4s$$
$$L_2\colon x=-2+4t,\ y=-3t,\ z=-1+t$$

Solution Recall from Section 9.5 that two skew lines do not intersect and are not parallel, but do lie in two parallel planes. Thus, the distance d between the lines L_1 and L_2 is the same as the distance between the parallel planes p_1 and p_2 that contain them (see Figure 9.66). Our strategy for finding the distance d is to find the distance from a point Q on p_1 to the plane p_2.

First, find a point Q on p_1; set $s=0$ in the parametric equations for L_1. Then $x=1, y=-1$, and $z=2$, so $Q(1,-1,2)$ is such a point. Next, vectors $\mathbf{v}_1=\langle 2,1,4\rangle$ and $\mathbf{v}_2=\langle 4,-3,1\rangle$ are parallel to L_1 and L_2, respectively, and the cross product $\mathbf{N}=\mathbf{v}_1\times\mathbf{v}_2$ is orthogonal to both p_1 and p_2. We find that

$$\mathbf{N}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 2 & 1 & 4\\ 4 & -3 & 1\end{vmatrix}=13\mathbf{i}+14\mathbf{j}-10\mathbf{k}$$

Setting $t=0$, we see that the point $P(-2,0,-1)$ is on p_2, so an equation for p_2 is

$$13(x+2)+14(y-0)-10(z+1)=0$$
$$13x+14y-10z+16=0$$

The distance d between the lines L_1 and L_2 is the same as the distance from the point $Q(1,-1,2)$ to the plane p_2. Thus

$$d=\frac{|13(1)+14(-1)-10(2)+16|}{\sqrt{13^2+14^2+(-10)^2}}=\frac{5}{\sqrt{465}}\approx 0.2319$$

In the final example of this section, we find the distance from a point P to a line L in $\mathbb{R}^3$. Let Q be a point on L and let $\mathbf{v}$ be a vector parallel to L. Then, as shown in Figure 9.67, the distance from P to L is given by

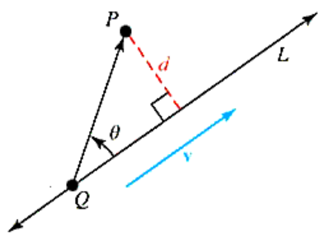


Figure 9.67 The distance from a point to a line in $\mathbb{R}^3$

$$d=\|\mathbf{QP}\|\,|\sin\theta|$$

where θ is the angle between $\mathbf{v}$ and the vector $\mathbf{QP}$.

This reminds us of the cross product $\mathbf{v}\times\mathbf{QP}$, and because

$$\|\mathbf{v}\times\mathbf{QP}\|=\|\mathbf{v}\|\,\|\mathbf{QP}\|\sin\theta$$

we have

$$d=\|\mathbf{QP}\|\sin\theta=\frac{\|\mathbf{v}\times\mathbf{QP}\|}{\|\mathbf{v}\|}$$

These observations are summarized in the following theorem.

Theorem 9.10 Distance from a point to a line

The distance from point P to the line L is given by the formula

$$d = \frac{\|\mathbf{v} \times \mathbf{QP}\|}{\|\mathbf{v}\|}$$

where $\mathbf{v}$ is a vector parallel to L and Q is any point on L.

Proof: A sketch of the proof precedes the statement of the theorem.

Example 9 Distance from a point to a line

Find the distance from the point $P(3, -8, 1)$ to the line

$$\frac{x-3}{3} = \frac{y+7}{-1} = \frac{z+2}{5}$$

Solution We need to find a point Q on the line. We see that $Q(3, -7, -2)$ is on the line and that $\mathbf{QP} = \langle 0, -1, 3\rangle$. A vector parallel to L is $\mathbf{v} = \langle 3, -1, 5\rangle$, so that

$$\mathbf{v} \times \mathbf{QP} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 5 \\ 0 & -1 & 3 \end{vmatrix} = 2\mathbf{i} - 9\mathbf{j} - 3\mathbf{k}$$

Thus,

$$d = \frac{\|\mathbf{v} \times \mathbf{QP}\|}{\|\mathbf{v}\|} = \frac{\sqrt{(2)^2 + (-9)^2 + (-3)^2}}{\sqrt{3^2 + (-1)^2 + 5^2}} = \frac{\sqrt{94}}{\sqrt{35}} \approx 1.64$$

PROBLEM SET 9.6

Level 1

1. ■ What does this say? Describe the relationship between normal vectors and planes.
2. ■ What does this say? Describe a procedure for finding the indicated distances
   a. from a point to a plane.
   b. from a point to a line in $\mathbb{R}^3$.

Write each equation for a plane given in Problems 3-6 in standard form.

3. $-2(x+1) + 4(y-3) - 8z = 0$
4. $4(x+1) - 2(y+1) + 6(z-2) = 0$
5. $5(x-2) - 3(y+2) + 4(z+3) = 0$
6. $-3(x-4) + 2(y+1) - 2(z+1) = 0$

Sketch each plane in Problems 7-10.

7. $4(x-5) + 3(y-4) + 2(z-7) = 0$
8. $-3(x+4) + 5(y-2) + 4(z-1) = 0$
9. $3(x-4) + 2(y-7) + 2(z+4) = 0$
10. $-4(x-5) + 2(y+3) + 8(z-4) = 0$

Find an equation for the plane that contains the point P and has the normal vector $\mathbf{N}$ given in Problems 11-16.

11. $P(0, -7, 1)$; $\mathbf{N} = -\mathbf{i} + \mathbf{k}$
12. $P(0, -3, 0)$; $\mathbf{N} = -2\mathbf{j} + 3\mathbf{k}$
13. $P(1, 1, -1)$; $\mathbf{N} = -\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$
14. $P(0, 0, 0)$; $\mathbf{N} = \mathbf{k}$
15. $P(0, 0, 0)$; $\mathbf{N} = \mathbf{i}$
16. $P(-1, 3, 5)$; $\mathbf{N} = 2\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$
17. Find two unit vectors perpendicular to the plane $5x - 3y + 2z = 15$
18. Find two unit vectors perpendicular to the plane $2x + 4y - 3z = 4$

Find the distance between the point and the plane given in Problems 19-22. Assume $a \neq 0$.

19. $P(0, 0, 0)$; $2x - 3y + 5z = 10$
20. $P(1, 0, -1)$; $x + y - z = 1$
21. $P(a, 2a, 3a)$; $3x - 2y + z = -\dfrac{1}{a}$
22. $P(a, -a, 2a)$; $2ax - y + az = 4a$

Find the distance from the point $(-1,2,1)$ *to each plane given in Problems* 23-26.

23. the plane through the points $(-1,1,1)$, $(4,3,7)$ and $(3,-1,0)$

24. the plane through the points $(0,0,0)$, $(1,2,4)$, and $(-2,-1,1)$

25. the plane through the point $(1,0,1)$ with normal vector $2\mathbf{i}-\mathbf{j}+2\mathbf{k}$

26. the plane through the point $(-3,5,1)$ with normal vector $3\mathbf{i}+\mathbf{j}+5\mathbf{k}$

Find the distance from the point P to the line L in Problems 27-32.

27. $P(9,-3)$; $3x-4y+8=0$

28. $P(4,5)$; $3x-4y+8=0$

29. $P(4,-3)$; $12x+5y-2=0$

30. $P(1,0,1)$; $\dfrac{x}{3}=\dfrac{y-1}{2}=\dfrac{z}{1}$

31. $P(1,0,-1)$; $\dfrac{x-2}{3}=\dfrac{y+1}{1}=\dfrac{z-1}{2}$

32. $P(1,-2,2)$; $\dfrac{x}{1}=\dfrac{2y}{1}=\dfrac{z}{-1}$

Find the distance between the lines given in Problems 33-36.

33. $x=2-t$, $y=5+2t$, $z=3t$ and $x=2t$, $y=-1-t$, $z=1+2t$

34. $\dfrac{x+1}{3}=\dfrac{y-2}{-2}=\dfrac{z-1}{1}$ and $\dfrac{x-2}{5}=\dfrac{y+1}{1}=\dfrac{z}{3}$

35. $x=-1+t$, $y=-2t$, $z=3$ and the line passing through $(0,-1,2)$ and $(1,-2,3)$

36. $\dfrac{x+1}{1}=\dfrac{y-3}{2}=\dfrac{z+2}{3}$ and the line passing through $(1,3,-2)$ and $(0,1,-1)$

Level 2

37. a. Show that the line

$$\frac{x-1}{3}=\frac{y}{-2}=\frac{z+1}{1}$$

is parallel to the plane $x+2y+z=1$.

b. Find the distance from the line to the plane in part **a**.

38. Three of the four vertices of a parallelogram $QRTS$ in $\mathbb{R}^3$ are $Q(-1,3,5)$, $R(6,-3,2)$, and $S(2,4,-3)$. (Note that the fourth vertex of the parallelogram may be located in one of several locations.) What is the area of the parallelogram?

39. Find an equation for the set of all points $P(x,y,z)$ such that the distance from P to the point $P_0(-1,2,4)$ is the same as the distance from P to the plane $2x-5y+3z=7$. (Do not expand binomials.)

40. Find an equation for the set of all points $P(x,y,z)$ such that the distance from P to the line

$$\frac{x-1}{4}=\frac{y+1}{-1}=\frac{z}{3}$$

is 5. (Do not expand.)

41. In Figure 9.68, $\mathbf{N}$ is normal to the plane p and L is a line that intersects p.

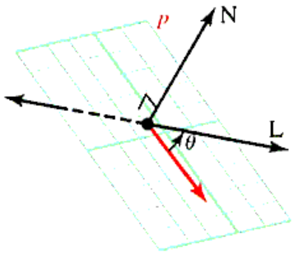


Figure 9.68 Problem 41

Assume $\mathbf{N}=a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$ and that L is given by $x=x_0+At$, $y=y_0+Bt$, $z=z_0+Ct$

a. Let $\phi=90^\circ-\theta$, where θ is the angle between L and the plane p. Find $\cos\phi$.

b. Find the angle between the plane $x+y+z=10$ and the line

$$\frac{x-1}{2}=\frac{y+3}{3}=\frac{z-2}{-1}$$

42. Find the equation of the sphere with center $C(-2,3,7)$ that is tangent to the plane $2x+3y-6z=5$.

43. Find a vector that is parallel to the line of intersection of the planes $2x+3y=0$ and $3x-y+z=1$.

44. Find an equation for the line of intersection of the planes $2x-y+z=8$ and $x+y-z=5$.

45. Find the direction cosines of a vector determined by the line of intersection of the planes $x+y+z=3$ and $2x+3y-z=4$.

46. Find an equation for the plane that passes through $P(1,-1,2)$ and is normal to $\mathbf{PQ}$ where Q is $Q(2,1,3)$.

47. Find an equation for the line that passes through the point $(1,-5,3)$ and is orthogonal to the plane $2x-3y+z=1$.

48. Find two unit vectors that are parallel to the line of intersection of the planes $x+y=1$ and $x-2z=3$.

49. Find two unit vectors that are parallel to the line of intersection of the planes $x+y+z=3$ and $x-y+z=1$.

50. Find an equation for the plane that contains the point $(2,1,-1)$ and is orthogonal to the line

$$\frac{x-3}{3}=\frac{y+1}{5}=\frac{z}{2}$$

51. Find a plane that passes through the point $(1,2,-1)$ and is parallel to the plane $2x-y+3z=1$.

52. Show that the line
$$\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-2}{4}$$
is parallel to the plane $x-2y+z=6$.

53. Find the point where the line
$$\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z}{3}$$
intersects the plane $3x+2y-z=5$.

54. The *angle between two planes* is defined to be the acute angle between their normal vectors. Find the angle between the planes $2x+y-4z=3$ and $x-y+z=2$, rounded to the nearest degree.

55. Find an equation for the line that passes through the point $P(2,3,1)$ and is parallel to the line of intersection of the planes $x+2y-3z=4$ and $x-2y+z=0$.

Level 3

56. The planes $Ax+By+Cz=D_1$ and $Ax+By+Cz=D_2$ are parallel.
   a. Is the distance between the planes $|D_1-D_2|$? If not, what is the correct formula?
   b. Find the distance between the parallel planes $x+y+2z=2$ and $x+y+2z=4$.

57. Show that the angle between the planes $A_1x+B_1y+C_1z+D_1=0$ and $A_2x+B_2y+C_2z+D_2=0$ is $\pi/2$ if and only if $A_1A_2+B_1B_2+C_1C_2=0$.

58. Show that a plane with x-intercept a, y-intercept b, and z-intercept c has the equation
$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$
assuming a,b, and c are all nonzero.

59. Suppose planes p_1 and p_2 intersect. If $\mathbf{v}_1$ and $\mathbf{w}_1$ are nonzero vectors on p_1 and $\mathbf{v}_2$ and $\mathbf{w}_2$ are nonzero vectors on plane p_2, then show that
$$(\mathbf{v}_1\times\mathbf{w}_1)\times(\mathbf{v}_2\times\mathbf{w}_2)$$
is parallel to the line of intersection of the planes.

60. Let L_1 and L_2 be skew lines that contain the points P_1 and P_2, respectively, and are parallel to the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. Show that the distance between the lines is given by
$$d=\left|\frac{(\mathbf{v}_1\times\mathbf{v}_2)\cdot\mathbf{P_1P_2}}{\|\mathbf{v}_1\times\mathbf{v}_2\|}\right|$$

9.7 Quadric Surfaces

IN THIS SECTION: *Catalog of quadric surfaces; methods for graphing quadric surfaces*

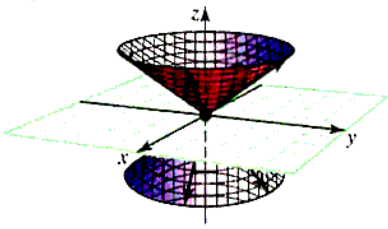
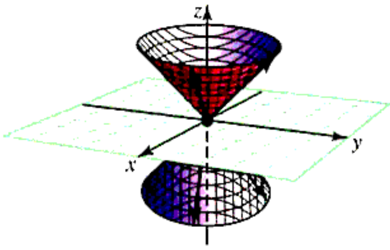
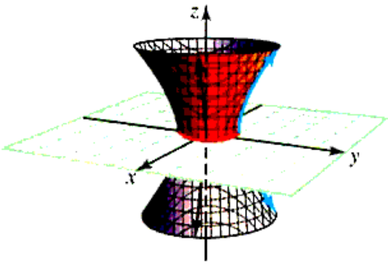
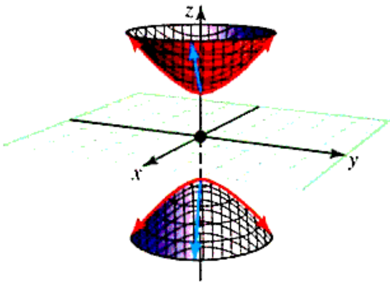
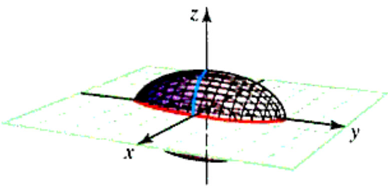
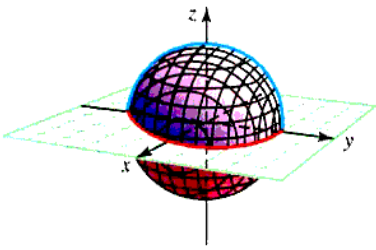
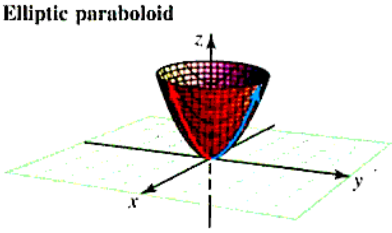
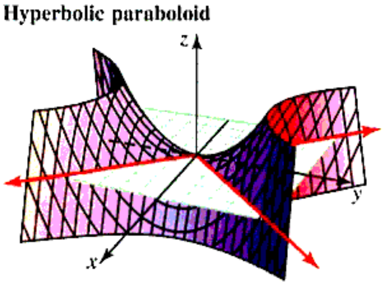
In this section we develop some of the most common three-dimensional surfaces, which are the three-dimensional counterparts for the conic sections, namely those two-dimensional curves whose equations are of the form
$$Ax^2+Bxy+Cy^2+Dx+Ey+F=0$$
Conic sections are graphs in $\mathbb{R}^2$ that can be represented by a second degree equation in two variables, and quadric surfaces are graphs in $\mathbb{R}^3$ which can be represented by a second-degree equation in three variables.

Catalog of Quadric Surfaces

Spheres and elliptic, parabolic, and hyperbolic cylinders are examples of **quadric surfaces**. In general, such a surface is the graph of an equation of the form
$$Ax^2+By^2+Cz^2+Dxy+Exz+Fyz+Gx+Hy+Iz+J=0$$
where A,B,C,D,E,F,G,H,I, and J are constants. By using rotations and translations, it can be shown that any such equation can be expressed in one of the following standard forms:
$$z=Mx^2+Ny^2 \qquad \text{or} \qquad Px^2+Qy^2+Rz^2=S$$
where M,N,P,Q,R, and S are constants. We will concentrate on examining graphs of the standard forms.

The quadric surfaces are shown in Table 9.2.

Table 9.2 Quadric Surfaces

Surface	Description	Surface	Description
Elliptic cone 	The trace in the xy-plane is a point; in planes parallel to the xy-plane, it is an ellipse. Traces in the xz- and yz-planes are intersecting lines; in planes parallel to these, they are hyperbolas: $$z^2 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$ The axis of this cone is the z-axis.	**Circular cone** 	The circular cone is a special kind of elliptic cone for which $a = b = r$. $$z^2 = \frac{x^2}{r^2} + \frac{y^2}{r^2}$$ The axis of this cone is the z-axis.
Hyperboloid of one sheet 	The trace in the xy-plane is an ellipse; in planes parallel to the xz- and yz-planes the traces are hyperbolas: $$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$ One negative on equation; the axis of this one is the z-axis.	**Hyperboloid of two sheets** 	There is no trace in the xy-plane. In planes parallel to the xy-plane which intersect the surface, the traces are ellipses. Traces in the xz- and yz-planes are hyperbolas. $$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$ Two negatives; the axis of this one is the z-axis.
Ellipsoid 	The traces in the coordinate planes are ellipses. $$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$	**Sphere** 	The sphere is a special kind of ellipsoid for which $a = b = c = r$ $x^2 + y^2 + z^2 = r^2$
Elliptic paraboloid 	The trace in the xy-plane is a point; in planes parallel to the xy-plane, it is an ellipse. Traces in the xz- and yz-planes are parabolas $$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$ Axis is z-axis (z first degree). If $a = b = r$, then the cross-section is a circle, and it is called a *circular paraboloid*.	**Hyperbolic paraboloid** 	The trace in the xy-plane is two intersecting lines; in the plane parallel to the xy-plane, the traces are hyperbolas. Traces in the xz-plane and yz-planes are parabolas. $$z = \frac{y^2}{b^2} - \frac{x^2}{a^2}$$ This is also called a **saddle.**

Example 1 Identifying a quadric surface

Describe the given surfaces.

a. $7x^2 + 4y^2 - 28z^2 = 0$ **b.** $3x^2 + 5y^2 - 15z = 0$

Solution

a. We rewrite the equation in one of the standard forms given in Table 9.2:

$$z^2 = \frac{x^2}{4} + \frac{y^2}{7}$$

We see this is an elliptic cone.

b. Rewrite as

$$z = \frac{x^2}{5} + \frac{y^2}{3}$$

We see that the surface is an elliptic paraboloid.

Methods for Graphing Quadric Surfaces

The intersection of a surface with a plane is called the **trace** of the surface in that plane. The trace of a curve is found by setting one of the variables equal to a constant and then graphing the resulting curve. If $x = k$ (k a constant), the resulting curve is drawn in the plane $x = k$, which is parallel to the yz-plane; if $y = k$ the resulting curve is drawn in the plane $y = k$ which is parallel to the xz-plane; and if $z = k$, the curve is drawn in the plane $z = k$, parallel to the xy-plane.

Example 2 Using traces to sketch the graph of an ellipsoid

Sketch the graph of the quadric surface given by

$$\frac{x^2}{4} + \frac{y^2}{5} + \frac{z^2}{9} = 1$$

Solution Consider the trace of the given surface in the plane $z = 0$. We substitute $z = 0$ into the equation to obtain

$$\frac{x^2}{4} + \frac{y^2}{5} = 1$$

which we recognize as an ellipse. More generally, the trace in the plane $z = k$ is, by substitution,

$$\frac{x^2}{4} + \frac{y^2}{5} + \frac{k^2}{9} = 1 \qquad \text{or} \qquad \frac{x^2}{4} + \frac{y^2}{5} = 1 - \frac{k^2}{9}$$

which is also an ellipse for $k^2 \leq 9$, since the right hand side is a constant. Similarly, the trace of the surface in a plane of the form $y = k$ is

$$\frac{x^2}{4} + \frac{k^2}{5} + \frac{z^2}{9} = 1 \qquad \text{or} \qquad \frac{x^2}{4} + \frac{z^2}{9} = 1 - \frac{k^2}{5}$$

which is an ellipse for $k^2 \leq 5$, as is the trace in the plane $x = k$ (for $k^2 \leq 4$):

$$\frac{k^2}{4} + \frac{y^2}{5} + \frac{z^2}{9} = 1 \qquad \text{or} \qquad \frac{y^2}{5} + \frac{z^2}{9} = 1 - \frac{k^2}{4}$$

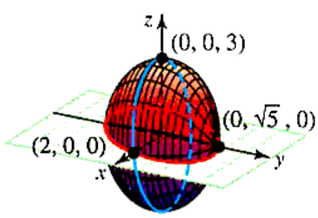


Figure 9.69 Graph of an ellipsoid

Sketch this ellipse in $\mathbb{R}^3$ as shown in Figure 9.69. Since the traces in all three principal directions are ellipses, we call this surface an **ellipsoid**.

Example 3 Using traces to sketch the graph of a hyperboloid

Sketch the graph of the surface $9x^2 + 36y^2 - 4z^2 + 36 = 0$.

Solution We begin by setting $z = k$, to obtain

$$9x^2 + 36y^2 = 4k^2 - 36$$

which is impossible for $k^2 < 9$, and represents an ellipse for $k^2 \geq 9$. For $x = k$, we have

$$4z^2 - 36y^2 = 36 + 9k^2$$

and for $y = k$

$$4z^2 - 9x^2 = 36 + 36k^2$$

both of which represent hyperbolas for all k.

To summarize, the surface has hyperbolic traces for planes parallel to the yz-plane and the xz-plane and elliptical traces for planes parallel to the xy-plane, except for a "gap" corresponding to the planes of the form $z = k$ with $|k| < 3$. This surface is a hyperboloid of two sheets. The graph is shown in Figure 9.70.

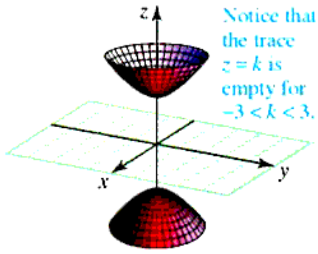


Figure 9.70 Graph of a hyperboloid of two sheets

Example 4 Identifying and sketching a quadric surface

Identify and sketch the surface with equation $9x^2 - 16y^2 + 144z = 0$.

Solution Look at Table 9.2 and note that the equation is second degree in x and y but first degree in z. This means it is an elliptic paraboloid or a hyperbolic paraboloid. Solve the equation for z:

$$\begin{aligned} 9x^2 - 16y^2 + 144z &= 0 \\ 144z &= 16y^2 - 9x^2 \\ z &= \frac{y^2}{9} - \frac{x^2}{16} \end{aligned}$$

We recognize this as a hyperbolic paraboloid. Next, we take cross sections of $9x^2 - 16y^2 + 144z = 0$.

Cross Section	Chosen Value	Equation	Description
xy-plane	$z = 0$	$\frac{y^2}{9} - \frac{x^2}{16} = 0$	Two intersecting lines
Parallel to the xy-plane	$z = 4$	$\frac{y^2}{36} - \frac{x^2}{64} = 1$	Hyperbola
xz-plane	$y = 0$	$z = -\frac{x^2}{16}$	Parabola opens downward
Parallel to the xz-plane	$y = 10$	$z - \frac{100}{9} = -\frac{x^2}{16}$	Parabola opens downward
yz-plane	$x = 0$	$z = \frac{y^2}{9}$	Parabola opens upward
Parallel to the yz-plane	$x = 5$	$z + \frac{25}{16} = \frac{y^2}{9}$	Parabola opens upward

These traces are shown in Figure 9.71.

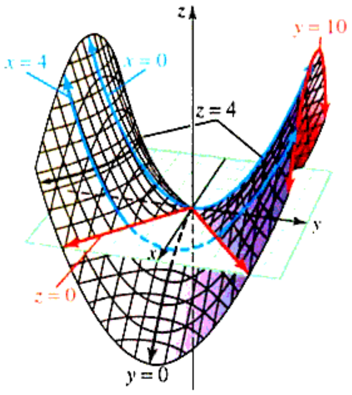
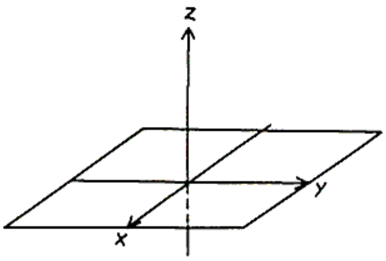


Figure 9.71 Graph of quadric surface

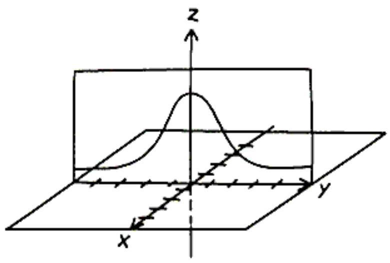
You can use Figure 9.71 to visualize the surface in Example 4, but that does not help you to draw a quadratic surface on your own paper. The following drawing lesson may help you do to this. For example, suppose you seek to sketch $z = \dfrac{1}{1 + x^2 + y^2}$.

Drawing Lesson: Sketching a surface in $\mathbb{R}^3$

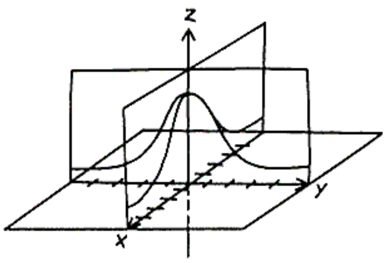
Draw the xy-plane in three dimensions adding the z-axis.



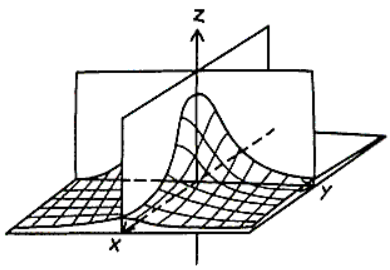
Draw a trace in one of the coordinate planes (in this case, the plane is $x = 0$). If necessary, adjust the z-scale to show the trace more clearly.



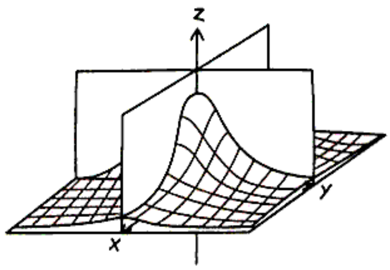
Draw a trace in another coordinate plane (in this case, the plane $y = 0$).



Draw several additional trace curves to reveal the contours of the surface.



Erase all hidden lines. Use highlighters to color the surface and the xy-plane. Plot (4,10,0) in the xy-plane.



PROBLEM SET 9.7

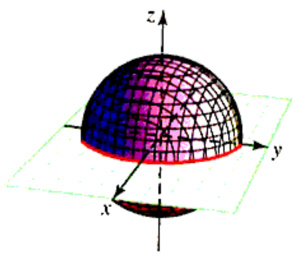
Level 1

1. ■ What does this say? What is a quadric surface?
2. ■ What does this say? What is the trace of a surface?

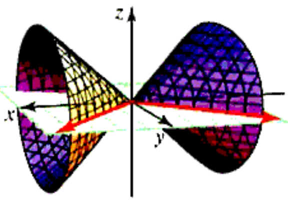
In Problems 3-14, match the equation with its graph (A-L).

3. $x^2 = z^2 + y^2$
4. $z^2 = \dfrac{x^2}{4} + \dfrac{y^2}{9}$
5. $x^2 + y^2 + z^2 = 9$
6. $y = x^2 + z^2$
7. $9z = x^2 + y^2$
8. $z = \dfrac{x^2}{16} - \dfrac{y^2}{4}$
9. $x = \dfrac{y^2}{25} + \dfrac{z^2}{16}$
10. $y = \dfrac{z^2}{4} - \dfrac{x^2}{9}$
11. $y^2 + z^2 - x^2 = -1$
12. $\dfrac{x^2}{9} + \dfrac{y^2}{16} - \dfrac{z^2}{4} = 1$
13. $\dfrac{x^2}{2} + \dfrac{y^2}{4} + \dfrac{z^2}{9} = 1$
14. $\dfrac{x^2}{4} - \dfrac{y^2}{9} + \dfrac{z^2}{9} = -1$

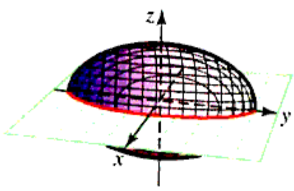
A.



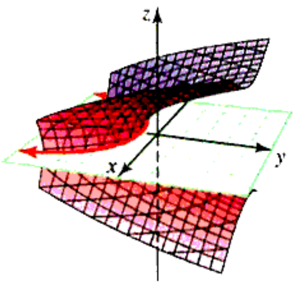
B.



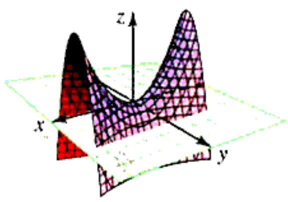
C.



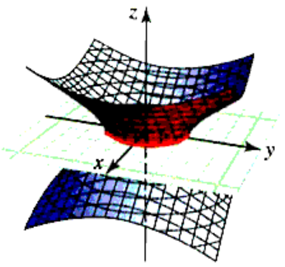
D.



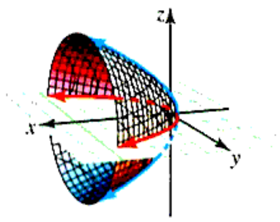
E.



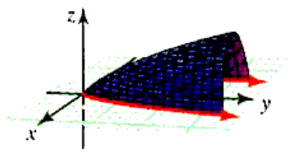
F.



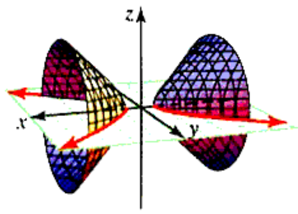
G.



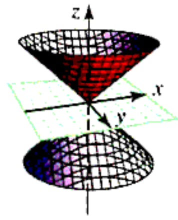
H.



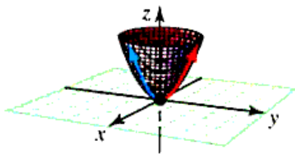
I.



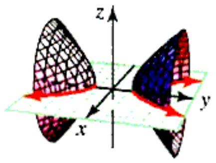
J.



K.



L.



In Problems 15-26, sketch the graph of each equation in $\mathbb{R}^3$.

15. $2x + y + 3z = 6$
16. $x = 4$
17. $x + 2y + 5z = 10$
18. $x + y + z = 1$
19. $3x + 2y + z = 6$
20. $y = x^2$
21. $z = x - 1$
22. $x^2 + y^2 = z$
23. $z = e^y$
24. $y = \ln x$
25. $2z + 5y^2 = 1$
26. $z = \sin y$

Sketch the graph of each surface given in Problems 27-30,

27. $y = \dfrac{1}{1 + x^2 + z^2}$
28. $x = \dfrac{1}{1 + y^2 + z^2}$
29. $x^2 + y^2 + z^2 = 1$
30. $y = x^2 + y^2$

In Problems 31-40, identify the quadric surface and describe the traces. Sketch the graph.

31. $9x^2 + 4y^2 + z^2 = 1$
32. $\dfrac{x^2}{4} + \dfrac{y^2}{9} - z^2 = 1$
33. $\dfrac{x^2}{4} + \dfrac{y^2}{9} - z^2 = -1$
34. $\dfrac{x^2}{9} - y^2 - z^2 = 1$
35. $z^2 = x^2 + \dfrac{y^2}{4}$
36. $z = x^2 + \dfrac{y^2}{4}$
37. $z = \dfrac{x^2}{9} - \dfrac{y^2}{16}$
38. $x^2 - 2y^2 = 9z$
39. $x^2 + 2y^2 = 9z^2$
40. $z^2 = 1 + \dfrac{x^2}{9} + \dfrac{y^2}{4}$

The equations given in Problems 41-44 represent quadric surfaces whose axis of symmetry is either the x-axis or the y-axis. Identify and sketch the surface.

41. $y - x^2 - 9z^2 = 0$ **42.** $x = \frac{y^2}{4} - \frac{z^2}{9}$

43. $x^2 = y^2 + 3z^2$ **44.** $y^2 - x^2 + z^2 = 1$

Level 2

Describe each of the translated quadric surfaces in Problems 45-50.

45. $\frac{(x-1)^2}{4} + (y+2)^2 + \frac{(z-3)^2}{9} = 1$

46. $z = (2x+1)^2 + \frac{(y-1)^2}{9} + 3$

47. $7x^2 + y^2 + 3z^2 - 9x + 4y + 7z = 1$

48. $z^2 = x^2 - y^2 + 2x + y - z + 5$

49. $y = x^2 - x + z^2 + z + 5$

50. $\frac{(x-1)^2}{2} - \frac{(y+1)^2}{4} - \frac{z^2}{9} = 1$

Describe the curve of intersection of the quadric surfaces in Problems 51 and 52.

51. $\frac{x^2}{9} + \frac{y^2}{4} + \frac{z^2}{5} = 1$ and $z^2 = \frac{x^2}{3} + \frac{y^2}{2}$

52. $z = \frac{y^2}{9} - \frac{x^2}{16}$ and $\frac{x^2}{16} + \frac{y^2}{2} - \frac{z^2}{3} = 1$

53. Use the method of cross-sections (Section 6.2) to show that the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

has volume $V = \frac{4}{3}\pi abc$. *Hint:* You may assume that the ellipse

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$$

has area πAB.

54. Use the method of cross-sections (Section 6.2) to find the volume bounded by the plane $z = 5$ and the paraboloid $9z = x^2 + y^2$.

55. What is the intersection of the surfaces $z = x^2 + y^2$ and $z = 1 - y^2$?

56. What is the intersection of the surfaces $z = x^2 + y^2$ and $z = y$?

Level 3

57. Rotational forces cause the earth to flatten at the poles so its shape is really an ellipsoid rather than a true sphere. Assume that the equatorial radius of the earth is 6,378.2 km and that its polar radius is 6,356.5 km.

a. Assuming that the earth is an ellipsoid with the general equation

$$\frac{x^2}{a^2} + \frac{y^2}{a^2} + \frac{z^2}{b^2} = 1$$

find a and b.

b. Use the formula in Problem 53 to find the volume of the earth.

58. The line

$$x = 1 - t,\ y = 2 + t,\ z = 3t$$

intersects the hyperboloid

$$\frac{z^2}{9} - \frac{x^2}{1} + \frac{y^2}{4} = 1$$

in two points P and Q. Find the distance between P and Q.

59. Find an equation for the surface S comprised of all points (x, y, z) the sum of whose distances from the points $A(1, 0, 0)$ and $B(-1, 0, 0)$ is 3. Is S a quadric surface?

60. Find an equation for the surface S comprised of all points (x, y, z) whose distance from the origin is the same as their distance from the plane $x + y + z = 5$. Is S a quadric surface?

CHAPTER 9 REVIEW

Time is said to have only ***one dimension****, and space to have* ***three dimensions****…*
The mathematical ***quaternion*** *partakes of* ***both*** *of these elements; in technical language it may be said to be "time plus space," or "space plus time"; and in the sense it has, or at least involves a reference to,* ***four dimensions****.*

W. R. Hamilton
Graves' *Life of Hamilton* (New York, 1882-1889)
Vol. 4, p. 635

Even though quaternions are not particularly important today, they preceded the modern concept of a vector. Our use of **i**, **j**, and **k** derives from Hamilton's quaternions.

Proficiency Examination

Concept Problems

1. **a.** What is a vector?
   b. What is a scalar?
   c. Give both an algebraic and a geometric interpretation of multiplication of a vector by a scalar.
   d. What is the parallelogram rule for vector sums?
2. State each of the following properties of vector operations:
   a. commutativity of vector addition
   b. associativity of vector addition
   c. identity for vector addition
   d. inverse property for vector addition
   e. magnitude of a vector for dot product
   f. commutativity for dot product
   g. multiple of a dot product
   h. distributivity for dot product over addition
   i. cross product of a zero vector
   j. anticommutativity for cross product
   k. distributivity for cross product over addition
3. **a.** Define cross product.
   b. Give a geometric interpretation of cross product.
   c. What is the formula for the magnitude of a cross product?
   d. If **u** and **v** are parallel vectors, what can be said about $\mathbf{u} \times \mathbf{v}$?
4. **a.** How do you find the length of a vector?
   b. What is a unit vector?
   c. How do you find a unit vector **u** in the direction of a given vector **v**?
   d. State the triangle inequality.
   e. What are the standard basis vectors in $\mathbb{R}^3$?
5. **a.** What is the standard-form equation of a sphere?
   b. What is a cylinder?
6. **a.** What is Lagrange's identity?
   b. What is the vector formula for work?
7. **a.** Define dot product.
   b. What are the direction cosines of a vector in $\mathbb{R}^3$?
8. **a.** What is the formula for the angle between two vectors?
   b. What is meant by orthogonal vectors?
   c. What is the algebraic condition for orthogonality?
9. **a.** What is a vector projection?
   b. Give a formula for finding the vector projection of **v** onto **w**.
   c. What is a scalar projection?
   d. Give a formula for finding the scalar projection of **v** onto **w**.
10. What is the right-hand rule for a coordinate system?
11. **a.** What is the determinant form for the scalar triple product?
    b. How do you find the volume of a parallelepiped?

12. a. What is the parametric form of a line in $\mathbb{R}^3$?
b. What is the symmetric form of a line in $\mathbb{R}^3$?
13. a. What is the normal vector of a plane in $\mathbb{R}^3$?
b. What is the point-normal form of a plane in $\mathbb{R}^3$?
c. What is the standard form of a plane in $\mathbb{R}^3$?
14. a. What is the distance between two points in $\mathbb{R}^3$?
b. What is the formula for the distance from a point to a plane?
c. What is the formula for the distance from a point to a line?
15. What is a quadric surface?

Practice Problems

16. Given $\mathbf{v} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$, $\mathbf{w} = 3\mathbf{i} - 2\mathbf{j}$. Find each of the following vectors or scalars.
a. $2\mathbf{v} + 3\mathbf{w}$ **b.** $\|\mathbf{v}\|^2 - \|\mathbf{w}\|^2$ **c.** $\mathbf{v} \cdot \mathbf{w}$ **d.** $\mathbf{v} \times \mathbf{w}$
e. vector projection of $\mathbf{v}$ onto $\mathbf{w}$ **f.** scalar projection of $\mathbf{w}$ onto $\mathbf{v}$
17. Given $\mathbf{u} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$, $\mathbf{v} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$, and $\mathbf{w} = 3\mathbf{i} + 5\mathbf{k}$. In each of the following cases, either perform the indicated computation or explain why it is not defined.
a. $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ **b.** $(\mathbf{u} \cdot \mathbf{v}) \times \mathbf{w}$ **c.** $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$ **d.** $(\mathbf{u} \cdot \mathbf{v}) \cdot \mathbf{w}$

Find the equations for the lines and planes in Problems 18-21.
18. the line through the points $P(-1, 4, -3)$ and $Q(0, -2, 1)$
19. the plane that contains the point $P(1, 1, 3)$ and is normal to the vector $\mathbf{v} = 2\mathbf{i} + 3\mathbf{k}$
20. the line of intersection of the planes $2x + 3y + z = 2$ and $y - 3z = 5$
21. the plane that contains the points $P(0, 2, -1)$, $Q(1, -3, 5)$ and $R(3, 0, -2)$
22. Find the direction cosines and the direction angles of the vector $\mathbf{u} = -2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$. Round to the nearest degree.
23. In each case, determine whether the lines intersect, are parallel, or are skew. If they intersect, find the point of intersection.
a. $x = 2t - 3$, $y = 4 - t$, $z = 2t$; and $\dfrac{x+2}{3} = \dfrac{y-3}{5}$; $z = 3$
b. $\dfrac{x-7}{5} = \dfrac{y-6}{4} = \dfrac{z-8}{5}$; and $\dfrac{x-8}{6} = \dfrac{y-6}{4} = \dfrac{z-9}{6}$
24. Let $\mathbf{u} = 2\mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{i} - \mathbf{j} - \mathbf{k}$, and $\mathbf{w} = 3\mathbf{i} + \mathbf{k}$. Find the volume of the parallelepiped determined by these vectors.
25. For the vectors in the previous problem, find a positive number A that guarantees that the tetrahedron determined by $A\mathbf{u}$, $A\mathbf{v}$, and $\mathbf{w}$ has volume that is twice the volume of the tetrahedron determined by $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$. *Hint:* See Problem 50, Section 9.4.
26. Find the following distances:
a. from $P(-1, 1, 4)$ to $2x + 5y - z = 3$ **b.** from $P(4, 5, 0)$ to $\dfrac{x-2}{3} = \dfrac{y}{5} = \dfrac{z+1}{-1}$
c. between the skew lines $x = t, y = 2t, z = 3t - 1$ and $x = 1 - t, y = t + 2, z = t$
27. An airplane flies at 200 mi/h parallel to the ground at an altitude of 10,000 ft. If the plane flies due south and the wind is blowing toward the northeast at 50 mi/h, what is the ground speed of the plane (that is, effective speed)?
28. Sketch the graph of $x = 2^t$, $y = 2^{t+1}$ for $t > 0$ by
a. using the parametric form **b.** eliminating the parameter
29. a. Graph the line $\dfrac{x}{4} = -\dfrac{y}{9}$
i. using rectangular form **ii.** by parameterizing
b. Graph the line $\dfrac{x}{4} = -\dfrac{y}{9} = \dfrac{z}{1}$
i. using rectangular form **ii.** by parameterizing
30. Identify the graph of each given equation.
a. $x^2 = z^2 + y^2$ **b.** $x^2 = z + y^2$ **c.** $\frac{x^2}{2} - \frac{y^2}{4} - \frac{z^2}{9} = 1$
d. $\frac{x^2}{2} + \frac{y^2}{4} - \frac{z^2}{9} = 1$ **e.** $\frac{x}{2} - \frac{y}{4} - \frac{z}{9} = 1$ **f.** $\frac{x}{2} = \frac{y}{4} = \frac{z}{9}$
g. $x^2 + y^2 + z^2 = 9$ **h.** $y = x^2 + z^2$ **i.** $y^2 + z^2 - x^2 = -1$

Supplementary Problems*

In Problems 1-4, a triangle in $\mathbb{R}^3$ has vertices $A(0, 2, -1)$, $B(1, 1, 3)$, and $C(1, 0, -4)$.

1. Find the perimeter of the triangle.
2. Find the area of the triangle.
3. Find the three vertex angles of the triangle. (Round to the nearest degree.)
4. Find a number p such that the points A, B, C and $D(p, p, 0)$ form a tetrahedron of volume $V = 100$ cubic units. *Hint:* See Problem 50, Section 9.4.

Find equations, in both parametric and symmetric forms, of the lines described in Problems 5-10. Find two additional points on each line.

5. passing through $A(4, -3, 2)$ and perpendicular to $2x - y + z = 4$
6. passing through $A(1, -2, 3)$, $B(4, -1, 2)$
7. passing through $P(1, 4, 0)$ with direction numbers $[2, 0, 1]$
8. passing through $P(2, 1, -3)$ orthogonal to the plane $5x - 2y + z = 1$
9. passing through $P(3, 4, -1)$ and parallel to the line of intersection of the planes $x + 2y + 2z + 5 = 0$ and $2x + y - 3z - 6 = 0$
10. passing through the point $P(0, 1, -1)$ and is parallel to the line of intersection of the planes $2x + y - 2z = 5$ and $3x - 6y - 2z = 7$

Find an equation for the plane satisfying the conditions given in Problems 11-18.

11. the xy-plane
12. parallel to the xz-plane passing through $(4, 3, 7)$
13. passing through $(1, -3, 4)$ with attitude numbers $[3, 4, -1]$
14. passing through $(-1, 4, 5)$ and orthogonal to a line with direction numbers $[4, 4, -3]$
15. passing through $(4, -3, 2)$ and parallel to the plane $5x - 2y + 3z - 10 = 0$
16. containing the lines $\dfrac{x-3}{4} = \dfrac{z-1}{2}; y = -2$ and $\dfrac{x-3}{3} = \dfrac{y+2}{1} = \dfrac{z-1}{-2}$
17. passing through $P(4, 1, 3)$, $Q(-4, 2, 1)$ and $R(1, 0, 2)$
18. passing through $(4, -1, 2)$ and parallel to the lines $\dfrac{x+2}{3} = \dfrac{y-2}{-1} = \dfrac{z+1}{2}$ and $\dfrac{x-2}{1} = \dfrac{y-3}{2} = \dfrac{z-4}{3}$

Given $\mathbf{v} = 4\mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $\mathbf{w} = 2\mathbf{i} + \mathbf{j} - 5\mathbf{k}$, *find the vectors or scalars in Problems 19-25.*

19. $\|\mathbf{v}\|$
20. $\mathbf{v} - \mathbf{w}$
21. $2\mathbf{v} + 3\mathbf{w}$
22. $5\mathbf{v} - 3\mathbf{w}$
23. $\|2\mathbf{v} - \mathbf{w}\|$
24. vector projection of $\mathbf{v}$ onto $\mathbf{w}$
25. scalar projection of $\mathbf{w}$ onto $\mathbf{v}$

Graph the curves, planes, or surfaces in $\mathbb{R}^3$defined by the equations in Problems 26-33.

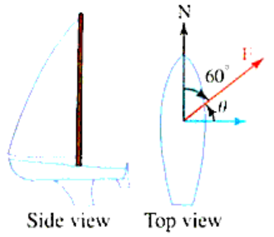
26. $x + 3y + 2z = 6$
27. $x + y = 3$
28. $x = 4$
29. $x = 3 - t$, $y = 4 + 2t$, $z = 6 - 3t$
30. $3(x - 2) - 2y + 3z = 0$
31. $z = \dfrac{1}{1 + x^2 + y^2}$
32. $x = 2\cos t$, $y = 2\sin t$, $0 \le t < 2\pi$
33. $x = t$, $y = t^3 + t^2$, $0 \le t \le 2$

Find the distance between the point and the line or the point and the plane given in Problems 34-37.

34. $P(1, 1, -1)$; $x - y + 2z = 4$
35. $P(2, 1, -2)$; $3x - 4y + z = -1$
36. $P(-4, 3, 2)$ to the plane containing $A(0, 0, 0)$, $B(1, 4, -1)$, and $C(2, -1, 3)$
37. $P(3, -2, 1)$; $\dfrac{x-3}{2} = \dfrac{y+1}{1} = \dfrac{z-2}{-1}$

*The supplementary probelms are presented in a somewhat random order, not necessarily in order of difficulty.

38. Suppose that the wind is blowing with a 1,000-lb magnitude force **F** in the direction of N60°E over a boat's sail. How much work does the wind perform in moving the boat in an easterly direction a distance of 50 ft? Give your answer in foot-pounds.



39. Given $\mathbf{u} = \mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{v} = 3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$, and $\mathbf{w} = \mathbf{i} + \mathbf{j} - \mathbf{k}$, find $(\mathbf{u} - \mathbf{v}) \cdot \mathbf{w}$ and $(2\mathbf{u} + \mathbf{v}) \times (\mathbf{u} - \mathbf{w})$.
40. If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal unit vectors, show that $(\mathbf{u} \times \mathbf{v}) \times \mathbf{u} = \mathbf{v}$. What is $(\mathbf{u} \times \mathbf{v}) \times \mathbf{v}$?
41. Show that three planes whose normals $\mathbf{u}, \mathbf{v}, \mathbf{w}$ satisfy $\mathbf{u} \times \mathbf{v} \cdot \mathbf{w} \neq 0$ intersect in exactly one point.
42. Show that $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$ if and only if $\mathbf{v} \times (\mathbf{w} \times \mathbf{u}) = \mathbf{0}$.
43. Find the direction cosines for $\mathbf{v} = (2\mathbf{i} + \mathbf{j}) \times (\mathbf{i} + \mathbf{j} - 3\mathbf{k})$.
44. Find the direction angles of the vector $\mathbf{u} = -2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$.
45. Find the (acute) angle, rounded to the nearest degree, between the intersecting lines

$$\frac{x-1}{3} = \frac{y-3}{-1} = \frac{z+5}{2} \quad \text{and} \quad \frac{x-1}{2} = \frac{y-3}{-1} = \frac{z+5}{-2}$$

46. Find the area of the parallelogram determined by $3\mathbf{i} - 4\mathbf{j}$ and $-\mathbf{i} - \mathbf{j} + \mathbf{k}$.
47. Find the center and the radius of the sphere

$$4x^2 + 4y^2 + 4z^2 + 12y - 4z + 1 = 0$$

48. Find the points of intersection of the line $x = 6 + 3t, y = 10 - 2t, z = 5t$ with each of the coordinate planes.
49. Find the point of intersection of the planes $3x - y + 4z = 15, 2x + y - 3z - 1 = 0$, and $x + 3y + 5z = 2$.
50. Find the equation of the plane determined by the intersecting lines

$$\frac{x+3}{3} = \frac{y}{-2} = \frac{z-7}{6} \quad \text{and} \quad \frac{x+6}{1} + \frac{y+5}{-3} = \frac{z-1}{2}$$

51. Find an equation for the set of all points that are equidistant from the planes $3x - 4y + 12z = 6$ and $4x + 3z = 7$.
52. Vertices B and C of $\triangle ABC$ lie along the line

$$\frac{x+2}{2} = \frac{y-1}{1} = \frac{z}{4}$$

Find the area of the triangle given that A has coordinates $(1, -1, 2)$ and line segment $\overline{BC}$ has length 5.
53. Find two unit vectors that are parallel to the line

$$\frac{x}{6} = \frac{y}{2} = \frac{z-1}{6}$$

54. A boy mowing the lawn wants to impress his dad by calculating the amount of work he is doing (see Figure 9.72).

Figure 9.72 Work mowing the lawn

The *displacement*, d, the distance the lawn mower is moved, is 2,450 ft and the angle in which the boy is pressing down on the handle is 40° with a force of 48.0 lb. What is the amount of work done by the boy?

55. A dad pulls his child in a wagon (see Figure 9.73) for 150 ft by exerting a constant force of 25 lb along the handle. How much work is done?

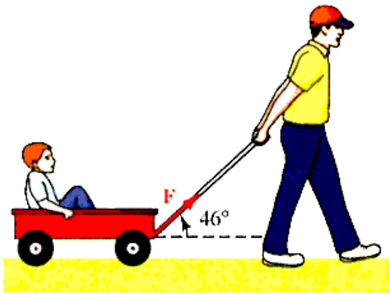


Figure 9.73 Work pulling a wagon

56. A girl pulls a sled 50 ft on level ground with a rope inclined at an angle of 30° with the horizontal (the ground). If she applies 3 lb of tension to the rope, how much work is performed on the sled? Give the exact answer.

57. Find the work done by the constant force $\mathbf{F} = 5\mathbf{i} + 4\mathbf{j} + \mathbf{k}$ in moving a particle along the line from $P(2,1,-1)$ to $Q(4,1,2)$.

58. How much work does it take to move a container 25 m along a horizontal loading platform onto a truck using a constant force of 100 newtons at an angle of $\frac{\pi}{6}$ from the horizontal?

59. Find a formula for the surface area of the tetrahedron determined by vectors **u**, **v**, and **w**. Assume the vectors do not all lie in the same plane.

60. Find the parametric and symmetric equations for the line passing through the point $(-1,1,6)$ perpendicular to $3x + y - 2z = 5$.

61. Use vectors to show that the sum of the squares of the lengths of the sides of a parallelogram equals the sum of the squares of the lengths of the diagonals.

62. Suppose **v** and **w** are nonzero vectors. Show that $\|\mathbf{v}\|\,\mathbf{w} + \|\mathbf{w}\|\,\mathbf{v}$ and $\|\mathbf{v}\|\,\mathbf{w} - \|\mathbf{w}\|\,\mathbf{v}$ are orthogonal vectors.

63. Let $\mathbf{v} = \cos\theta\mathbf{i} + \sin\theta\mathbf{j}$ and $\mathbf{w} = \cos\phi\mathbf{i} + \sin\phi\mathbf{j}$. Find $\mathbf{v} \times \mathbf{w}$. Interpret this cross product geometrically, and use it to derive a well-known trigonometric identity.

64. Find the three vertex angles (rounded to the nearest degree) of the triangle whose vertices are $A(1,-2,3)$, $B(-1,2,-3)$, and $C(2,1,-3)$.

65. Find a relationship between the numbers a_1, b_1, and c_1 so that the angle between the vectors $\mathbf{v} = a_1\mathbf{i} + b_1\mathbf{j} + c_1\mathbf{k}$ and $\mathbf{w}_1 = \mathbf{i} - 2\mathbf{j}$ is the same as the angle between **v** and $\mathbf{w}_2 = 2\mathbf{i} + \mathbf{k}$.

66. Find an equation of the plane that passes through $(a,0,0)$, $(0,a,0)$, and $(0,0,a)$.

67. Find the area of the triangle with vertices $A(0,-1,2)$, $B(1,2,-1)$, and $C(3,-1,2)$.

68. Find an equation for the surface comprised of all points equidistant from $D(0,0,6)$ and the xy-plane. Is this a quadric surface?

69. Find the area of the triangle determined by the vectors $\mathbf{v} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{w} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

70. Find an equation for the plane that passes through the origin and is parallel to the vectors $\mathbf{v} = \mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$ and $\mathbf{w} = -\mathbf{i} + \mathbf{j} + 2\mathbf{k}$.

71. Find an equation for the plane that passes through the origin and whose normal vector is parallel to the line of intersection of the planes $2x - y + z = 4$ and $x + 3y - z = 2$.

72. Find a number A such that the angle between the planes $2Ax + 3y + z = 1$ and $x - Ay + 3z = 5$ is $\pi/2$.

73. The lines L_1 and L_2 are parallel to the vectors $\mathbf{v}_1 = \mathbf{i} - \mathbf{j}$ and $\mathbf{v}_2 = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$, respectively. Find an equation for the line L that passes through the point $(-1,2,0)$ and is orthogonal to both L_1 and L_2.

74. A parallelepiped is determined by the vectors $\mathbf{u} = \mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, and $\mathbf{w} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$. Find the altitude from the tip of **w** to the side determined by **u** and **v**.

75. In the next chapter, we will show that $\mathbf{T} = \mathbf{i} + 2x\mathbf{j}$ is a vector in the direction of the tangent line at each point $P(x,x^2)$ on the parabola $y = x^2$. Find a unit vector normal to the parabola at the point $(3,9)$.

76. For nonzero vectors **u**, **v**, and **w**, show that the vector

$$(\mathbf{u} \times \mathbf{v}) \times (\mathbf{u} \times \mathbf{w})$$

is parallel to **u**.

77. Let $\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ and $\mathbf{w} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$, where a, b, c, A, B, and C are constants. Describe the set of vectors $\mathbf{v} + t\mathbf{w}$, where t is any scalar.

78. A 40-lb child sits on a seesaw, 3 ft from the fulcrum, as shown in Figure 9.74. What torque is exerted when the child is 2 ft above the horizontal?

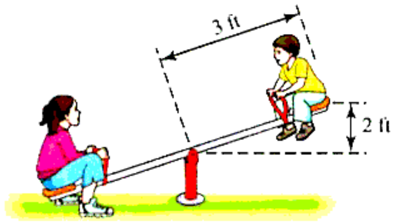


Figure 9.74 Seesaw torque

79. Let $\mathbf{u}_0 = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ and let $\mathbf{u} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Describe the set of points in $\mathbb{R}^3$ defined by

$$\|\mathbf{u}_0 - \mathbf{u}\| < r$$

where $r > 0$ and a, b, c, are constants.

80. The vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are said to be *linearly independent* in $\mathbb{R}^3$ if the only solution to the equation $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = \mathbf{0}$ is $a = b = c = 0$. Otherwise, the are *linearly dependent*. The simplest example of a set of linearly independent vectors is $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$. Determine whether the vectors $\mathbf{u} = -\mathbf{i} + 2\mathbf{k}$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{w} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ are linearly independent or dependent.

81. Figure 9.75 shows a parallelogram $ABCD$. In Problem 82, we show that if M is the midpoint of side $\overline{AB}$, then the line $\overline{CM}$ intersects diagonal $\overline{BD}$ at a point P located one-third of the distance from B to D. In order to complete this, we must first show the following:

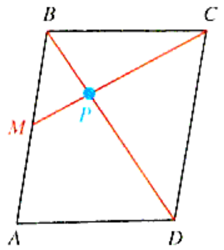


Figure 9.75 Parallelogram *ABCD*

Let a and b be scalars such that $\mathbf{MP} = a\mathbf{MC}$ and $\mathbf{BP} = b\mathbf{BD}$. Show that

$$\frac{1}{2}\mathbf{AB} + b(\mathbf{AD} - \mathbf{AB}) = a(\mathbf{AD} - \frac{1}{2}\mathbf{AB})$$

82. Use the fact that $\mathbf{AB}$ and $\mathbf{AD}$ in Figure 9.75 are linearly independent along with the results of Problem 81 to show that

$$\frac{1}{2} - b - \frac{1}{2}a = 0 \text{ and } a - b = 0$$

Solve this system of equations to show that P has the required location.

83. Show that $\mathbf{u} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{v} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ and $\mathbf{w} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$ are linearly dependent (see Problem 80), if and only if

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

84. Show that

$$\begin{vmatrix} \mathbf{u}_1 \cdot \mathbf{v}_1 & \mathbf{u}_1 \cdot \mathbf{v}_2 \\ \mathbf{u}_2 \cdot \mathbf{v}_1 & \mathbf{u}_2 \cdot \mathbf{v}_2 \end{vmatrix} = (\mathbf{u}_1 \times \mathbf{u}_2) \cdot (\mathbf{v}_1 \times \mathbf{v}_2)$$

85. Let $A(-2,3,7), B(1,5,-3)$, and $C(2,8,-1)$ be the vertices of a triangle in $\mathbb{R}^3$. What are the coordinates of the point M where the medians of the triangle meet (the centroid)?

86. The medians of a triangle meet at a point (the centroid) located two-thirds of the distance from each vertex to the midpoint of the opposite side. Generalize this result by showing that the four lines that join each vertex of a tetrahedron to the centroid of the opposite face meet at a point located three-fourths of the distance from the vertex to the centroid.

87. A triangle in $\mathbb{R}^3$ is determined by the vectors $\mathbf{v}$ and $\mathbf{w}$ as shown in Figure 9.76.

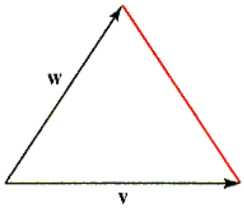


Figure 9.76 Area of a triangle

Show that the triangle has area

$$A = \frac{1}{2}\sqrt{\|\mathbf{v}\|^2 \|\mathbf{w}\|^2 - (\mathbf{v} \cdot \mathbf{w})^2}$$

88. Think Tank Problem

a. If $\mathbf{u} \times \mathbf{w} = \mathbf{v} \times \mathbf{w}$, does it follow that $\mathbf{u} = \mathbf{v}$? **b.** If $\mathbf{u} \cdot \mathbf{w} = \mathbf{v} \cdot \mathbf{w}$, does it follow that $\mathbf{u} = \mathbf{v}$?

Think Tank Problems *In Figure 9.77, $\triangle ABC$ is equilateral and the points M, N, and O are located so that*

$$\mathbf{AM} = \frac{1}{3}\mathbf{AB} \quad \mathbf{BN} = \frac{1}{3}\mathbf{BC} \quad \mathbf{CO} = \frac{1}{3}\mathbf{CA}$$

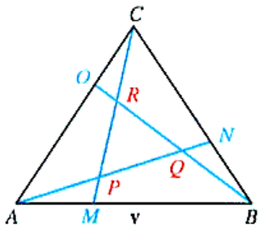


Figure 9.77 Problems 89-91

It can be shown that $\triangle PQR$ is also equilateral, and

$$\|\boldsymbol{PM}\| = \|\boldsymbol{QN}\| = \|\boldsymbol{RO}\| \text{ and } \|\boldsymbol{AP}\| = \|\boldsymbol{BQ}\| = \|\boldsymbol{CR}\|$$

Use this information in Problems 89-91.

89. Show that $\mathbf{PQ} = \frac{3}{7}\mathbf{AN}$, then show that $\|\mathbf{AN}\|^2 = \frac{7}{9}\|\mathbf{AB}\|^2$. *Hint:* Use the law of cosines.

90. Show that $\triangle PQR$ has area $\frac{1}{7}$ that of $\triangle ABC$.

91. Do you think the results from Problems 89 and 90 would hold if $\triangle ABC$ were not equilateral? Investigate your conjecture.

Think Tank Problems The Gram-Schmidt orthogonalization process *In Problems 92-93, let* **u**, **v**, *and* **w** *be nonzero vectors in* $\mathbb{R}^3$ *that do not lie on the same plane. Define vectors* $\boldsymbol{\alpha}$ *and* $\boldsymbol{\beta}$ *as follows:*

$$\boldsymbol{\alpha} = \mathbf{v} - \left[\frac{\mathbf{v}\cdot\mathbf{u}}{\|\mathbf{u}\|^2}\right]\mathbf{u} \quad \text{and } \boldsymbol{\beta} = \mathbf{w} - \left[\frac{\mathbf{w}\cdot\mathbf{u}}{\|\mathbf{u}\|^2}\right]\mathbf{u} - \left[\frac{\mathbf{w}\cdot\boldsymbol{\alpha}}{\|\boldsymbol{\alpha}\|^2}\right]\boldsymbol{\alpha}$$

92. Show that $\mathbf{u}$, $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ are mutually orthogonal (any pair is orthogonal).

93. If $\boldsymbol{\gamma}$ is any vector in $\mathbb{R}^3$, show that

$$\boldsymbol{\gamma} = \left[\frac{\boldsymbol{\gamma}\cdot\mathbf{u}}{\|\mathbf{u}\|^2}\right]\mathbf{u} + \left[\frac{\boldsymbol{\gamma}\cdot\boldsymbol{\alpha}}{\|\boldsymbol{\alpha}\|^2}\right]\boldsymbol{\alpha} + \left[\frac{\boldsymbol{\gamma}\cdot\boldsymbol{\beta}}{\|\boldsymbol{\beta}\|^2}\right]\boldsymbol{\beta}$$

94. Suppose $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are nonzero vectors in $\mathbb{R}^3$ with $\mathbf{u}\times\mathbf{v} = \mathbf{w}$ and $\mathbf{u}\cdot\mathbf{v} = 0$. Show that $\mathbf{v} = s(\mathbf{w}\times\mathbf{u})$ and $\mathbf{u} = t(\mathbf{v}\times\mathbf{w})$ for scalars s and t.

95. **Think Tank Problem** In Figure 9.78, $ABCD$ is a rectangle, with M the midpoint of side $\overline{CD}$ and $\mathbf{AR}_k = \dfrac{1}{2k+1}\mathbf{AB}$. If P is the intersection of $\overline{AM}$ and $\overline{CR_k}$, and R_{k+1} is the foot of the perpendicular drawn from P to $\overline{AB}$, show that

$$\mathbf{AR}_{k+1} = \frac{1}{2k+3}\mathbf{AB}$$

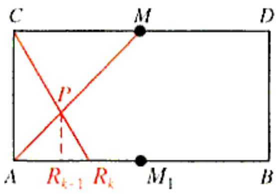


Figure 9.78 Problem 95

96. **Think Tank Problem** It is easy to subdivide a line segment line $\overline{AB}$ into halves, or fourths, or eighths, etc. However, dividing it into thirds or fifths, $\cdots$ is more difficult. Use the result obtained in Problem 95 to describe a procedure for subdividing $\overline{AB}$ into an odd number of equal parts.

97. **Putnam Examination Problem** Find the equations of two straight lines, each of which cuts all four of the following lines:

$$L_1\colon x = 1, y = 0 \qquad L_2\colon y = 1, z = 0 \qquad L_3\colon z = 1, x = 0 \qquad L_4\colon x = y = -6z$$

98. **Putnam Examination Problem** Find the equation of the smallest sphere that is tangent to both the lines

$$L_1\colon x = t+1,\ y = 2t+4,\ z = -3t+5 \qquad L_2\colon x = 4t-12, y = t+8, z = t+17$$

99. **Putnam Examination Problem** The hands of an accurate clock have lengths 3 cm and 4 cm. Find the distance between the tips of the hands when the distance is increasing most rapidly.

Chapter 9 Group Research Project*

Working in small groups is typical of most work environments, and this book seeks to develop skills with group activities. We present a group project at the end of each chapter. These projects are to be done in groups of three or four students.

Star Trek Quest

The starship *Enterprise* has been captured by the evil Romulans and is being held in orbit by a Romulan tractor beam. The orbit is elliptical with the planet Romulus at one focus of the ellipse. Repeated efforts to escape have been futile and have almost exhausted the fuel supplies. Morale is low and food reserves are dwindling.

In searching the ship's log, Lieutenant Commander Data discovers that the Enterprise had been captured long ago by a Romulan tractor beam and had escaped. The key to that escape was to fire the ship's thrusters at exactly the right position in the orbit. Captain Picard gives the command to feed the required information into the computer to find that position. But, alas, a Romulan virus has rendered the computer useless for this task. Everyone turns to you and asks for your help in solving the problem.

Here is what Data discovered. If F represents the focus of the ellipse and P is the position of the ship on the ellipse, then the vector **FP** can be written as a sum $\mathbf{T}+\mathbf{N}$ where **T** is tangent to the ellipse and **N** is normal to the ellipse (not necessarily unit vectors). The thrusters must be fired when the ratio $\|\mathbf{T}\|/\|\mathbf{N}\|$ is equal to the eccentricity of the ellipse.

Extended paper for further study. Your mission is to save the starship from the evil Romulans. Write a paper describing your plan.

*Adapted from *MAA Notes*: 17 (1991) "Priming the Calculus Pump: Innovations and Resources," Marcus S. Cohen, Edward D. Gaughan, R. Arthur Knoebel, Douglas S. Kurtz, and David J. Pengelley.